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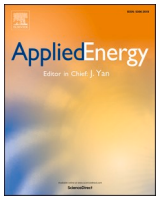
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Analytical dispatch strategies for pumped storage hydro: A conditional dynamic programming approach to discontinuous multi-period optimization problems

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HIGHLIGHTS

- Developed two PSH economic dispatch models with non-zero lower bounds.
- Applied dynamic programming to derive optimal multi-period PSH dispatch policies.
- Used auxiliary variables to address discontinuities in reward and value functions.
- Non-zero limits increase idle probability, reducing PSH profitability.
- Validated model with synthetic and real data, demonstrating practical applications.

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ABSTRACT

The increasing integration of renewable energy sources like wind and solar poses significant challenges to secure and stable grid operation. Energy storage systems, particularly pumped storage hydro (PSH), play a crucial role in balancing power supply and demand. Traditional analytical studies of PSH economic dispatch problems often assume zero lower bounds for generating and pumping rates to simplify analysis and derive analytical solutions for multi-period optimization problems. However, the inherent mechanical design constraints of PSH require non-zero minimum flow rates for efficient operation. We analyze two scenarios, merchants having PSH only and merchants having both PSH and wind farms. In the PSH-only scenario, four analytically determined State of Charge (SOC) reference points divide the SOC range into five sub-regions, each corresponding to a specific optimal decision. In the co-optimized PSH and wind farms scenario, five SOC reference points split the SOC range into six sub-regions. By simply comparing the current SOC with these reference points in the next period, users can derive unique, analytically optimal economic dispatch decisions for each period. Our findings indicate that non-zero minimum capacity constraints significantly influence the optimal dispatch strategy, increasing the likelihood of PSH systems remaining idle and potentially reducing arbitrage profits. We validate our proposed approach using both synthesized data and real data from the Midcontinent Independent System Operator, demonstrating its practical applicability. This methodology represents a significant advancement in energy storage dispatch strategies by providing scalable, analytically optimal solutions for multi-period optimization problems, enhancing economic efficiency in the management of energy storage systems.

1. Introduction

Amid growing concerns about global warming and the environmental impact of thermal fossil-fired power generation [1], advancements in energy storage technology have enhanced the viability of increased penetration of renewable energy sources like wind and photovoltaic power, which have seen substantial growth over the past several decades. However, renewable energy generation depends on external environmental conditions, making it intermittent, volatile, and

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Nomenclature

Abbreviations

PSH	Pumped Storage Hydro
SOC	State of Charge
MISO	Midcontinent Independent System Operator

Parameters and sets

$\underline{Q^g}$	The minimum PSH's generating capacity
$\underline{Q^p}$	The minimum PSH's pumping capacity
$\overline{E}$	Maximum energy capacity
$\underline{E}$	Minimum energy capacity
$\underline{Q^p}$	Minimum charging constraint
$\overline{Q^p}$	Maximum charging constraint
$\overline{Q^g}$	The upper limit for discharging
$\underline{Q^g}$	The lower limit for discharging
α	The efficiency of the pumping/charging mode, $\alpha \in (0,1]$
β	The efficiency of the generating/discharging mode, $\beta \in (0,1]$
ρ	The transmission line efficiency, $\rho \in (0,1]$
η_t	Accounts for the energy loss due to factors like evaporation and leakage, $\eta_t \in (0,1]$
P_t	The electricity price at time t
c	The unit operating cost for PSH generating or pumping
$S(t)$	The state at period t
π^*	The optimal decision rule
$V_t(S(t))$	The value function in period t
c_w	The unit wind power generating cost
T	Total numbers of periods

Decision variables

q_t^p	Quantity of pumping/charging at time t
q_t^g	Quantity of generating/discharging at time t
E_t	State of charge (SOC) level in the PSH at the beginning of period t
U_t^p	Unit commitment to power pumping in period t $[t, t+1)$
U_t^g	Unit commitment to power generating in period t $[t, t+1)$
α_t	Action for each period t
w_t	Wind power generation at time t

unpredictable [2,3]. Because the power grid requires a real-time balance between supply and demand, the high uncertainty of renewable energy sources poses challenges to power system operations [4].

Energy storage systems are pivotal in the electricity market, managing power surpluses by storing excess electricity and mitigating deficits by releasing stored energy. With advancements in energy storage technology, various systems have been developed, including thermal energy storage [5,6], pumped storage hydropower (PSH) plants [7,8], chemical energy storage [9,10], and battery storage [11,12]. Among these, PSH stands out due to its significant capacity, accounting for over 96 % of installed grid storage capacity in the United States [13] and 94 % of energy storage capacity in the world [14,15], which highlights its critical role in energy management and grid stability.

This paper focuses on PSH systems, which are subject to physical constraints, including storage capacity, charging/pumping and discharging/generating efficiencies, and the capacities for pumping/charging and generating/discharging in each period. Based on forecasted electricity prices, the PSH owner makes economic dispatch decisions—whether to buy or sell power—and determines the quantity of electricity to dispatch in each period in the day-ahead wholesale electricity market.

A variety of methodologies have been employed in the domain of optimizing energy storage economic dispatch strategies, including dynamic programming [16,17], convex optimization [18,19], heuristic algorithms [20,21], stochastic optimization [22,23], and game theory [24,25]. In most of these methods, numerical simulation is carried out to obtain numerical solutions [24,25]. Unlike these methods, using dynamic programming allows us to obtain an analytical solution to the multiple-period energy storage economic dispatch optimization problem. In the analytical solution, we can derive the mathematical relationship between the decision variables and the parameters via closed-form functions, which deliver not only the optimal outcomes but also the logic and reasons behind decisions based on the current state. Analytical solutions allow for rapid parameter adjustments without complete recalculation, significantly reducing computational burdens. This makes it easier for asset owners or operators to optimize their dispatch strategies effectively. While it is possible to obtain optimal solutions using traditional numerical methods, such as modern solvers like Gurobi, even when changing certain parameters, decision-makers cannot gain direct insights into how these parameter changes affect the optimal solution and often have to resort to numerical sensitivity studies.

In this paper, we specifically adopt the dynamic programming approach to delve into the optimization of energy storage dispatch strategies. However, achieving analytical closed-form optimal results for multi-period energy storage economic dispatch optimization requires continuous reward and value functions [17,26–29]. To meet this requirement and achieve globally optimal solutions, researchers typically assume zero lower limits for storage or PSH discharging/generating and charging/pumping capacities in each period [17,26,28–30]. This assumption facilitates the differentiation of both the reward and value functions and simplifies the analytical solution procedure. While suitable for battery energy storage systems, this approach is inadequate for PSH systems, which face inherent operational and mechanical design constraints that impose non-zero lower boundaries for charging/pumping and discharging/generating [31,32]. These limits, dictated by the physical requirements of the hydro turbines and pumps, necessitate maintaining a minimum flow rate to operate efficiently and prevent mechanical wear or damage. Additionally, energy conversion efficiency in PSH systems significantly drops at lower operation levels, making very low or near-zero flow rates economically and technically impractical. These operational characteristics compel the inclusion of non-zero pumping and generating minimum capacity limits in the optimization models for PSH, distinctly differentiating these systems from typical battery storage operations [6,33].

Traditional research adopted a zero lower bound assumption, studied continuous problems, and allowed merchants to dispatch power at any scale, providing high flexibility. In contrast, this paper adopts a non-zero lower bound constraint and studies discontinuous problems, where, once the merchants decide to charge or discharge, there is a minimum limit, which undoubtedly reduces the flexibility of decision-making. Meanwhile, the discontinuity of the reward functions, value functions, and range of decision variables makes it difficult for us to obtain an analytical solution directly through dynamic programming. For merchants who only have PSH, considering the discontinuity makes the research more realistic and more valuable, while adjusting the lower bound constraint can make this framework applicable to PSH facilities of different scales. Motivated by these challenges, this paper aims to achieve analytically closed-form optimal solutions for the discontinuous multiple-periods energy storage economic dispatch problems, which guide the merchants to make the best decisions and also understand how the parameters impact the results. This study not only introduces a novel

solution framework, but also enhances understanding of the implications of non-zero lower bounds on optimal dispatch strategies, enabling practitioners to make more informed decisions in managing energy resources.

Given the substantial market share and operational peculiarities of PSH systems [13], this paper aims to optimize economic dispatch strategies specifically tailored for PSH. A critical focus is placed on addressing the unique challenges posed by non-zero lower limit constraints, which are not typically encountered by traditional battery storage systems. The imposition of non-zero minimum generating and pumping capacity constraints introduces discontinuities in both the reward and value functions, complicating the application of conventional methods [16,17] for deriving analytical solutions. To tackle this issue, we propose a novel approach that effectively addresses the discontinuous multi-period optimization problems caused by these non-zero lower bound constraints. Theoretically, our innovation lies in the introduction of a new set of auxiliary decision variables and functions that redefine the treatment of constraints in optimizing PSH systems. This innovative modeling technique involves the construction of continuous auxiliary functions—specifically redesigned reward and value functions that integrate seamlessly into traditional dynamic programming frameworks.

We employ continuous auxiliary functions to circumvent challenges arising from discontinuities in reward and value functions due to non-zero lower-limit constraints. This allows us to achieve global optimal solutions for the auxiliary functions using standard dynamic programming techniques. Moreover, we can map these solutions back to the original discontinuous problem, ensuring that optimal solutions are obtained, which would otherwise be unattainable with traditional approaches. In essence, this study introduces a transformative framework that utilizes auxiliary decision variables to convert the traditional discontinuous multi-period economic dispatch optimization problem into a continuous format. This advancement not only enhances the tractability of dynamic programming applications in PSH systems but also significantly contributes to the broader understanding and optimization of economic dispatch strategies in the context of renewable energy integration.

Co-optimization of renewable power plants and energy storage [26,29,34] can generate potential future value by mitigating the intermittent nature of renewable energy generation through storing electricity when there is surplus renewable power and releasing it when the grid needs power at profitable prices. The benefits of co-optimization of storage and wind power generation have been reported by Braff et al. (2016) [35]. One example of such a system is the Wilmot Energy Center, which includes a 100-MW solar array and a 30-MW battery energy storage system [36]. Another system is Consumers Energy and DTE Energy, which features a Ludington PSH and a 100-MW Lake Winds Energy Park system [37]. In this paper, we also explore the integration of PSH facilities with renewable energy sources, especially wind power. PSH, as an energy storage facility, can mitigate the volatility of renewable energy sources [38], and thus co-optimization of PSH with wind power can reduce the cost of operation [39], obtain higher revenues [40], ensure the stability of the grid [30], and reduce carbon emissions [41]. For the joint optimization problem of PSH and wind power plants, the introduction of discontinuity would fundamentally change the merchant's actions.

To capture a broad application, we analyze two scenarios. In the first scenario, similar to previous research [28,29,42,43], the storage owner has only a PSH system with an upper reservoir and an effectively unlimited lower reservoir (e.g., a large lake or river). The merchant has three actions: (a) purchase power to store, (b) remain idle, or (c) discharge the stored power and sell it back to the grid. In the second scenario, we investigate a storage owner with a wind farm and a PSH [28,30]. Similar to Liu et al. (2022) [30] and Zhou et al. (2019) [28], a co-optimization electricity merchant who operates both a wind farm and a PSH can take four actions: (a) using wind power and/or purchased electricity to pump water to the upper reservoir, (b) using partial wind power to pump water and selling the remaining wind power to the market, (c) releasing water from the upper reservoir and selling all wind power to the market, and (d) keeping the system idle. To address this, we first develop a PSH economic dispatch optimization model that incorporates these physical constraints and then design auxiliary decision variables, reward functions, and value functions to transform the discontinuous multi-period problem into a continuous one. We employ dynamic programming to solve this refined problem and derive an analytical solution. This method enables us to explore optimal economic dispatch decision-making by electricity merchants or PSH owners based on current state of charge (SOC) in the storage, forecasted electricity prices, and minimum generating and pumping capacity limits at each period to maximize expected profits.

This paper addresses two pivotal research questions: (1) How do the model discontinuities caused by non-zero lower limit constraints (i.e., discontinuous reward functions and value functions) on generating and pumping capacities at each period influence the economic dispatch strategies and profitability of merchants? (2) What differences emerge between dispatch strategies that consider non-zero lower limits on generating and pumping capacities and those that adhere to traditional approaches with zero lower limits on charging and discharging capacities? Therefore, the primary theoretical innovation of this study centers on this methodological advancement: transforming a discontinuous multi-period optimization problem into a tractable continuous form through auxiliary decision variables. This breakthrough facilitates the application of well-established optimization techniques and enhances the practical implementation of these strategies in managing PSH operations effectively.

Our study makes three principal contributions: first, we develop a novel approach that redefines decision variables and their domains by using auxiliary decision variables and functions. This approach transforms the discontinuous multi-period economic dispatch optimization problem into a continuous multi-period optimization problem, simplifying the attainment of optimal solutions via conditional dynamic programming and the corresponding Bellman Equation. Subsequently, by mapping these auxiliary variables and functions back to the original problem, we derive analytically optimal solutions for the discontinuous multi-period optimization problem.

Second, for an electricity merchant with storage only, the optimal trading strategy at each period is determined by four reference points $E^{p^*}_{t+1} - \underline{Q^p}$, $E^{p^{**}}_{t+1}$, $E^{g^{**}}_{t+1}$, and $E^{g^*}_{t+1} + \underline{Q^g}$, which have a unique ordered relationship $E^{p^*}_{t+1} - \underline{Q^p} \le E^{p^{**}}_{t+1} \le E^{g^{**}}_{t+1} \le E^{g^*}_{t+1} + \underline{Q^g}$ for any positive forecasted electricity prices. The feasible SOC of PSH is divided into five subregions by these four reference points. If the current SOC is low enough and below the first reference point (i.e., $E_t < E^{p^*}_{t+1} - \underline{Q^p}$), the merchant should purchase power from the market and increase the SOC to the first reference point as closely as possible. In contrast, if the current SOC is high enough and above the fourth reference point (i.e., $E_t > E^{g^*}_{t+1} + \underline{Q^g}$), the merchant should sell power to the market and reduce the SOC to the fourth reference point as closely as possible. However, when the stored energy inventory falls between these two reference points (i.e. $E^{p^*}_{t+1} - \underline{Q^p} \le E_t \le E^{g^*}_{t+1} + \underline{Q^g}$), the merchant has three options: if the stored energy is less than the second reference point (i.e., $E^{p^*}_{t+1} - \underline{Q^p} \le E_t < E^{p^{**}}_{t+1}$), the merchant should buy minimum required power to pump/charge; if the current SOC exceeds the third reference point (i.e., $E^{g^{**}}_{t+1} < E_t \le E^{g^*}_{t+1} + \underline{Q^g}$), the merchant should sell minimum required power to reduce SOC; and if the current SOC is between both the second and third reference points, the merchant should remain idle, neither buying nor selling power. This approach significantly advances traditional studies by providing precise and economically efficient operation strategies that account for realistic operational constraints.

Third, in the context of managing both PSH and wind power plants,

merchants follow a sophisticated strategy involving five reference points for SOC: $E_{t+1}^{(1)^*} - \alpha w_t$, $E_{t+1}^{(2)^*} - \underline{Q^p}$, $E_{t+1}^{(2)^{**}}$, $E_{t+1}^{(3)^{**}}$, and $E_{t+1}^{(3)^*} + \underline{Q^g}$. The SOC is segmented into six distinct subregions by these five reference points, which dictate the merchant's actions to optimize expected profits. For instance, when the SOC is below the first reference point (i.e., $E_{t+1}^{(1)^*} - \alpha w_t$), the strategy is to store all available wind power and purchase additional power if necessary to increase the SOC up to that point. Between the first and second (i.e., $E_{t+1}^{(2)^*} - \underline{Q^p}$) reference points, the merchant stores part of the wind power and sells the rest to reach the second reference point. If the SOC falls between the second and third (i.e., $E_{t+1}^{(2)^{**}}$) reference points, the merchant should buy the minimum required power $\underline{Q^p}$. Crucially, if the SOC falls between the third and fourth (i.e., $E_{t+1}^{(3)^{**}}$) reference points, the merchant should remain idle, neither buying nor selling power, to manage the energy balance efficiently. If the SOC lies between the fourth and fifth (i.e., $E_{t+1}^{(3)^*} + \underline{Q^g}$) reference points, the strategy shifts to selling the minimum required power $\underline{Q^g}$. If the SOC exceeds the fifth reference point, the merchant should sell all available wind power and release stored power to decrease the SOC to that point as closely as possible. By understanding and implementing this new economic dispatch policy, merchants can effectively navigate the complexities of integrating renewable energy sources with PSH or storage, enhancing the profitability of their energy management systems.

To visually summarize these strategies and highlight the distinctions between our approach and existing studies, Fig. 1 illustrates the decision-making processes for merchants under the two scenarios.

Fig. 1 illustrates the decision-making processes for merchants under two scenarios: PSH-only optimization and PSH-Wind co-optimization, both influenced by non-zero economic dispatch minimum constraints. The decisions highlighted in grey represent additional actions necessitated by the discontinuities in reward functions. These greyed-out decisions are guided by new reference points, labeled as $E_{t+1}^{p^{**}}$ and $E_{t+1}^{g^{**}}$, which are specific to the algorithm introduced in this paper. This differentiation in decision-making is a significant distinction between our approach and existing studies.

The rest of the paper is organized as follows: Section 2 reviews the literature related to energy storage modeling and energy storage economic dispatch. Section 3 develops the economic dispatch for the scenario where electricity merchants with storage only consider non-zero minimum generating and pumping capacity constraints and presents the optimal solution. In Section 4, we study the economic dispatch model for the electricity merchant with both PSH and wind farms. Section 5 validates the proposed approach in detail using a case study with three periods of synthesized data. Section 6 uses real-world data from the Midcontinent Independent System Operator (MISO) to conduct numerical simulations over a 336-period (two-week) case, showcasing the results of the proposed approach. Finally, we summarize our findings and recommend future research in Section 7.

2. Literature review

Our work focuses on two aspects of energy storage: modeling and dispatch optimization. We will provide a detailed introduction to each of these aspects in the following subsections.

2.1. Energy storage modeling

Several studies have employed stochastic optimization and artificial intelligence (AI)-based methods to enhance energy storage operations [40–43], but they often overlook the non-zero lower bound constraints that are critical in PSH systems. For example, Cao et al. (2020) [44] explored wind power producers' bidding strategies in energy and reserve markets, employing an asynchronous advantage actor-critic method and neural networks to significantly mitigate profit losses through optimized bidding strategies. Toubeau et al. (2021) [45] proposed a hybrid AI-based multi-stage method to solve the complex unit commitment problem in power systems integrating thermal, wind, and solar power, demonstrating both robustness and cost-effectiveness. Similarly, Martinez-Rico et al. (2021) [46] examined a hybrid renewable system combining wind, solar, and battery storage, proposing a multi-objective cost function optimized via particle swarm optimization, to enhance both battery longevity and system efficiency. Favaro et al. (2024) [47] proposed a multi-fidelity method combining mixed-integer linear programming and an agent-based optimization algorithm to solve the low-fidelity problem of traditional MILP. The proposed method was proven to be more efficient through numerical simulations. Ochoa et al. (2022) [48] developed efficient bidding strategies for photovoltaic plants with storage in multi-timescale electricity markets, employing a novel multi-intelligent deep reinforcement learning framework, which outperformed traditional methods and yielded higher profits.

Liu et al. (2023) [27] analyzed optimal scheduling strategies for electricity merchants operating both PSH and wind farms, considering

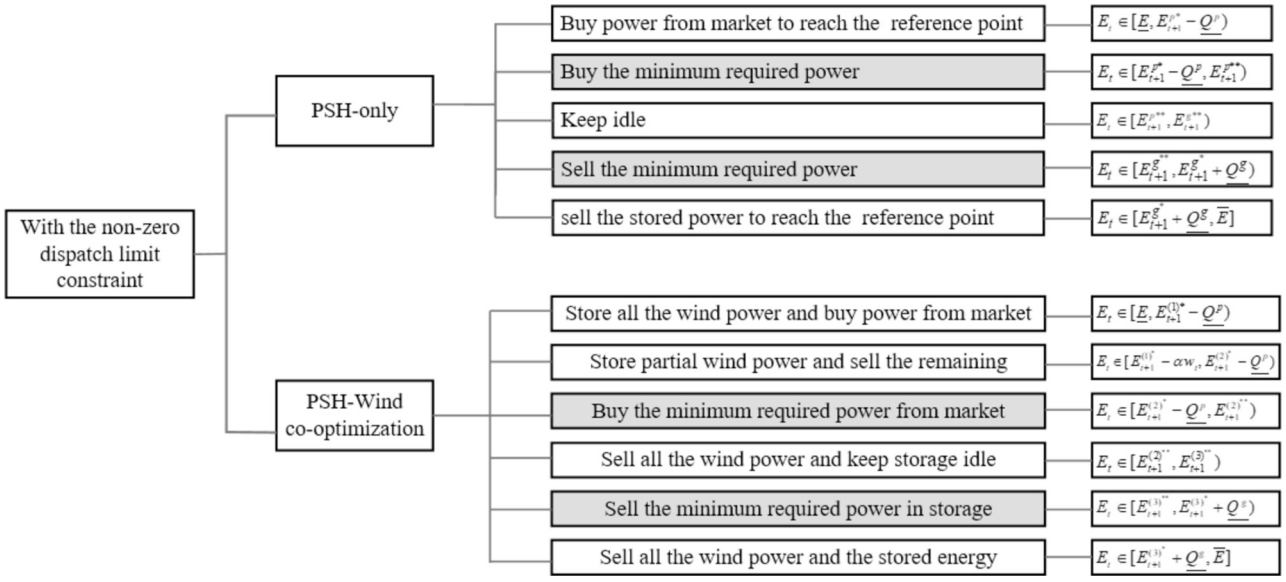


Fig. 1. Decision-making process structure for electricity merchants under two scenarios.

the market impact of large-scale PSH on the electricity market. Shi et al. (2024) [41] developed a multi-stage stochastic optimization model for coordinated green-seaport energy-logistics systems, demonstrating through simulations that their strategy minimized both operation costs and renewable energy curtailment. Li et al. (2024) [49] focused on a hybrid hydro-wind-solar energy system and proposed a risk management model based on the "risk identification-evaluation-control" framework. Their three-stage optimization method reduced the risk of part-load operation by 57.1 % on average, improving the system's risk management capability.

Tian et al. (2020) [50] utilized stochastic programming combined with downside risk constraints to manage the bidding strategy of PSH, showing that the strategy achieves a zero-risk condition by reducing profit by 33 %. Cerejo et al. (2020) [38] focused on the joint production optimization problem of a hydro-wind power system, primarily addressing the neutralization challenge considering the uncertainty of wind power forecasting. The results indicate a depreciation in optimal revenues from hydro power when wind forecasting is uncertain. Li et al. (2021) [53] proposed a new stochastic P-robust optimization method to optimize the bidding strategy for pumped storage systems, accounting for the financial risk of day-ahead price uncertainty. Although profits decreased, the relative maximum regret was significantly reduced. Topalovic et al. (2024) [51] investigated the economic feasibility of PSH and lithium batteries when used for price arbitrage. They analyzed the arbitrage of the two energy systems separately. Their results showed that PSH is economically more advantageous, and they also found that the profit from price arbitrage depends on the number of full-load hours, the tariff spread, and the available flexibility options, rather than on the installed share of renewable energy. Wang et al. (2024) [52] developed a short-term synergistic dispatch model for a hydro-wind-solar-PSHP hybrid energy system, which retained emergency demand capabilities while pursuing peak load balancing and significantly improved the flexibility of PSH. The case study results showed that the system effectively reduces power fluctuations, improves load factor, and reduces the number of necessary operating units.

Zhang et al. (2022) [54] proposed a method to determine the regulating capacity of a cascaded hydro-PV-PSH complementary joint power system. Through a case study, they determined the configuration between PV and PSH regulating capacities, and the data analysis showed that the method is robust. Gong et al. (2024) [55] considered a combined unit commitment framework for hydropower and hydrogen plants to accommodate large-scale photovoltaic power generation. Simulation results showed that the proposed method improves the operational efficiency of the system and reduces water consumption. Shi et al. (2024) [41] combined a hydro-wind-solar supplemental renewable energy system with both PSH and battery storage and studied the nested storage capacity optimization problem across multiple time scales. The proposed three-battery grouping strategy effectively saved cost, reduced battery degradation, and decreased carbon emission. Additionally, Zhang et al. (2024) [39] proposed a multi-attribute decision-making method to study the capacity planning problem of PSH in a hybrid operating system considering uncertainty of wind power and obtained the optimal PSH capacity under different degrees of risk aversion.

However, these studies primarily use numerical simulations to obtain optimal solutions, while fewer focus on analytical solutions. In this paper, we choose to use dynamic programming to obtain analytical solutions. Unlike numerical solutions, analytical solutions utilize closed-form mathematical functions to analyze the relationship between the parameters and the solutions, thus understanding the effects of each parameter on outputs.

2.2. Energy storage economic dispatch

Numerous studies have focused on energy arbitrage activities within the electricity market, aiming to maximize the economic benefits, with one of the most important being the optimization of economic dispatch strategies. Agarwal and Pileggi (2022) [56] utilized a differential dynamic programming scheme to solve the multi-period optimal power flow problem for renewable energy storage systems, considering the generation ramp constraint. Khaloie and Vallée (2024) [57] studied liquid air energy storage and proposed a day-ahead scheduling model for LAES coupled with a liquefied natural gas regasification process. The authors also proposed a two-stage stochastic dispatch model for the realistic market environment, and the cost-effectiveness of the facility was verified through a case study. Liang et al. (2022) [58] tackled the economic dispatch problem for regionally integrated energy systems, incorporating pipeline dynamics and proposing an enhanced approximate dynamic programming algorithm to increase system revenue. Lauinger et al. (2024) [59] addressed the problem of selling frequency regulation power in the day-ahead trading market by quantifying power through an implicit function and deriving an analytical solution. Their findings showed that the benefit of frequency regulation is inversely proportional to the time spent in regulating power.

Dynamic programming has been less commonly applied to PSH due to the complexities introduced by non-zero lower limits on generating and pumping capacities. Our study seeks to fill this void by adapting dynamic programming to accommodate these specific constraints, offering a novel approach to model and solve the economic dispatch problem for PSH systems. Alvarez et al. (2019) [60] focused on the long-term energy dispatching problem of PSH using the stochastic dual dynamic programming method to achieve the optimal operating cost while satisfying end-point conditions for the long-term plan. Karhinen and Huuki (2019) [40] investigated the optimal operation of PSH for different wind penetration levels using dynamic programming, finding that increased wind penetration increases the profitability of energy storage. Toufani et al. (2022) [61] assessed the potential benefits of converting a conventional cascade hydropower station to a pumped storage system. They formulated the problem as stochastic dynamic programming and demonstrated through numerical validation that the PSH system can yield greater benefits in the presence of frequent negative tariffs. Pan et al. (2024) [62] proposed an improved state-space approximate dynamic programming algorithm to study the economic dispatch of offshore-onshore regional integrated energy systems, considering the stochastic fluctuations of offshore gas sources and wind power output. Our research advances beyond these studies by delivering analytical solutions that offer more generalizable and deeper insights into PSH dispatch strategies, underscoring our methodological innovation in transforming discontinuous problems into analytically tractable forms.

Liu et al. (2021) [17] focused on the market impact of large-scale energy storage, particularly PSH, on the market, employing dynamic programming theory to explore the economic dispatch problem of price-making energy storage, thus providing methodological support for electricity merchants. Zheng et al. (2022) [63] proposed an efficient stochastic dynamic programming algorithm to study energy storage arbitrage with variable charging and discharging efficiencies, which improved energy storage revenues. Liu et al. (2022) [30] considered the market impact of energy storage and the uncertainty of wind power generation, utilizing dynamic programming to optimize the co-optimization economic dispatch of merchants with wind farms and energy storage. Notably, given that the government offers production tax credits (PTC) for wind power generation, Liu et al. (2022) [26] incorporated this policy into the economic dispatch problem of wind power plants, comparing the optimal dispatch strategies under two different PTC scenarios using dynamic programming, thus providing strategic decision-making guidance for merchants. Baldick et al. (2023) [64] studied the scheduling optimization of pumped storage, presenting a general yet basic PSH model. By solving a relaxed version of this model, they explained how the combination of mutual exclusivity constraints and SOC limits results in a complex and challenging model.

All the aforementioned studies applied dynamic programming methods to explore energy storage economic dispatch strategies and

obtained optimal analytical solutions. However, these papers typically assume that the minimum capacity constraints for charging/pumping and discharging/generating are zero to maintain the continuity of the reward functions and value functions. This assumption simplifies the model but does not account for the physical realities of systems like PSH, where such constraints cannot be zero due to operational requirements.

Our research bridges this gap by incorporating non-zero minimum generating and pumping capacity constraints inherent to PSH systems into dynamic programming models. We have developed a PSH dispatch model that effectively manages discontinuous reward and value functions using an innovative auxiliary function, which converts these into a continuous format. This method simplifies the modeling process while accurately reflecting the system's physical constraints. Consequently, we have derived an optimal analytical economic dispatch strategy that maximizes profits under these realistic constraints, providing a more accurate depiction of PSH operations. Additionally, our proposed methodology has broad applicability to address the joint dispatch optimization problem for PSH and wind power plants. Our approach enhances practical implementation and underscores the importance of integrating realistic constraints into theoretical models, thus providing transformative solutions for PSH system operators.

3. Model set-up and optimization analysis

This section uses dynamic programming to discuss the optimal energy dispatch strategy for profit-maximizing PSH owners within the day-ahead wholesale electricity market. Merchants profit by releasing stored energy when electricity prices are high and purchasing energy to recharge their storage when prices are low. We employ dynamic programming to maximize expected profits by analyzing these strategies. In this scenario, the merchant has three options: (a) purchase power to store, (b) remain idle, or (c) discharge stored power and sell it back to the grid.

We consider a PSH system consisting of an upper reservoir with finite capacity and a lower reservoir that is effectively unlimited, such as a large lake or river, such as the Ludington PSH facility in Michigan (see https://www.consumersenergy.com/ for details). This means the storage capacity constraints pertain only to the upper reservoir. The lower reservoir's capacity does not impose limitations on the system's operation. We assume that the merchant's PSH storage has a finite maximum energy capacity $\overline{E}$ and a minimum energy capacity $\underline{E}$, where, $\overline{E} > \underline{E} \geq 0$. This simplification allows us to focus on the impact of non-zero lower bounds on charging and discharging rates without the added complexity of modeling both reservoirs' capacities. Beyond these limits, the storage cannot operate. In addition to the storage capacity, the charging and discharging capacities are also constrained; the minimum and maximum charging constraints are $\underline{Q^p}$ and $\overline{Q^p}$, respectively, and for discharging, the upper and lower limits are $\overline{Q^g}$ and $\underline{Q^g}$. Unlike traditional assumptions [33,53] that often set $\underline{Q^p} = \underline{Q^g} = 0$, this assumption does not hold when studying PSH. Our study acknowledges non-zero lower limits for dispatch, i.e., $\underline{Q^p} > 0$ and $\underline{Q^g} > 0$.

Efficiency considerations include α and β, denoting the efficiencies of the pumping/charging and generating/discharging modes, respectively, with $\alpha, \beta \in (0,1]$. Another efficiency, $\rho \in (0,1]$, represents the transmission line efficiency, indicating the ratio of power out to the power in. The third efficiency type, $\eta_t \in [0,1]$, accounts for the energy loss due to factors like evaporation and leakage, affecting the stored energy over each period.

Based on the above discussion, the net pumping power capacity and gross generating power capacity are represented by $\overline{Q^p}/\alpha\rho$ and $\overline{Q^g}\cdot\beta\rho$, respectively. We differentiate between slow and fast storage capabilities based on the values of $\overline{Q^p}/\alpha\rho$ and $\overline{Q^g}\cdot\beta\rho$. A "slow" storage system is characterized by $\overline{Q^g}\cdot\beta\rho < \overline{E} - \underline{E}$ or $\overline{Q^p}/\alpha\rho < \overline{E} - \underline{E}$ [43], which means that the energy storage cannot be fully filled up or emptied within one decision period. When $\overline{Q^g}\cdot\beta\rho \geq \overline{E} - \underline{E}$ and $\overline{Q^p}/\alpha\rho \geq \overline{E} - \underline{E}$, the storage is considered "fast" because it can be fully charged and discharged within one decision period. Since fast storage is a special case of slow storage, we focus on deriving the optimal policy structure for slow storage systems.

The electricity price at time t is represented by P_t ($/MWh), and the sequence of prices is represented by the vector $P = (P_1, P_2, P_3, \ldots, P_t)$ ($/MWh). The operating cost when the energy storage is discharged or charged is expressed in c ($/MWh). The decision for each period t is donated by q_t^p or q_t^g (MWh) to represent the energy change between period t and $t+1$ before considering efficiency losses. To maintain the tractability of our model, we assume the energy storage has a linear operating cost for both discharging/generating and pumping/charging. Thus, the rewards function $R(q_t^g, q_t^p, P_t)$ from energy storage dispatch decision (q_t^g, q_t^p) at time t and the forecast prices P_t are defined as follows:

$$R(q_t^g, q_t^p, P_t) = \begin{cases} P_t q_t^g \beta\rho - c q_t^g \beta\rho & \left(\text{the storage is discharged and } \overline{Q^g} \geq q_t^g \geq \underline{Q^g}\right) \\ 0 \quad (\text{the storage is neither charged nor discharged}) & \\ -P_t q_t^p \Big/ \alpha\rho - c q_t^p \Big/ \alpha\rho & \left(\text{the storage is charged and } \overline{Q^p} \geq q_t^p \geq \underline{Q^p}\right) \end{cases} \tag{1}$$

The first line in Eq. (1) represents the rewards when the merchant sells the stored electricity to the market. Here, $P_t q_t^g \beta\rho$ and $c q_t^g \beta\rho$ represent the revenue obtained and operating cost paid at time t for $q_t^g \beta\rho$ units energy, respectively. The second line corresponds to the merchant who decides to do nothing at time t. The third line details the cost of electricity purchased from the market, where $P_t q_t^p/\alpha\rho$ is the cost for the merchant buying electricity to pump into storage, and $c q_t^p/\alpha\rho$ is the operating cost.

Denote E_t as the SOC level in the PSH storage at the beginning of period t, and the sequential SOC levels as $\widehat{E} = (E_1, E_2, \cdots, E_t)$, where $E_i \in \left[\underline{E}, \overline{E}\right]$, $\forall i \in \{1, 2, \cdots, T\}$. We define the feasible trading decisions set based on the current SOC level $E_t \in E$ as:

$$Action(E_t) := \left\{ \begin{array}{l} (q_t^g q_t^p) \in \mathbb{R}: \\ \underline{Q^p} U_t^p \leq q_t^p \leq \overline{Q^p} U_t^p, q_t^p \leq \overline{E} - E_t, \underline{Q^g} U_t^g \leq q_t^g \leq \overline{Q^g} U_t^g, q_t^g \leq E_t - \underline{E} \end{array} \right\} \tag{2}$$

here, Eq. (2) represents the maximum and minimum quantity of energy that can be pumped and generated. The first two constraints define the upper and lower pumping boundaries due to the limit and PSH storage capacity, respectively. The third and fourth constraints show the upper and lower generating boundaries because of the limit and available energy in the PSH storage. Both binary decision variables U_t^p and U_t^g

represent the unit commitment to power pumping and generating in $[t, t+1)$. Without loss of generality, $U_t^p + U_t^g \leqslant 1$, where $U_t^p \in \{0,1\}$ and $U_t^g \in \{0,1\}$, means the PSH cannot pump and generate simultaneously. If $U_t^p + U_t^g = 0$, it indicates the merchant decides to do nothing in this period.

Furthermore, Eq. (2) reveals that the merchant's action space at each period in our proposed optimization model is non-continuous due to the non-zero lower bounds (i.e., $\underline{Q^p} > 0$ and $\underline{Q^g} > 0$). This discontinuity challenges traditional models that assume continuous decision variables for simplicity. Addressing these discontinuities is crucial for accurately modeling real-world energy dispatch scenarios. In this paper, we address this challenge using a mathematical approach that includes an auxiliary function, enhancing the model's robustness and applicability by ensuring continuity, which is crucial for effectively applying dynamic programming and deriving optimal decisions.

At the beginning of period t, the merchant knows the SOC (MWh) of the PSH and the predicted electricity price P_t ($/MWh). Based on this information, the merchant makes electricity dispatch decisions for this period. We use q_t^p and q_t^g to represent the actions that occur during the period. At the end of the time t, the PSH undergoes a storage self-discharge. Therefore, the following formula outlines the state transition or energy balance from time t to $t+1$:

$$E_{t+1} = (E_t + q_t^p - q_t^g)\eta_t \tag{3}$$

In Eq. (3), q_t^p represents the increase in SOC due to pumping, q_t^g represents the decrease in SOC due to generating, and $\eta_t \in [0,1]$ accounts for the self-discharge efficiency over the period. The PSH system operates by pumping water to the upper reservoir during periods of low electricity prices (charging) and releasing water to generate electricity during periods of high prices (discharging). Since the lower reservoir is assumed to be a large natural body of water with unlimited capacity, the constraints of storage capacity apply only to the upper reservoir. Therefore, the SOC of the PSH system refers to the amount of energy stored in the upper reservoir.

3.1. Optimization and analysis

To maximize expected profits, we assume that all forecasted electricity prices are known in advance [26–29]. This assumption allows us to focus specifically on the storage dispatch problem, making the storage control process deterministic for the merchant. Trading decisions are made periodically within a finite horizon of each period $t \in \{1,2,\ldots,T\}$. Following a previous study [29], this paper models the merchant's dispatch strategy using Markov dynamic programming with a finite horizon. The state variables in each stage t are E_t and P_t, and the state at period t is defined by $S(t) = S_t(E_t, P_t)$. The merchant aims to find the optimal decision rule π^* to maximize the expected value function at the initial stage (Stage 1) over the limited period $t \in \{1,2,\ldots,T\}$.

$$\max_{\pi} \sum_{t=1}^{T} E\left[\left(q_t^g \beta\rho(P_t - c) - (P_t + c)q_t^p/\alpha\rho\right)|S(1)\right] \tag{4}$$

In this paper, Eq. (4) ignores the discount factor, and E is the expectation concerning E_1, P_1. Both E_1 and P_1 are the given initial SOC and the initial price, respectively. Let $V_t(S(t))$ denote the value function in period t and state $S(t) = S_t(E_t, P_t) \in \widehat{E} \times P$. This function satisfies the Bellman Equation:

$$V_t(S(t)) = \max_{Action(E_t)} \left[R\left(q_t^g, q_t^p, P_t\right) + E(V_{t+1}(S(t+1)|S(t)))\right] \tag{5}$$

Based on previous studies [29,43], we generally assume that the value of electricity in PSH becomes zero at terminal period $T+1$, thus $V_{T+1}(S(T+1)) = V(E_{T+1}, P_{T+1}) = 0$.

We have modeled the dispatch problem of merchants as Eq. (4), which is a binary integer Markov dynamic programming function that is too complex to obtain the closed-form optimal analytical solutions directly. To overcome this complexity, we first replace the binary variables with equivalent discontinuous decision variables, then construct an auxiliary function to transform the discontinuous decision variables into continuous ones, thereby converting Eq. (4) into traditional Markov dynamic programming problem.

First, we let $E_{t+1} = (E_t + q_t^p - q_t^g)\eta_t = (E_t + \alpha_t)\eta_t$ (i.e., $\alpha_t = q_t^p - q_t^g$) be the decision variable [65]. We use the action α_t for each period t to replace the previous decisions q_t^g and q_t^p, representing the SOC change between periods t to $t+1$ before considering energy loss. We can easily see that when $\alpha_t \geq \underline{Q^p}$, SOC increases, meaning the merchant buys electricity from the market and pumps it into the storage. When $\alpha_t \leq -\underline{Q^g}$, SOC decreases, and the merchant releases electricity to sell to the market. When $\alpha_t = 0$, the merchant neither buys nor sells electricity.

To obtain the optimal dispatch solutions, we further refine the optimization problem detailed in Eq. (5) by decomposing it into three distinct sub-optimization problems, each tailored to address different actionable scenarios:

$$V_t(S(t)) = \begin{cases} -(P_t + c)\alpha_t \Big/ \alpha\rho + E[V_{t+1}(S(t+1)|S(t))] & \left(\overline{Q^p} \geq \alpha_t \geq \underline{Q^p}\right) \\ E[V_{t+1}(S(t+1)|S(t))] & (\alpha_t = 0) \\ -(P_t - c)\alpha_t \beta\rho + E[V_{t+1}(S(t+1)|S(t))] & \left(-\overline{Q^g} \leq \alpha_t \leq -\underline{Q^g}\right) \end{cases} \tag{6}$$

The primary challenge encountered in Eq. (6) is the discontinuity of the value function caused by the non-zero lower limits on generating and pumping capacities. Since dynamic programming requires a continuous value function to effectively solve the Bellman equation and achieve globally optimal results, this discontinuity poses a significant obstacle. To address this challenge, we introduce an innovative auxiliary function designed to transform the discontinuous decision variables into continuous ones, thereby restoring the necessary continuity.

Next, we introduce the auxiliary function in detail:

First, we define the following new auxiliary decision variables:

$$\begin{aligned} y_t &= \begin{cases} \alpha_t - \underline{Q^p} & \left(\overline{Q^p} \geq \alpha_t \geq \underline{Q^p}\right) \\ \alpha_t + \underline{Q^g} & \left(-\overline{Q^g} \leq \alpha_t \leq -\underline{Q^g}\right) \end{cases} \Rightarrow \alpha_t \\ &= \begin{cases} y_t + \underline{Q^p} & \left(\overline{Q^p} - \underline{Q^p} \geq y_t \geq 0\right) \\ y_t - \underline{Q^g} & \left(-\overline{Q^g} + \underline{Q^g} \leq y_t \leq 0\right) \end{cases} \end{aligned} \tag{7}$$

here, y_t is the new decision variable constructed through the auxiliary function. The original decision variable α_t has a discontinuous range $\left[-\overline{Q^g}, -\underline{Q^g}\right] \cup \{0\} \cup \left[\underline{Q^p}, \overline{Q^p}\right]$, while y_t has a continuous range $\left[-\overline{Q^g} + \underline{Q^g}, \overline{Q^p} - \underline{Q^p}\right]$. By introducing the new auxiliary decision variable, the originally discontinuous reward function and value function become continuous. We can solve this problem through conventional dynamic programming techniques to obtain the optimal solution for y_t, and then map these solutions back to the original discontinuous

problem to obtain the optimal solution for α_t.

Using the substitution from Eq. (7), Eq. (6) can be reformulated as:

$$V_t(S(t)) = \begin{cases} -(P_t + c)\left(y_t + \underline{Q^p}\right) \Big/ \alpha\rho + E[V_{t+1}(S(t+1)\,|S(t))] & \left(\overline{Q^p} - \underline{Q^p} \geq y_t \geq 0\right) \\ -(P_t - c)\left(y_t - \underline{Q^g}\right)\beta\rho + E[V_{t+1}(S(t+1)\,|S(t))] & \left(-\overline{Q^g} + \underline{Q^g} \leq y_t \leq 0\right) \end{cases} \tag{8}$$

It is easy to notice that there are three formulas in Eq. (6) while there are only two in Eq. (8). The fact that Eq. (6) has three formulas is due to the discontinuous range of the original decision variable α_t. The three formulas correspond to the value equations for the three decisions: buying electricity, idling, and selling electricity. However, when we introduce the new continuous auxiliary decision variable y_t, there are only two formulas. At this point, $y_t = 0$ corresponds to three trading decisions, which are buying the minimum amount of electricity, idling, and selling the minimum amount of electricity. When we obtain the optimal solution for y_t and map it back to the original problem, we need to choose the optimal decision among the original discrete options. We will discuss this part in the next section.

This reformulation leverages the relationship between the newly introduced auxiliary decision variables and the SOC of the PSH system:

$$y_t = \begin{cases} E_{t+1} \Big/ \eta_t - E_t - \underline{Q^p} & \left(\overline{Q^p} - \underline{Q^p} \geq y_t \geq 0\right) \\ E_{t+1} \Big/ \eta_t - E_t + \underline{Q^g} & \left(-\overline{Q^g} + \underline{Q^g} \leq y_t \leq 0\right) \end{cases}$$

Then, by letting E_{t+1} be the new decision variable, the $V_t(S(t))$ in Eq. (8) can be rewritten as follows:

$$\begin{cases} V_t^p(S(t)) = -(E_{t+1}/\eta_t - E_t)(P_t + c) \Big/ \alpha\rho + E[V_{t+1}(S(t+1)\,|S(t))] & \left(\overline{Q^p} - \underline{Q^p} \geq y_t \geq 0\right) \\ V_t^g(S(t)) = (c - P_t)\beta\rho(E_{t+1}/\eta_t - E_t) + E[V_{t+1}(S(t+1)\,|S(t))] & \left(-\overline{Q^g} + \underline{Q^g} \leq y_t \leq 0\right) \end{cases} \tag{9}$$

Eq. (9) enhances the model's tractability and aligns it with the dynamics of SOC. On this basis, the optimization problem can be segmented into two sub-problems $\underset{E_{t+1}}{max} V_t^g(S(t))$ and $\underset{E_{t+1}}{max} V_t^p(S(t))$, subject to the constraints of $max\left\{-\overline{Q^g} + \underline{Q^g}, \underline{E} - E_t + \underline{Q^g}\right\} \leq y_t \leq min\left\{\overline{Q^p} - \underline{Q^p}, \overline{E} - E_t - \underline{Q^p}\right\}$.

Maximizing Eq. (9) enables us to derive the optimal results for the specified conditions by removing the given state $S(t)$ (i.e., which includes the values for E_t and P_t). Following the Bellman Equation [66], the optimal solutions for E_{t+1} under these constraints should satisfy the conditions for the two actions: charging/pumping and discharging/generating, as illustrated in:

$$\begin{cases} V_t^{p^*}(S(t)) = max_{\underline{E} \leq E_{t+1} \leq \overline{E}}\left\{-(E_{t+1}/\eta_t - E_t)(P_t + c)/\alpha\rho + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ V_t^{g^*}(S(t)) = max_{\underline{E} \leq E_{t+1} \leq \overline{E}}\left\{(c - P_t)\beta\rho(E_{t+1}/\eta_t - E_t) + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \end{cases} \tag{10}$$

In the next section, we conduct a thorough analysis of the optimal results derived from Eq. (10). This analysis utilizes the principles of dynamic programming to streamline complexity and systematically identify optimal decisions for each period, considering the two possible actions of charging and discharging separately. Subsequently, we compare these outcomes to determine the optimal economic dispatch decision that maximizes expected profit based on forecasted prices while adhering to system constraints.

3.2. Profit maximization and optimal decision rule

To obtain the optimal decision rule in Eq. (10), following the previous studies [29,42], we establish that for any $t \in \{1, 2, \ldots, T\}$, in each period t, the value functions $V_t(S(t))$ and $E[V_{t+1}(S(t+1)\,|S(t))]$ are concave in $E_t \in \left[\underline{E}, \overline{E}\right]$ for each given state $S(t) = S_t(E_t, P_t)$ and $P_t < \infty$ holds. Thus, we derive the following relationship:

$$\partial^2 E[V_{t+1}(S(t+1)\,|S(t))]/\partial {E_{t+1}}^2 \leq 0 \tag{11}$$

From this, we infer that the second derivative of these two sub-optimization problems in Eq. (10) is negative. This allows us to find a unique optimal solution through first-order conditions. The best results for the SOC are presented as follows:

Lemma 1. *Let $E_{t+1}^{p^*}$ and $E_{t+1}^{g^*}$ be the optimal SOC reference points in Eq.*

(12) *for electricity merchants when charging and discharging, respectively, as shown:*

$$\begin{cases} E_{t+1}^{p^*} = argmax_{\underline{E}\le E_{t+1}\le \overline{E}}\{-(E_{t+1}/\eta_t - E_t)(P_t + c)/\alpha\rho + E[V_{t+1}^*(S(t+1)|S(t))]\} \\ or\ \partial^2 E[V_{t+1}^*(S(t+1)|S(t))]/\partial {E_{t+1}}^2\big|_{E_{t+1}=E_{t+1}^{p^*}} = (P_t + c)/\alpha\rho\eta_t \\ E_{t+1}^{g^*} = argmax_{\underline{E}\le E_{t+1}\le \overline{E}}\{(c - P_t)\beta\rho(E_{t+1}/\eta_t - E_t) + E[V_{t+1}^*(S(t+1)|S(t))]\} \\ or\ \partial^2 E[V_{t+1}^*(S(t+1)|S(t))]/\partial {E_{t+1}}^2\big|_{E_{t+1}=E_{t+1}^{g^*}} = (P_t - c)\beta\rho/\eta_t \end{cases} \tag{12}$$

From Eq. (12), it is evident that the values of $E_{t+1}^{p^*}$ and $E_{t+1}^{g^*}$ are determined by the current electricity price P_t. This indicates that for the profit-maximizing electricity merchant, the optimal economic dispatch policy at each decision time depends on the forecasted price.

Before further discussing the optimal economic dispatch strategy for energy storage, we introduce an assumption to ensure the smooth operation of energy dispatching.

Assumption. We assume that $E_{t+1}^{g^*} \ge \underline{E} + \underline{Q^g}$ and $E_{t+1}^{p^*} \le \overline{E} - \underline{Q^p}$.

Without this assumption, the following scenario may occur: When $E_{t+1}^{g^*} < \underline{E} + \underline{Q^g}$ and $E_t \in \left[E_{t+1}^{g^*}, \underline{E} + \underline{Q^g}\right]$, we would be compelled to $\alpha_t = 0$, because at that point, choosing to discharge would result in $E_{t+1} < \underline{E}$, violating the storage minimum capacity constraint. In Appendix A, we also discuss three scenarios where this assumption is not satisfied. The results demonstrate that the optimal economic dispatch strategy maintains a high degree of consistency even if the assumption cannot be satisfied.

Focusing solely on the positive electricity price scenario for simplicity, we present the following conclusions and insights based on Eq. (12). The corresponding optimal result is given in the following proposition (the proof is provided in Appendix A):

Proposition 1. For every stage $t \in \{1, 2, \ldots, T\}$, and a positive forecasted electricity price $P_t \in P$, there exists a unique optimal storage level relationship $\underline{E} \le E_{t+1}^{p^*} \le E_{t+1}^{g^*} \le \overline{E}$, which depends on the state of $S(t)$.

In Proposition 1, we find $\underline{E} \le E_{t+1}^{p^*} \le E_{t+1}^{g^*} \le \overline{E}$, which plays an important role in formulating the optimal dispatch strategy. The entire SOC is segmented by these two reference points into three sub-regions.

(1) When $E_t \in \left[\underline{E}, E_{t+1}^{p^*}\right]$, merchants need to buy power from the market to fill the storage SOC up to $E_{t+1}^{p^*}$ as closely as possible.
(2) When $E_t \in (E_{t+1}^{p^*}, E_{t+1}^{g^*})$, the optimal scheduling decision is to remain idle, i.e., neither charging/pumping nor discharging/generating.
(3) When $E_t \in \left[E_{t+1}^{g^*}, \overline{E}\right]$, the merchant should release stored energy and sell it on the market so that the storage SOC is reduced to $E_{t+1}^{g^*}$ as closely as possible.

$$\begin{cases} V_t^p(S(t)) = -(P_t + c)\alpha_t/\alpha\rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right)|S(t)\right)\right] \quad \left(\alpha_t = \underline{Q^p}\right) \\ V_t^i(S(t)) = E[V_{t+1}(S_{t+1}(E_t, P_{t+1})|S(t))] \quad (\alpha_t = 0) \\ V_t^g(S(t)) = -(P_t - c)\alpha_t\beta\rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t - \underline{Q^g}, P_{t+1}\right)|S(t)\right)\right] \quad \left(\alpha_t = -\underline{Q^g}\right) \end{cases}$$

It is evident that $E_{t+1}^{p^*} \le E_{t+1}^{g^*}$. If $E_{t+1}^{p^*} > E_{t+1}^{g^*}$, the merchant will find that the SOC level is greater than $E_{t+1}^{g^*}$ after charging/pumping and must discharge/generate.

This paper, diverging from the traditional research, assumes that the lower limit value of dispatching is not zero, i.e., $\underline{Q^p} > 0$ and $\underline{Q^g} > 0$, and therefore the storage capacity constraints of this problem differ from those in tradition research.

The economic dispatch strategy of electricity merchants is divided into three scenarios based on different SOC reference points (the proof is given in Appendix A):

$$y_t^* = \begin{cases} min\left\{\overline{Q^p} - \underline{Q^p}, E_{t+1}^{p^*} - E_t - \underline{Q^p}\right\} & E_t \in \left[\underline{E}, E_{t+1}^{p^*} - \underline{Q^p}\right) \\ 0 & E_t \in \left[E_{t+1}^{p^*} - \underline{Q^p}, E_{t+1}^{g^*} + \underline{Q^g}\right] \\ max\left\{-\overline{Q^g} + \underline{Q^g}, E_{t+1}^{g^*} - E_t + \underline{Q^g}\right\} & E_t \in \left(E_{t+1}^{g^*} + \underline{Q^g}, \overline{E}\right] \end{cases} \tag{13}$$

Clearly, when the auxiliary variable $y_t^* = 0$, the corresponding α_t has three cases: $\alpha_t = \underline{Q^p}$, $\alpha_t = -\underline{Q^g}$, and $\alpha_t = 0$. To obtain the correct optimal scheduling policy, a more detailed discussion of the decisions when the current SOC $E_t \in \left[E_{t+1}^{p^*} - \underline{Q^p}, E_{t+1}^{g^*} + \underline{Q^g}\right]$ is necessary.

Comparison with Traditional Research: Unlike traditional studies that assume zero lower bounds for charging and discharging capacities—resulting in only one actionable decision, which is to remain idle—our model introduces three possible actions within the two reference points: (a) remain idle, (b) charge at the minimum lower bound, and (c) discharge at the minimum lower bound. By comparing these three feasible solutions, we can determine the optimal dispatch decision that maximizes profitability while adhering to the non-zero lower bound constraints.

From Eq. (6) we derive the value equations corresponding to these three decisions:

The first line represents the value equation for charging at the minimum lower bound, the second line applies when the merchant remains idle, and the last line corresponds to discharging at the minimum lower bound.When $E_t \in \left[E_{t+1}^{p^*} - \underline{Q^p}, E_{t+1}^{p^*}\right]$, the merchant faces two decisions: $\alpha_t = \underline{Q^p}$ (charge at the minimum lower bound) and $\alpha_t = 0$ (idle). We need to compare the corresponding value functions to obtain a new reference point $E_{t+1}^{p^{**}}$ to guide the merchant's decision. Similarly, when $E_t \in \left[E_{t+1}^{g^*} + \underline{Q^g}, E_{t+1}^{g^*}\right]$, the merchant chooses between $\alpha_t = -\underline{Q^g}$ (discharge at the minimum lower bound) and $\alpha_t = 0$ (idle). A new reference point $E_{t+1}^{g^{**}}$ is needed to maximize profit. To further refine our understanding of the optimal decisions under non-zero lower bounds, we present the following lemma (see Appendix A for proof).

Lemma 2. *Let $E_{t+1}^{p^{**}}$ and $E_{t+1}^{g^{**}}$ be the new reference points in Eq. (14) for electricity merchants' charging and discharging decisions, respectively, as shown*:

$$\begin{cases} E_{t+1}^{p^{**}} = argmax_{\underline{E} \le E_{t+1} \le \overline{E}} \left\{ (P_t + c)\underline{Q^p} \Big/ \alpha\rho + E[V_{t+1}(S_{t+1}(E_t, P_{t+1}) \,|S(t))\,] - E\left[V_{t+1}\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right) |S(t)\right)\right] \right\} \\ or \left\{ (P_t + c)\underline{Q^p} \Big/ \alpha\rho + E[V_{t+1}(S_{t+1}(E_t, P_{t+1}) \,|S(t))\,] - E\left[V_{t+1}\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right) |S(t)\right)\right] \right\} \Big|_{E_{t+1} = E_{t+1}^{p^{**}}} = 0 \\ E_{t+1}^{g^{**}} = argmax_{\underline{E} \le E_{t+1} \le \overline{E}} \left\{ (P_t - c)\underline{Q^g}\beta\rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t - \underline{Q^g}, P_{t+1}\right) |S(t)\right)\right] - E[V_{t+1}(S_{t+1}(E_t, P_{t+1}) \,|S(t))\,] \right\} \\ or \left\{ (P_t - c)\underline{Q^g}\beta\rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t - \underline{Q^g}, P_{t+1}\right) |S(t)\right)\right] - E[V_{t+1}(S_{t+1}(E_t, P_{t+1}) \,|S(t))\,] \right\} \Big|_{E_{t+1} = E_{t+1}^{g^{**}}} = 0 \end{cases} \tag{14}$$

In Eq. (14), it becomes clear that the values of $E_{t+1}^{p^{**}}$ and $E_{t+1}^{g^{**}}$ are determined by the current SOC E_t, the current electricity price P_t, and the minimum power dispatch constraints $\underline{Q^p}$ and $\underline{Q^g}$. For every stage $t \in \{1, 2, \ldots, T\}$, with a positive forecasted electricity price $P_t \in P$, there exists a unique optimal storage level relationship $\underline{E} + \underline{Q^g} \le E_{t+1}^{p^{**}} \le E_{t+1}^{g^{**}} \le \overline{E} - \underline{Q^p}$. Since $E_{t+1}^{p^{**}}$ and $E_{t+1}^{g^{**}}$ are new reference points derived from $E_{t+1}^{p^*}$ and $E_{t+1}^{g^*}$ in Proposition 1, it is straightforward to ascertain that $E_{t+1}^{p^{**}} \le E_{t+1}^{p^*}$ and $E_{t+1}^{g^{**}} \ge E_{t+1}^{g^*}$. Having established the relationship between $E_{t+1}^{p^*}$ and $E_{t+1}^{g^*}$ as $E_{t+1}^{p^*} \le E_{t+1}^{g^*}$ in Proposition1, we can further deduce that $E_{t+1}^{p^{**}} \le E_{t+1}^{g^{**}}$. Furthermore, we can deduce that the discontinuity of the model leads to a larger range of choices to remain idle in the optimal dispatch policy than in traditional studies. The likelihood of a merchant choosing to remain idle rises, which implies a lower frequency of dispatch (discharge and charge) may lead to a loss in profit. Thus, discontinuity in the model leads to lower profits.

In traditional studies, merchants rely on the SOC reference point to guide their decision-making processes and develop optimal dispatch policies. Their objective is to adjust the SOC level as closely as possible to this reference point through trading decisions. However, the introduction of a non-zero dispatch lower bound constraints $\underline{Q^p}$ and $\underline{Q^g}$ complicates this approach. When the current SOC level is very close to the reference point, any charging or discharging action will cause the SOC to exceed the reference point.

Consequently, it becomes necessary to compare scenarios where merchants make trading decisions under lower-bound constraints versus remaining idle. This comparison is essential for establishing a new reference point and developing an economic dispatch policy that enables merchants to maximize profits through optimal decisions despite the presence of discontinuous multiple-period optimization problems. This represents one of the primary contributions of this paper.

Based on the above discussion, we can finally determine the optimal dispatching policy as follows:

Proposition 2. *For every stage $t \in \{1, 2, \ldots, T\}$, with a positive forecasted electricity price $P_t \in P$, the optimal dispatch policy for PSH with non-zero dispatch lower limit constraint is shown below*:

$$\alpha_t^* = \begin{cases} min\left\{E_{t+1}^{p^*} - \underline{Q^p} - E_t, \overline{Q^p}\right\} & E_t \in \left[\underline{E}, E_{t+1}^{p^*} - \underline{Q^p}\right) \\ \underline{Q^p} & E_t \in \left[E_{t+1}^{p^*} - \underline{Q^p}, E_{t+1}^{p^{**}}\right) \\ 0 & E_t \in [E_{t+1}^{p^{**}}, E_{t+1}^{g^{**}}] \\ -\underline{Q^g} & E_t \in \left(E_{t+1}^{g^{**}}, E_{t+1}^{g^*} + \underline{Q^g}\right] \\ max\left\{E_{t+1}^{g^*} + \underline{Q^g} - E_t, -\overline{Q^g}\right\} & E_t \in \left(E_{t+1}^{g^*} + \underline{Q^g}, \overline{E}\right] \end{cases} \tag{15}$$

Specifically:

(1) When $E_t \in \left[\underline{E}, E_{t+1}^{p^*} - \underline{Q^p}\right)$, the merchant buys electricity from the market and charges it into the storage, aiming to raise the SOC level up to $E_{t+1}^{p^*} - \underline{Q^p}$ as closely as possible.

(2) When $E_t \in \left[E_{t+1}^{p^*} - \underline{Q^p}, E_{t+1}^{p^{**}}\right)$, the merchant opts for the lowest level of charging (i.e., $\alpha_t = \underline{Q^p}$).

(3) When $E_t \in \left[E_{t+1}^{p^{**}}, E_{t+1}^{g^{**}}\right)$, the merchant chooses to remain idle, which means $\alpha_t = 0$.

(4) When $E_t \in \left[E_{t+1}^{g^{**}}, E_{t+1}^{g^*} + \underline{Q^g}\right)$, the merchant opts for the lowest level of discharging (i.e., $\alpha_t = -\underline{Q^g}$).

When $E_t \in \left[E_{t+1}^{g^*} + \underline{Q^g}, \overline{E}\right]$, the merchant discharges electricity from the storage and sells it in the market, decreasing the SOC down to $E_{t+1}^{g^*} + \underline{Q^g}$ as closely as possible.

These four optimal SOC reference points depend on the current available energy E_t in storage, the forecasted electricity price P_t, and the minimum PSH's generating and pumping capacities $\underline{Q^p}$ and $\underline{Q^g}$. The findings highlight the operational challenges and strategic decisions necessary for managing non-zero lower dispatch limits.

Managerial Insight 1: *To effectively navigate these constraints, merchants must develop sophisticated economic dispatch strategies that dynamically adjust to fluctuating energy levels within defined limits. The presence of*

non-zero minimum bounds increases the probability of the electricity merchants opting to remain idle. It decreases the probability of electricity merchants' charging/pumping and discharging/pumping actions, thereby reducing expected profits compared to scenarios with zero lower boundary constraints. However, such strategic idleness is crucial for maximizing economic efficiency, necessitating advanced modeling and real-time decision-making by merchants.

By deriving analytical closed-form solutions that overcome the challenges posed by non-zero lower bound constraints in PSH systems, our research provides decision-makers with precise guidelines for optimal dispatching. This allows operators to determine exactly when to charge, discharge, or remain idle based on specific SOC reference points and forecasted electricity prices, ensuring maximum profitability within the physical constraints of their systems. The analytical nature of our solutions means that decision-makers can quickly adjust strategies in response to changing market conditions without the need for complex numerical simulations, thereby enhancing responsiveness and operational efficiency.

For electricity merchants, this translates into significant value by improving profitability and reducing computational and operational costs. Our contributions lie in providing a novel, practical framework that bridges the gap between theoretical optimization and real-world application, offering actionable strategies that enhance both the economic and operational performance of PSH systems. One innovative aspect of this approach is achieving optimal solutions without needing to rerun the entire model for varying initial SOC levels or forecasted prices. This is accomplished through the derivation of closed-form functions that provide optimal solutions under different conditions. By leveraging these analytical functions, merchants can efficiently adapt their strategies to real-time market fluctuations without the computational burden of recalculating from scratch. This advancement not only streamlines decision-making but also enhances the precision of operational adjustments.

Building upon Managerial Insight 1, we recognize that owners of PSH facilities experience profit losses due to the physical constraints that impose non-zero minimum limits on scheduling. Considering the additional constraints, merchants must sometimes remain idle, which reduces profitability. One approach to improve profits is to invest in new technologies that relax these physical constraints. Our methodology enables companies to better understand the impact of non-zero lower bounds on their operations, allowing them to make informed decisions about investing in technological advancements that could relax these constraints.

This section established the optimal dispatch strategy for merchants with PSH only, highlighting the influence of SOC levels and non-zero lower bounds on decision-making. The next section extends this analysis to merchants operating both PSH and wind farms, exploring joint optimization strategies.

4. Co-optimization of PSH with wind farms

This section extends the analysis by examining the joint optimization of systems combining both wind power plants and PSH. We assess how non-zero minimum dispatch constraints influence economic dispatch decisions in this integrated context.

The economic dispatch optimization now incorporates additional parameters due to the inclusion of wind farms, specifically wind power generation and its associated costs. Wind power generation at any given time t is denoted by w_t, represented sequentially by the vector $W = (w_1, w_2, w_3, \ldots, w_t)$.The unit wind power generating cost is c_w. The decision variable for each period t is donoted by q_t^p or q_t^g, reflecting the energy change between period t and $t+1$ before considering efficiency losses. Other parameters in this section remain the same as those in Section 3.

Thus, the rewards function $R(q_t^g, q_t^p, P_t, w_t)$ for energy storage dispatch decision (q_t^g, q_t^p) at time t, considering the forecast prices P_t and wind generation w_t, is defined as follows:

Case 1 for $\alpha w_t < \underline{Q^p}$

$$R(q_t^g, q_t^p, P_t, w_t) = \begin{cases} P_t(w_t + q_t^g\beta)\rho - cq_t^g\beta\rho - c_w w_t & \left(\text{the storage is discharged and } \overline{Q^g} \geq q_t^g \geq \underline{Q^g}\right) \\ P_t w_t \rho - c_w w_t & (\text{the storage is neither charged nor discharged}) \\ -P_t(q_t^p/\alpha - w_t)\Big/\rho - cq_t^p\Big/\alpha\rho - c_w w_t & \left(\text{the storage is charged and } \overline{Q^p} \geq q_t^p \geq \underline{Q^p} > \alpha w_t\right) \end{cases} \tag{16.1}$$

Case 2 for $\alpha w_t \geq \underline{Q^p}$

$$R(q_t^g, q_t^p, P_t, w_t) = \begin{cases} P_t(w_t + q_t^g\beta)\rho - cq_t^g\beta\rho - c_w w_t & \left(\text{the storage is discharged and } \overline{Q^g} \geq q_t^g \geq \underline{Q^g}\right) \\ P_t w_t \rho - c_w w_t & (\text{the storage is neither charged nor discharged}) \\ P_t(w_t - q_t^p/\alpha)\rho - cq_t^p\Big/\alpha\rho - c_w w_t & \left(\text{the storage is charged and } \alpha w_t > q_t^p \geq \underline{Q^p}\right) \\ -P_t(q_t^p/\alpha - w_t)/\rho - cq_t^p/\alpha\rho - c_w w_t & (\text{the storage is charged and } \overline{Q^p} \geq q_t^p \geq \alpha w_t) \end{cases} \tag{16.2}$$

when the wind power generation (w_t) is low (Case 1) and does not meet the minimum dispatch threshold even if all wind-generated power is stored, the electricity merchant has three primary decision options: 1) store all wind-generated power and buy additional power from the market to store; 2) remain idle, selling all wind-generated power to the market without engaging in any buying or selling; 3) sell all wind-generated power and release stored power to meet market demands. Conversely, when the wind-generated power (w_t) is high (Case 2), the merchant has four options: 1) store all wind-generated power and buy additional power from the market to store; 2) store a portion of wind-generated power and sell the reset to market; 3) remain idle, storage engaging in neither buying nor selling beyond wind-generated power; 4) sell all wind-generated power and release stored power to the market. In both cases, the reward functions account for revenues from wind power

sales, operating costs, and market power purchases, with Case 2 including additional revenue from selling excess wind power.

Compared to the scenario in Section 3, the introduction of wind farms adjusts the reward functions. However, the constraints on PSH's generating and pumping operations and its state transition function remain the same as in Eqs. (2) and (3) of Section 3. The state variables at each stage t are E_t, P_t, and w_t; the state at period t is defined by $S(t) = S_t(E_t, P_t, w_t)$. The merchant's goal is to discover the optimal decision rule π^* to maximize the expected value function at the initial stage (Stage 1) over the limited period $t \in \{1, 2, \ldots, T\}$, as outlined in the objective function:

$$\max_{\pi} \sum_{t=1}^{T} E\left[R\left(q_t^g, q_t^p, w_t, P_t\right) | S(1)\right] \tag{17}$$

Case 1 for $\alpha w_t < \underline{Q^p}$

$$V_t(S(t)) = \begin{cases} P_t(w_t - \alpha_t\beta)\rho + c\alpha_t\beta\rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & \left(-\overline{Q^g} \le \alpha_t \le -\underline{Q^g}\right) \\ P_t w_t \rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & (\alpha_t = 0) \\ -P_t(\alpha_t/\alpha - w_t)\Big/\rho - c\alpha_t\Big/\alpha\rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & \left(\overline{Q^p} \ge \alpha_t \ge \underline{Q^p} > \alpha w_t\right) \end{cases} \tag{19.1}$$

Case 2 for $\alpha w_t \ge \underline{Q^p}$

$$V_t(S(t)) = \begin{cases} P_t(w_t - \alpha_t\beta)\rho + c\alpha_t\beta\rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & \left(-\overline{Q^g} \le \alpha_t \le -\underline{Q^g}\right) \\ P_t w_t \rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & (\alpha_t = 0) \\ P_t(w_t - \alpha_t/\alpha)\rho - c\alpha_t\Big/\alpha\rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & \left(\alpha w_t > \alpha_t \ge \underline{Q^p}\right) \\ -P_t(\alpha_t/\alpha - w_t)/\rho - c\alpha_t/\alpha\rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & (\overline{Q^p} \ge \alpha_t \ge \alpha w_t) \end{cases} \tag{19.2}$$

The value function for period t and state $S(t) = S_t(E_t, P_t, w_t) \in \widehat{E} \times P \times W$, denoted as $V_t(S(t))$, adheres to the following Bellman Equation:

$$V_t(S(t)) = \max_{Action(E_t)} \left[R\left(q_t^g, q_t^p, w_t, P_t\right) + E\left(V_{t+1}(S(t+1)|S(t))\right)\right] \tag{18}$$

Consistent with previous studies [29,43], it is generally presumed that the value of electricity in PSH is zero at the terminal period. Hence, $V_{T+1}(S(T+1)) = V(E_{T+1}, P_{T+1}, w_{T+1}) = 0$.

Initially, we define $E_{t+1} = \left(E_t + q_t^p - q_t^g\right)\eta_t = (E_t + \alpha_t)\eta_t$ (i.e.,$\alpha_t = q_t^p - q_t^g$) as the decision variable [54]. Similar to Section 3, we use the action α_t for each period t to replace the previous decision variables q_t^g and q_t^p, representing the energy storage level/SOC change between periods t to $t+1$ before considering energy losses.

When wind generation is too low, even if all the generated wind power is stored, it may still be insufficient to meet the minimum dispatch requirement. With the integration of wind generation, non-zero dispatch lower limit constraints introduces additional complexities, as wind generation levels affect the feasibility of meeting minimum dispatch requirements. To obtain the optimization problem detailed in Eq. (19), we decompose it into sub-optimization problems as follows:

To address the discontinuity in the value function due to non-zero lower limits, we reintroduce the auxiliary decision variable y_t, as shown in Eq. (7), that transforms discontinuous decision variables into continuous ones. By leveraging the relationship between the newly introduced auxiliary decision variables and the SOC of the PSH system, we reformulate the Eq. (19) into the following form:

Case 1 for $\alpha w_t < \underline{Q^p}$

$$\begin{cases} V_t^{(1)}(S(t)) = -P_t((E_{t+1}/\eta_t - E_t)/\alpha - w_t)\Big/\rho - c(E_{t+1}/\eta_t - E_t)\Big/\alpha\rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & \left(\overline{Q^p} - \underline{Q^p} \ge y_t \ge 0\right) \\ V_t^{(3)}(S(t)) = P_t(w_t - (E_{t+1}/\eta_t - E_t)\beta)\rho + c(E_{t+1}/\eta_t - E_t)\beta\rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & \left(-\overline{Q^g} + \underline{Q^g} \le y_t \le 0\right) \end{cases} \tag{20.1}$$

Case 2 for $\alpha w_t \ge \underline{Q^p}$

$$\begin{cases} V_t^{(1)}(S(t)) = -P_t((E_{t+1}/\eta_t - E_t)/\alpha - w_t)\Big/\rho - c(E_{t+1}/\eta_t - E_t)\Big/\alpha\rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & \left(\overline{Q^p} - \underline{Q^p} \ge y_t \ge \alpha w_t - \underline{Q^p}\right) \\ V_t^{(2)}(S(t)) = P_t(w_t - (E_{t+1}/\eta_t - E_t)/\alpha)\rho - c(E_{t+1}/\eta_t - E_t)\Big/\alpha\rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & \left(\alpha w_t - \underline{Q^p} > y_t \ge 0\right) \\ V_t^{(3)}(S(t)) = P_t(w_t - (E_{t+1}/\eta_t - E_t)\beta)\rho + c(E_{t+1}/\eta_t - E_t)\beta\rho - c_w w_t + E[V_{t+1}(S(t+1)|S(t))] & \left(0 > y_t \ge -\overline{Q^g} + \underline{Q^g}\right) \end{cases} \tag{20.2}$$

The optimization problem can be segmented into three sub-optimization problems $\max_{E_{t+1}} V_t^{(1)}(S(t))$, $\max_{E_{t+1}} V_t^{(2)}(S(t))$ and $\max_{E_{t+1}} V_t^{(3)}(S(t))$, subject to $max\left\{-\overline{Q^g}+\underline{Q^g}, \underline{E}-E_t+\underline{Q^g}\right\} \le y_t \le min\left\{\overline{Q^p}-\underline{Q^p}, \overline{E}-E_t-\underline{Q^p}\right\}$. By maximizing Eq. (20), we derive the optimal results for the specified conditions by considering the specified state $S(t)$ (i.e., which includes the values for E_t, P_t and w_t).

Following the Bellman Equation [66], the optimal solutions for each stage E_{t+1} should satisfy the conditions for the two actions: charging and discharging, as demonstrated in:

Case 1 for $\alpha w_t < \underline{Q^p}$

$$\begin{cases} V_t^{(1)^*}(S(t)) = max_{\underline{E}\le E_{t+1}\le \overline{E}}\{-P_t((E_{t+1}/\eta_t - E_t)/\alpha - w_t)/\rho - c(E_{t+1}/\eta_t - E_t)/\alpha\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)|S(t))\right]\} \\ V_t^{(3)^*}(S(t)) = max_{\underline{E}\le E_{t+1}\le \overline{E}}\{P_t(w_t - (E_{t+1}/\eta_t - E_t)\beta)\rho + c(E_{t+1}/\eta_t - E_t)\beta\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)|S(t))\right]\} \end{cases} \tag{21.1}$$

Case 2 for $\alpha w_t \ge \underline{Q^p}$

$$\begin{cases} V_t^{(1)^*}(S(t)) = max_{\underline{E}\le E_{t+1}\le \overline{E}}\{-P_t((E_{t+1}/\eta_t - E_t)/\alpha - w_t)/\rho - c(E_{t+1}/\eta_t - E_t)/\alpha\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)|S(t))\right]\} \\ V_t^{(2)^*}(S(t)) = max_{\underline{E}\le E_{t+1}\le \overline{E}}\{P_t(w_t - (E_{t+1}/\eta_t - E_t)/\alpha)\rho - c(E_{t+1}/\eta_t - E_t)/\alpha\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)|S(t))\right]\} \\ V_t^{(3)^*}(S(t)) = max_{\underline{E}\le E_{t+1}\le \overline{E}}\{P_t(w_t - (E_{t+1}/\eta_t - E_t)\beta)\rho + c(E_{t+1}/\eta_t - E_t)\beta\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)|S(t))\right]\} \end{cases} \tag{21.2}$$

To obtain the optimal decision rule as indicated in Eq. (21), and consistent with previous studies [29,42], we established that for any $t \in \{1, 2, \ldots, T\}$, in each period t, the value function $V_t(S(t))$ and $E[V_{t+1}(S(t+1)|S(t))]$ are concave with respect to $E_t \in \left[\underline{E}, \overline{E}\right]$ for each given state $S(t) = S_t(E_t, P_t, w_t)$ and $P_t < \infty$ holds. Thus, we can deduce $\partial^2 E[V_{t+1}(S(t+1)|S(t))]/\partial {E_{t+1}}^2 \le 0$.

From this, we infer that the second derivatives of these two sub-optimization problems in Eq. (21) are negative. This allows us to find a unique optimal solution through first-order conditions. The best results of SOC are shown as (see Appendix A for proof):

Lemma 3. *Let $E_{t+1}^{(1)^*}$, $E_{t+1}^{(2)^*}$ and $E_{t+1}^{(3)^*}$ be the optimal SOC reference points in Eq. (22) for electricity merchants when charging, idling, and discharging, respectively, as shown*:

Case 1 for $\alpha w_t < \underline{Q^p}$

$$\begin{cases} E_{t+1}^{(1)^*} = argmax_{\underline{E}\le E_{t+1}\le \overline{E}}\{-P_t((E_{t+1}/\eta_t - E_t)/\alpha - w_t)/\rho - c(E_{t+1}/\eta_t - E_t)/\alpha\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)|S(t))\right]\} \\ or \quad \partial E\left[V_{t+1}^*(S(t+1)|S(t))\right]/\partial E_{t+1}\big|_{E_{t+1}=E_{t+1}^{(1)^*}} = (P_t + c)/\alpha\rho\eta_t \\ E_{t+1}^{(3)^*} = argmax_{\underline{E}\le E_{t+1}\le \overline{E}}\{P_t(w_t - (E_{t+1}/\eta_t - E_t)\beta)\rho + c(E_{t+1}/\eta_t - E_t)\beta\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)|S(t))\right]\} \\ or \quad \partial E\left[V_{t+1}^*(S(t+1)|S(t))\right]/\partial E_{t+1}\big|_{E_{t+1}=E_{t+1}^{(3)^*}} = (P_t - c)\beta\rho/\eta_t \end{cases} \tag{22.1}$$

Case 2 for $\alpha w_t \ge \underline{Q^p}$

$$\begin{cases} E_{t+1}^{(1)^*} = argmax_{\underline{E}\le E_{t+1}\le \overline{E}}\{-P_t((E_{t+1}/\eta_t - E_t)/\alpha - w_t)/\rho - c(E_{t+1}/\eta_t - E_t)/\alpha\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)|S(t))\right]\} \\ or \quad \partial E\left[V_{t+1}^*(S(t+1)|S(t))\right]/\partial E_{t+1}\big|_{E_{t+1}=E_{t+1}^{(1)^*}} = (P_t + c)/\alpha\rho\eta_t \\ E_{t+1}^{(2)^*} = argmax_{\underline{E}\le E_{t+1}\le \overline{E}}\{P_t(w_t - (E_{t+1}/\eta_t - E_t)/\alpha)\rho - c(E_{t+1}/\eta_t - E_t)/\alpha\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)|S(t))\right]\} \\ or \quad \partial E\left[V_{t+1}^*(S(t+1)|S(t))\right]/\partial E_{t+1}\big|_{E_{t+1}=E_{t+1}^{(2)^*}} = (c + P_t\rho^2)/\alpha\rho\eta_t \\ E_{t+1}^{(3)^*} = argmax_{\underline{E}\le E_{t+1}\le \overline{E}}\{P_t(w_t - (E_{t+1}/\eta_t - E_t)\beta)\rho + c(E_{t+1}/\eta_t - E_t)\beta\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)|S(t))\right]\} \\ or \quad \partial E\left[V_{t+1}^*(S(t+1)|S(t))\right]/\partial E_{t+1}\big|_{E_{t+1}=E_{t+1}^{(3)^*}} = (P_t - c)\beta\rho/\eta_t \end{cases} \tag{22.2}$$

Lemma 3 indicates that the optimal results, $E_{t+1}^{(1)^*}$, $E_{t+1}^{(2)^*}$, and $E_{t+1}^{(3)^*}$ for the electricity merchants are determined by the current electricity price P_t. This suggests that for a profit-maximizing electricity merchant, the optimal trading policy at each decision time significantly depends on the forecasted prices.

Proposition 3. Given the dependence on positive electricity prices, for each stage $t \in \{1, 2, \ldots, T\}$ with positive forecasted electricity prices $P_t \in P$, there is a distinctive optimal storage level relationship $\underline{E} \le E_{t+1}^{(1)^*} \le E_{t+1}^{(2)^*} \le E_{t+1}^{(3)^*} \le \overline{E}$, which depends on the current state of $S(t)$.

In Proposition 3, we find $\underline{E} \le E_{t+1}^{(1)^*} \le E_{t+1}^{(2)^*} \le E_{t+1}^{(3)^*} \le \overline{E}$. This relationship is foundational in crafting precise dispatch strategies for electricity merchants, guiding decisions across various several scenarios based on different SOC reference points, detailed as follows:

Case 1: For $\alpha w_t < \underline{Q^p}$: Wind-generated power is low.

Case 1.1: If $\alpha w_t < min\left\{E_{t+1}^{(1)^*}, \overline{Q^p} - \underline{Q^p}\right\}$, which means the wind-generated power is insufficient to reach the desired SOC $E_{t+1}^{(1)*}$. Therefore, the merchant may need to buy additional power from the market.

Case 1.1.1: If $E_{t+1}^{(1)^*} < \underline{Q^p} \Rightarrow \alpha w_t < E_{t+1}^{(1)^*} < \underline{Q^p}$, which means the minimum dispatch value is large, charging operations will lead to an SOC exceeding $E_{t+1}^{(1)^*}$. In such cases, merely storing all wind-generated power and buying power from the market is not feasible. The optimal policy for y_t in this case can be expressed as:

$$y_t^* = \begin{cases} 0 \quad E_t \in \left[\underline{E}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} \quad E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 1.1.2: If $E_{t+1}^{p^*} \ge \underline{Q^p} \Rightarrow E_{t+1}^{p^*} \ge \underline{Q^p} > \alpha w_t$, this indicates that the minimum dispatch value is small. It becomes feasible to store all wind-generated power and supplement it with market purchases. The optimal policy for y_t under these conditions can be implemented as:

$$y_t^* = \begin{cases} min\left\{E_{t+1}^{(1)^*} - E_t - \underline{Q^p}, \overline{Q^p} - \underline{Q^p}\right\} \quad E_t \in \left[\underline{E}, E_{t+1}^{(1)^*} - \underline{Q^p}\right) \\ 0 \quad E_t \in \left[E_{t+1}^{(1)^*} - \underline{Q^p}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} \quad E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 1.2: If $\alpha w_t \ge min\left\{E_{t+1}^{p^*}, \overline{Q^p} - \underline{Q^p}\right\}$, this indicates that the wind-generated power is substantial enough to exceed the reference point $E_{t+1}^{(1)*}$. The optimal policy involves storing a portion of wind power and selling the remainder to the market. Thus, the optimal policy for y_t in this case can be expressed as:

$$y_t^* = \begin{cases} 0 \quad E_t \in \left[\underline{E}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} \quad E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 2: For $\alpha w_t \ge \underline{Q^p}$: Wind-generated power is large.

Case 2.1: If $\alpha w_t < min\left\{E_{t+1}^{(1)^*}, \overline{Q^p} - \underline{Q^p}\right\}$, this indicates that there is minimal wind power generation that does not achieve the necessary SOC $E_{t+1}^{(1)*}$ alone, and we can get the following relationship: $\underline{Q^p} \le \alpha w_t < E_{t+1}^{(1)^*} \Rightarrow \underline{Q^p} < E_{t+1}^{(1)^*}$. This implies that storing all wind power and buying additional power from the market is a viable strategy. The optimal policy for y_t in this case can be expressed as:

$$y_t^* = \begin{cases} min\left\{E_{t+1}^{(1)^*} - E_t - \underline{Q^p}, \overline{Q^p} - \underline{Q^p}\right\} \quad E_t \in \left[\underline{E}, E_{t+1}^{(1)^*} - \underline{Q^p} - \alpha w_t\right) \\ min\left\{E_{t+1}^{(2)^*} - E_t - \underline{Q^p}, \alpha w_t\right\} \quad E_t \in \left[E_{t+1}^{(1)^*} - \underline{Q^p} - \alpha w_t, E_{t+1}^{(2)^*} - \underline{Q^p}\right) \\ 0 \quad E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} \quad E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 2.2: If $\alpha w_t \ge min\left\{E_{t+1}^{(1)^*}, \overline{Q^p} - \underline{Q^p}\right\}$, this indicates that the wind power generation is large enough to exceed the reference point $E_{t+1}^{(1)*}$. The optimal policy involves selling all available wind power and releasing stored power to decrease the SOC to $E_{t+1}^{(1)*}$ as closely as possible. Thus, the optimal policy for y_t in this case can be expressed as:

$$y_t^* = \begin{cases} min\left\{E_{t+1}^{(2)^*} - E_t - \underline{Q^p}, \alpha w_t\right\} \quad E_t \in \left[0, E_{t+1}^{(2)^*} - \underline{Q^p}\right) \\ 0 \quad E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} \quad E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

When the auxiliary variable $y_t^* = 0$ is introduced, it divides the potential actions into three scenarios: $\alpha_t = \underline{Q^p}$, $\alpha_t = -\underline{Q^g}$, and $\alpha_t = 0$. This represents a significant contribution to our paper, providing a more nuanced and realistic approach to economic dispatch strategies in PSH systems.

Further Analysis of Case 1 and Case 2:

For Case 1: Through the analysis of Eq. (19.1), we can obtain the value equations corresponding to these three decisions as follows:

$$\begin{cases} V_t^p(S(t)) = -P_t(\alpha_t/\alpha - w_t) \Big/ \rho - c\alpha_t \Big/ \alpha\rho - c_w w_t + E\left[V_{t+1}\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right)|S(t)\right]\right] \quad \left(\alpha_t = \underline{Q^p}\right) \\ V_t^i(S(t)) = P_t w_t \rho - c_w w_t + E[V_{t+1}(S_{t+1}(E_t, P_{t+1}) \,|S(t))] \quad (\alpha_t = 0) \\ V_t^g(S(t)) = P_t(w_t - \alpha_t\beta)\rho + c\alpha_t\beta\rho - c_w w_t + E\left[V_{t+1}\left(S_{t+1}\left(E_t - \underline{Q^g}, P_{t+1}\right)|S(t)\right)\right] \quad \left(\alpha_t = -\underline{Q^g}\right) \end{cases} \tag{23.1}$$

When $E_t \in \left[E_{t+1}^{(1)^*} - \underline{Q^p}, E_{t+1}^{(1)^*}\right]$, the merchant has two options: $\alpha_t = \underline{Q^p}$ and $\alpha_t = 0$. We compare the corresponding value functions to determine a new reference point $E_{t+1}^{(1)^{**}}$ to guide the merchant's decision-making. Similarly, when $E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, E_{t+1}^{(3)^*}\right]$, the merchant evaluates two options: $\alpha_t = -\underline{Q^g}$(discharging/generating at lower bound) and $\alpha_t = 0$

(idle). A new reference point $E_{t+1}^{(3)^{**}}$ is needed to maximize profit effectively.

For Case 2: Analysis from Eq. (19.2) suggests the value equations corresponding to these three decisions as follows:

$$\begin{cases} V_t^p(S(t)) = P_t(w_t - \alpha_t/\alpha)\rho - c\alpha_t \Big/ \alpha\rho - c_w w_t + E\left[V_{t+1}\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right)|S(t)\right)\right] \quad \left(\alpha_t = \underline{Q^p}\right) \\ V_t^i(S(t)) = P_t w_t \rho - c_w w_t + E[V_{t+1}(S_{t+1}(E_t, P_{t+1})\,|S(t))] \quad (\alpha_t = 0) \\ V_t^g(S(t)) = P_t(w_t - \alpha_t\beta)\rho + c\alpha_t\beta\rho - c_w w_t + E\left[V_{t+1}\left(S_{t+1}\left(E_t - \underline{Q^g}, P_{t+1}\right)|S(t)\right)\right] \quad \left(\alpha_t = -\underline{Q^g}\right) \end{cases} \tag{23.2}$$

When $E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(2)^*}\right]$, the merchant has two decisions to choose from: $\alpha_t = \underline{Q^p}$ (charging at lower bound) and $\alpha_t = 0$ (idle). We need to compare the corresponding value functions to obtain a new reference point $E_{t+1}^{(2)^{**}}$ to guide the merchant's decision-making. Similarly, when $E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, E_{t+1}^{(3)^*}\right]$, the merchant must decide between $\alpha_t = -\underline{Q^g}$ (discharging at lower bound) and $\alpha_t = 0$ (idle). Identifying a new reference point $E_{t+1}^{(3)^{**}}$ is critical to maximizing profit.

Based on this analysis, we can summarize the following insights:

Lemma 4. Let $E_{t+1}^{(1)^{**}}$, $E_{t+1}^{(2)^{**}}$ and $E_{t+1}^{(3)^{**}}$ be the new optimal reference points in Eq. (24) for electricity merchants, respectively, as shown:

Case 1. For $\alpha w_t < \underline{Q^p}$.

$$\begin{cases} E_{t+1}^{(1)^{**}} = argmax_{\underline{E} \le E_{t+1} \le \overline{E}} \left(\begin{array}{l} P_t\left(\underline{Q^p} \Big/ \alpha - w_t\right)\Big/\rho + c\underline{Q^p}\Big/\alpha\rho + P_t w_t \rho + E[V_{t+1}(S_{t+1}(E_t, P_{t+1})\,|S(t))] \\ -E\left[V_{t+1}\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right)|S(t)\right)\right] \end{array} \right) \\ E_{t+1}^{(3)^{**}} = argmax_{\underline{E} \le E_{t+1} \le \overline{E}} \left(\begin{array}{l} P_t\left(w_t - \underline{Q^g}\beta\right)\rho + c\underline{Q^g}\beta\rho - P_t w_t \rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t - \underline{Q^g}, P_{t+1}\right)|S(t)\right)\right] \\ -E[V_{t+1}(S_{t+1}(E_t, P_{t+1})\,|S(t))] \end{array} \right) \end{cases} \tag{24.1}$$

Case 2. For $\alpha w_t \ge \underline{Q^p}$.

$$\begin{cases} E_{t+1}^{(2)^{**}} = argmax_{\underline{E} \le E_{t+1} \le \overline{E}} \left(\begin{array}{l} -P_t\left(w_t - \underline{Q^p} \Big/ \alpha\right)\rho + c\underline{Q^p}\Big/\alpha\rho + P_t w_t \rho + E[V_{t+1}(S_{t+1}(E_t, P_{t+1})\,|S(t))] \\ -E\left[V_{t+1}\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right)|S(t)\right)\right] \end{array} \right) \\ E_{t+1}^{(3)^{**}} = argmax_{\underline{E} \le E_{t+1} \le \overline{E}} \left(\begin{array}{l} P_t\left(w_t - \underline{Q^g}\beta\right)\rho + c\underline{Q^g}\beta\rho - P_t w_t \rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t - \underline{Q^g}, P_{t+1}\right)|S(t)\right)\right] \\ -E[V_{t+1}(S_{t+1}(E_t, P_{t+1})\,|S(t))] \end{array} \right) \end{cases} \tag{24.2}$$

In Eq. (24), we can conclude that the values of $E_{t+1}^{(1)^{**}}$, $E_{t+1}^{(2)^{**}}$, and $E_{t+1}^{(3)^{**}}$ are determined by the current SOC E_t, the current electricity price P_t, the current wind generation w_t and the minimum power dispatch constraints $\underline{Q^p}$ and $\underline{Q^g}$. For every stage $t \in \{1, 2, \ldots, T\}$, with a positive forecasted electricity price $P_t \in P$, there exists a unique optimal storage level relationship $\underline{E} \le E_{t+1}^{(1)^{**}} \le E_{t+1}^{(2)^{**}} \le E_{t+1}^{(3)^{**}} \le \overline{E}$, which depends on the state of $S(t)$ and dispatch lower limit constraints $\underline{Q^g}$ and $\underline{Q^p}$. From this, we derive the optimal dispatching strategies (see Appendix A for proof).

Proposition 4. For every stage $t \in \{1, 2, \ldots, T\}$, with a positive forecasted electricity price $P_t \in P$, the optimal dispatch policy for PSH with non-zero dispatch lower limit constraint is shown below:

Case 1: For $\alpha w_t < \underline{Q^p}$: Wind is low.

Case 1.1: If $\alpha w_t < min\left\{E_{t+1}^{(1)^*}, \overline{Q^p}\right\}$, the optimal policy needs to be further discussed.

Case 1.1.1: If $E_{t+1}^{(1)^*} < \underline{Q^p} \Rightarrow \alpha w_t < E_{t+1}^{(1)^*} < \underline{Q^p}$, the optimal policy for α_t^* can be expressed as:

$$\alpha_t^* = \begin{cases} 0 & E_t \in \left[\underline{E}, E_{t+1}^{(1)^{**}}\right) \\ -\underline{Q^g} & E_t \in \left[E_{t+1}^{(1)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t, -\overline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases} \tag{25.1}$$

Case 1.1.2: If $E_{t+1}^{p^*} \geq \underline{Q^p} \Rightarrow E_{t+1}^{p^*} \geq \underline{Q^p} > \alpha w_t$, the optimal policy for α_t^* can be expressed as:

$$\alpha_t^* = \begin{cases} min\left\{E_{t+1}^{(1)^*} - E_t, \overline{Q^p}\right\} & E_t \in \left[\underline{E}, E_{t+1}^{(1)^*} - \underline{Q^p}\right) \\ \underline{Q^p} & E_t \in \left[E_{t+1}^{(1)^*} - \underline{Q^p}, E_{t+1}^{(1)^{**}}\right) \\ 0 & E_t \in \left[E_{t+1}^{(1)^{**}}, E_{t+1}^{(3)^{**}}\right) \\ -\underline{Q^g} & E_t \in \left[E_{t+1}^{(3)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases} \tag{25.2}$$

Case 1.2: If $\alpha w_t \geq min\{E_{t+1}^{p^*}, \overline{Q^p}\}$, the optimal policy for α_t^* in this case can be expressed as:

$$\alpha_t^* = \begin{cases} 0 & E_t \in \left[\underline{E}, E_{t+1}^{(1)^{**}}\right) \\ -\underline{Q^g} & E_t \in \left[E_{t+1}^{(1)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t, -\overline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases} \tag{25.3}$$

Case 2: For $\alpha w_t \geq \underline{Q^p}$: Wind is high.

Case 2.1: If $\alpha w_t < min\left\{E_{t+1}^{(1)^*}, \overline{Q^p}\right\}$, the optimal policy for α_t^* in this case can be expressed as:

$$\alpha_t^* = \begin{cases} min\left\{E_{t+1}^{(1)^*} - E_t, \overline{Q^p}\right\} & E_t \in \left[\underline{E}, E_{t+1}^{(1)^*} - \alpha w_t\right) \\ min\left\{E_{t+1}^{(2)^*} - E_t, \alpha w_t\right\} & E_t \in \left[E_{t+1}^{(1)^*} - \alpha w_t, E_{t+1}^{(2)^*} - \underline{Q^p}\right) \\ \underline{Q^p} & E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(2)^{**}}\right) \\ 0 & E_t \in \left[E_{t+1}^{(2)^{**}}, E_{t+1}^{(3)^{**}}\right) \\ -\underline{Q^g} & E_t \in \left[E_{t+1}^{(3)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t, -\overline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases} \tag{25.4}$$

Case 2.2: If $\alpha w_t \geq min\left\{E_{t+1}^{(1)^*}, \overline{Q^p}\right\}$, the optimal policy for α_t^* in this case can be expressed as:

$$\alpha_t^* = \begin{cases} min\left\{E_{t+1}^{(2)^*} - E_t, \alpha w_t\right\} & E_t \in \left[0, E_{t+1}^{(2)^*} - \underline{Q^p}\right) \\ \underline{Q^p} & E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(2)^{**}}\right) \\ 0 & E_t \in \left[E_{t+1}^{(2)^{**}}, E_{t+1}^{(3)^{**}}\right) \\ -\underline{Q^g} & E_t \in \left[E_{t+1}^{(3)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t, -\overline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases} \tag{25.5}$$

According to Proposition4, we identify up to five reference points for the joint optimization of PSH and wind farms. Specifically:

(1) When $E_t \in \left[\underline{E}, E_{t+1}^{(1)^*} - \underline{Q^p}\right)$, the merchant stores all available wind-generated power and purchase additional power to raise the SOC level up to $E_{t+1}^{(1)^*} - \underline{Q^p}$ as closely as possible.

(2) When $E_t \in \left[E_{t+1}^{(1)^*} - \alpha w_t, E_{t+1}^{(2)^*} - \underline{Q^p}\right)$, the merchant stores a portion of the wind power and sells the remaining wind power down to $E_{t+1}^{(2)^*} - \underline{Q^p}$.

(3) When $E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(2)^{**}}\right)$, the merchant opts for the lowest level of charging, i.e., $\alpha_t = \underline{Q^p}$.

(4) When $E_t \in \left[E_{t+1}^{(2)^{**}}, E_{t+1}^{(3)^{**}}\right)$, the merchant remains idle, which means $\alpha_t = 0$.

(5) When $E_t \in \left[E_{t+1}^{(3)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right)$, the merchant discharges at lowest level, that is, $\alpha_t = -\underline{Q^g}$.

(6) When $E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right]$, the merchant discharges electricity from the storage to sell to the market, decreasing the SOC down to $E_{t+1}^{(3)^*} + \underline{Q^g}$ as closely as possible.

These points are strategically set based on the current available energy in storage, forecasted electricity prices, wind-generated power, and the specific non-zero minimum generating and pumping capacities of PSH systems. In practice, we need to analyze the specific relationship between αw_t, $E_{t+1}^{(1)^*}$, and $\underline{Q^p}$. For example, when the wind power is high and $\underline{Q^p} < \alpha w_t < E_{t+1}^{(1)^*}$ is satisfied (e.g., Case 2.1), merchants can select actions from discharging, charging, or remaining idle, depending on market signals. Accordingly, when the wind power is low and $\alpha w_t < E_{t+1}^{(1)^*} < \underline{Q^p}$ is satisfied (e.g., Case 1.1.1), the merchants are more likely to choose between remaining idle or discharging, as charging under such conditions would not be economically viable.

Our analytical approach offers computational and strategic advantages over simulation methods, particularly in the joint optimization of storage and wind farms. It eliminates the need for iterative recalculations required by numerical methods, aligning operational strategies with the dynamic nature of energy markets. Additionally, this approach enhances the economic dispatch efficiency of hybrid energy systems, combining PSH and wind farms, thereby optimizing both energy storage and renewable energy resources. The establishment of SOC reference points provides a comprehensive decision-making framework that allows electricity merchants to dynamically adjust their strategies based on real-time market conditions, thus maximizing arbitrage profits.

Managerial Insight 2: *The co-optimization of PSH with wind farms provides electricity merchants with a robust strategic framework that enhances energy storage efficiency and maximizes profitability. This integration allows for dynamic adjustments of energy storage levels in response to real-time wind generation and fluctuating electricity prices, ultimately facilitating more responsive decision-making.*

Through Management Insight 2, we found that the co-optimization of PSH and wind farms results in a win-win situation. PSH can store excess wind power and provide power when wind power is insufficient, ensuring the grid stability. Wind power reduces the cost of energy storage and increases the profits of merchants. However, the introduction of renewable energy sources also complicates the dispatch strategy of PSH. The presence of non-zero lower dispatch limits shifts the decision-making paradigm from a single actionable choice—remaining idle—to a set of strategic actions: charging at the minimum lower bound, discharging at the minimum lower bound, and idling. Our approach provides merchants with precise control over energy storage

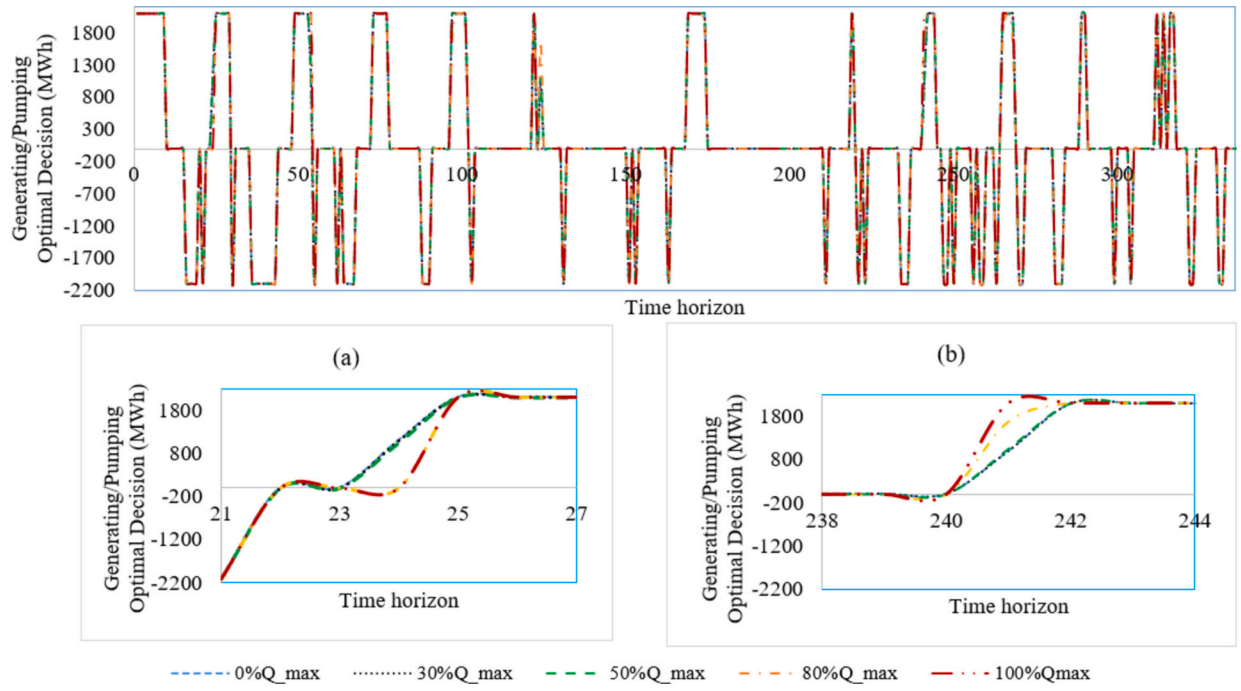


Fig. 2. The optimal action with PSH only.

levels, enabling them to navigate market fluctuations and enhance their overall economic efficiency.

In our study, we obtain the optimal analytic solution for PSH and wind farm co-optimization. Unlike the numerical solution, the relationship between the optimal results and the parameters can be explicitly obtained. Specifically, due to the uncertainty of wind power, the amount of wind power generated in each cycle varies. Using our method, the analytical solution can be updated quickly. However, the numerical simulation must recalculate the problem from scratch after each wind power generation, which is less efficient than our method.

For decision-makers, this means they can utilize our analytical solutions to quickly adjust their dispatch strategies in response to changes in wind generation, storage constraints, and electricity prices. By analyzing various parameters—such as storage charging and discharging constraints, forecasted prices, and wind farm capacity under the assumption of uniformly distributed wind—we identify all possible scenarios and their corresponding optimal dispatch strategies. This enables merchants to select the most profitable strategy under the given circumstances, thereby maximizing their profits.

Our analytical framework not only simplifies the decision-making process but also enhances the adaptability and responsiveness of energy merchants in the dynamic energy market. This provides decision-makers with a practical tool to optimize operations efficiently in the co-optimization of PSH and wind farms.

5. Synthetic data case study

To demonstrate the efficacy of our proposed method, we present two simplified case studies (detailed in Appendix B) that elucidate the conditions under which optimal results are derived.

5.1. Case 1-Merchant with only PSH

In this case, we consider three time periods (i.e., $T = 3$). At each period, the power price takes one of the values in set $P_t = \{P_1, P_2, P_3\} = \{5, 2, 10\}$. For simplicity, we assume that the energy storage capacity is sufficiently large to not be fully recharged in one period but can be recharged in less than two periods, it holds that $\underline{E} + \overline{Q^p} \le \overline{E}$ and $\underline{E} + 2\overline{Q^p} \ge \overline{E}$. Additionally, we assume the storage can be fully discharged in one period (i.e., $\overline{E} - \underline{E} \le \overline{Q^g}$).

Specifically, the storage capacity is set to 10 units (i.e., $\overline{E} = 10$ and $\underline{E} = 0$), the generating/discharging maximum capacity is 12 units and the minimum capacity is 2 units. The maximum and minimum pumping/charging capacities are 7 and 1 unit, respectively. We assume the operational cost is 1 unit, with the efficiencies for pumping/charging and generating/ discharging at 0.9, and self-discharging and transmission efficiencies at 1. Based on these parameters, we apply backward dynamic programming to derive the optimal policy and results:

Stage 3: The value function is defined and calculated based on our model parameters and constraints.

$$V_3 = -(P_3 - c)\left(\underline{E} - E_3\right)\beta\rho + V_4 = 8.1E_3 \quad E_3 \in [0, 10]$$

Stage 2: Utilizing Eq. (12), we derive the following optimal actions and corresponding value functions for this stage.

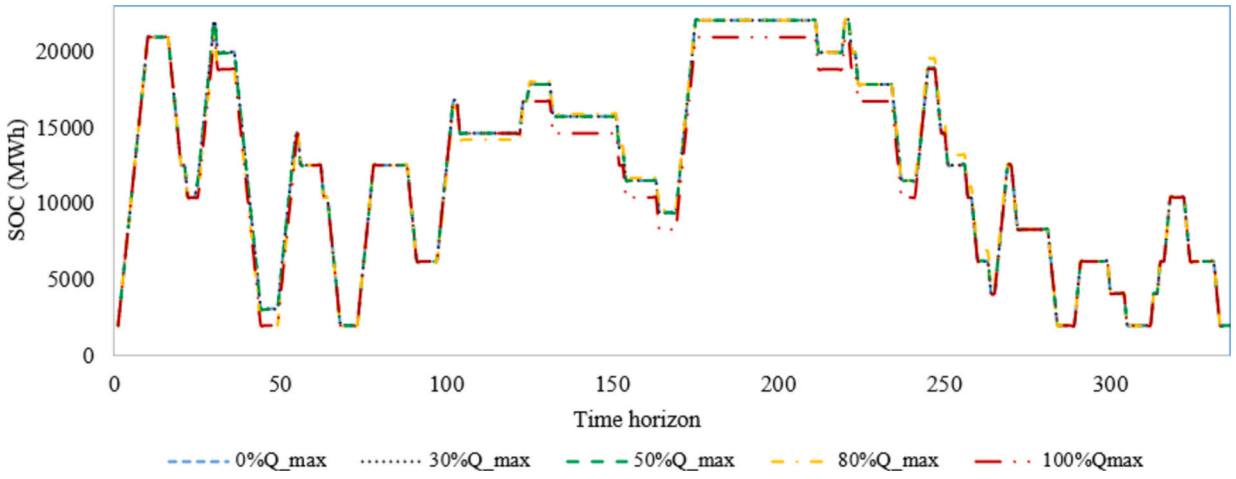


Fig. 3. The SOC curve with PSH only.

Table 1
Profit for PSH Only ($).

	0	30 % Q_max	50 % Q_max	80 % Q_max	Q_max
10/01–14/2023	3,974,700	3,974,700	3,974,600	3,972,200	3,966,000
11/06–19/2023	1,383,900	1,383,900	1,383,900	1,383,500	1,379,500
12/04–17/2023	1,163,200	1,163,200	1,163,200	1,162,300	1,158,400

$$\begin{cases} E_3^{g^*} = argmax_{\underline{E}\le E_3\le \overline{E}}\{-(P_2-c)E_3\beta\rho + V_3^*\} = 10 \\ E_3^{p^*} = argmax_{\underline{E}\le E_3\le \overline{E}}\{-(P_2+c)E_3/\alpha\rho + V_3^*\} = 10 \end{cases}$$

Following proposition 2, we get the following optimal action at stage 2.

$$\alpha_2^* = \begin{cases} 7 & E_2 \in [0,3] \\ 10-E_2 & E_2 \in (3,9] \\ 0 & E_2 \in (9,10] \end{cases}$$

Thus, the optimal value functions in stage 2 are shown as follows:

$$V_2^* = max(R_2 + V_3^*) = \begin{cases} 8.1E_2 + 33.37 & E_2 \in [0,3) \\ 3.33E_2 + 47.67 & E_2 \in [3,9) \\ 8.1E_2 & E_2 \in [9,10] \end{cases}$$

Stage 1: Similarly, by applying Eq. (12), we calculate the optimal actions and value functions.

$$\begin{cases} E_2^{g^*} = argmax_{\underline{E}\le E_2\le \overline{E}}\{-(P_1-c)E_2\beta\rho + V_2^*\} = 3 \\ E_2^{p^*} = argmax_{\underline{E}\le E_2\le \overline{E}}\{-(P_1+c)E_2/\alpha\rho + V_2^*\} = 3 \end{cases}$$

Following proposition 2, we determine the following optimal action at Stage 1.

$$\alpha_1^* = \begin{cases} min\left\{max\left\{E_2^{p^*} - E_1, \underline{Q^p}\right\}, \overline{Q^p}\right\} & E_1 \in \left[\underline{E}, E_2^{p^{**}}\right) \\ 0 & E_t \in [E_2^{p^{**}}, E_2^{g^{**}}] \\ max\left\{min\left\{E_2^{g^*} - E_1, -\underline{Q^g}\right\}, -\overline{Q^g}\right\} & E_1 \in (E_2^{g^{**}}, \overline{E}] \end{cases}$$

$$= \begin{cases} 3-E_1 & E_1 \in [0,2) \\ 1 & E_1 \in [2,2.3) \\ 0 & E_1 \in [2.3,4.88) \\ -2 & E_1 \in [4.88,5) \\ 3-E_1 & E_1 \in [5,10] \end{cases}$$

The outcomes are summarized as follows, given the initial SOC in storage: $E_1 = 1$.

Stage 1: If $E_1 = 1$, (Action 1: buy and pump, $\alpha_1^* = 2$), resulting in $E_2 = 3$;

Stage 2: If $E_2 = 3$, (Action 2: buy and pump, $\alpha_2^* = 7$), resulting in $E_3 = 10$;

Stage 3: If $E_3 = 10$, (Action 3: generate and sell, $\alpha_3^* = 10$), resulting in $E_4 = 0$.

The optimal value at Stage 1 is shown as: $V_1^* = 20E_1/3 + 37.67 = 44.34$.

5.2. Case 2-Merchant with PSH and wind farm

This case study extends our analysis to the joint optimization of a PSH and a wind power plant, with the base parameters identical to those in Case 1. We incorporate an operating cost of the PHS, denoted as $c = 0.1$, and assume the line transmission efficiency is $\rho = 0.9$. Additionally, wind power generation is introduced as $w_t = \{3,5,0\}$, with the cost of wind power generation is negligible. Using the backward dynamic

Table 2
Total full charge and discharge times for PSH only.

		0	30 % zQ_max	50 % Q_max	80 % Q_max	Q_max
10/01–14/2023	Full Charge Times	50	50	49	47	52
	Full Discharge Times	52	52	52	52	52
11/06–19/2023	Full Charge Times	36	36	36	34	36
	Full Discharge Times	36	36	36	34	36
12/04–17/2023	Full Charge Times	23	23	23	22	24
	Full Discharge Times	24	24	24	22	24

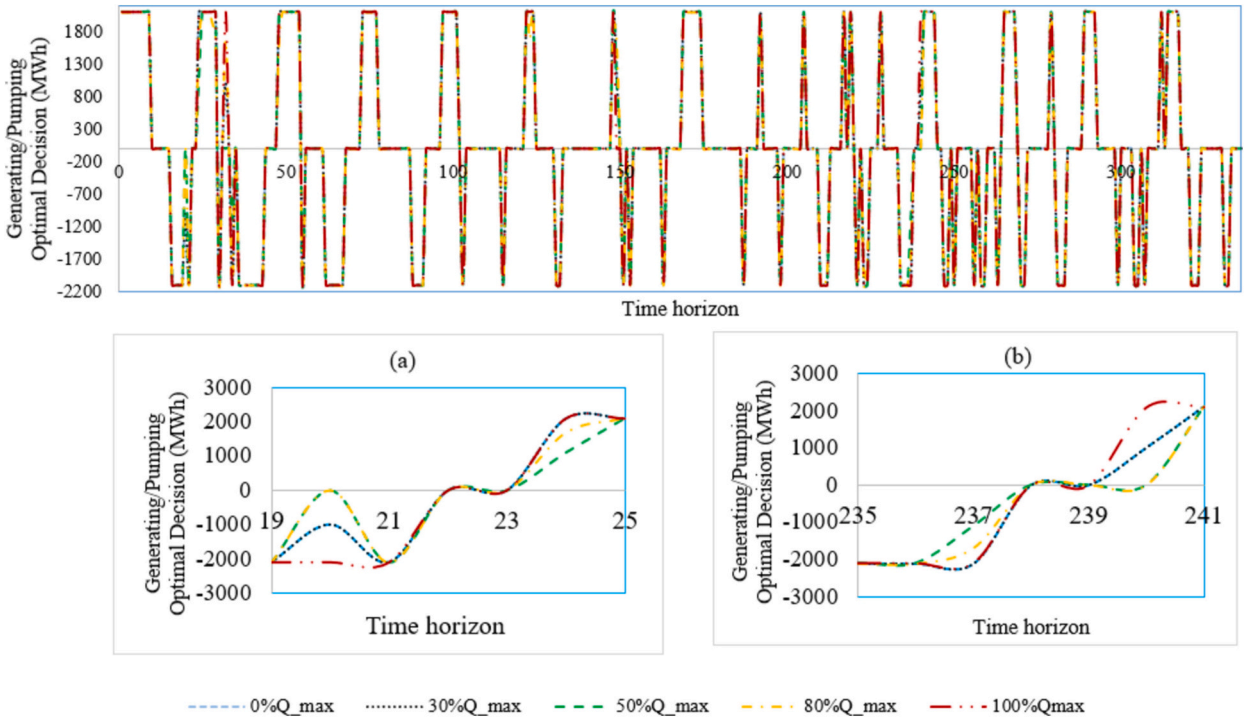


Fig. 4. The optimal action with PSH and wind power plants.

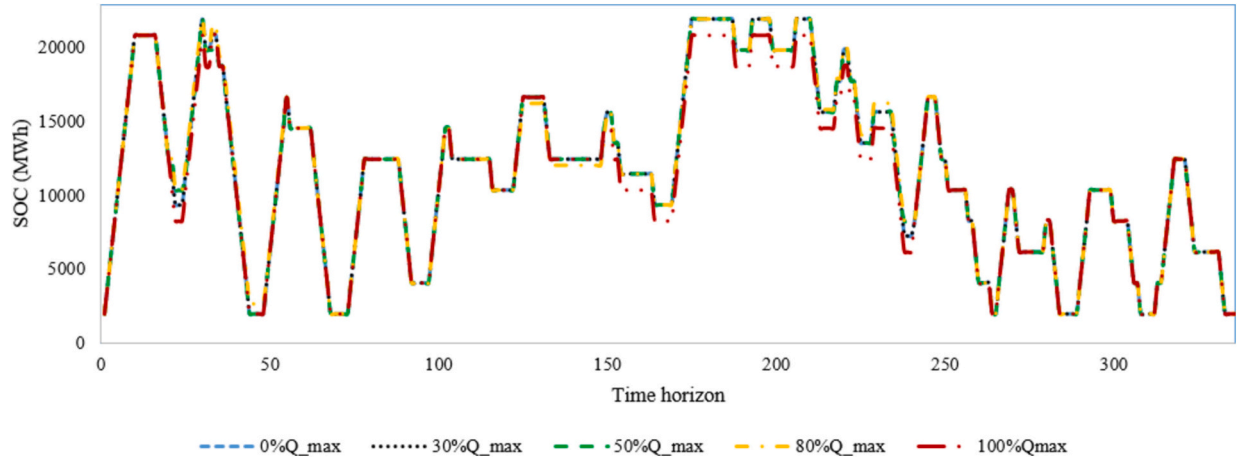


Fig. 5. The SOC curve with PSH and wind power plants.

programming, we derive the optimal policy and results across three stages:

Stage 3: The value function for this final stage is calculated and illustrated based on the current parameters and operational conditions:

$$V_3 = P_3(w_3 - (E_4/\eta - E_3)\beta)\rho + c(E_4/\eta - E_3)\beta\rho + V_4 = 8.02E_3$$

Stage 2: First, we calculate $\alpha w_2 = 0.9^{*}5 = 4.5 > \underline{Q^p}$ to determine wind power availability. Utilizing Eq. (25.2), we can get the following results:

$$\begin{cases} E_3^{(1)^*} = argmax_{\underline{E}\le E_3\le \bar{E}}\{ -P_2((E_3/\eta - E_2)/\alpha - w_2)/\rho - c(E_3/\eta - E_2)/\alpha\rho + V_3^*\} = 10 \\ E_3^{(2)^*} = argmax_{\underline{E}\le E_3\le \bar{E}}\{P_2(w_2 - (E_3/\eta - E_2)/\alpha)\rho - c(E_3/\eta - E_2)/\alpha\rho + V_3^*\} = 10 \\ E_3^{(3)^*} = argmax_{\underline{E}\le E_3\le \bar{E}}\{P_2(w_2 - (E_3/\eta - E_2)\beta)\rho + c(E_3/\eta - E_2)\beta\rho + V_3^*\} = 10 \end{cases}$$

Adhering to the guidelines of Proposition 4, we determine the necessary actions and corresponding value functions to establish the optimal action for this stage.

$$\alpha_2^* = \begin{cases} 7 & E_2 \in [0,3) \\ 10 - E_2 & E_2 \in [3,9) \\ 0 & E_2 \in [9,10] \end{cases}$$

Thus, the optimal value functions in stage 2 are shown as follows:

$$V_2^* = max(R_2 + V_3^*) = \begin{cases} 8.02E_2 + 49.13 & E_2 \in [0,3) \\ 2.59E_2 + 65.4 & E_2 \in [3,5.5) \\ 2.12E_2 + 68 & E_2 \in [5.5,9) \\ 8.02E_2 + 9 & E_2 \in [9,10] \end{cases}$$

Stage 1: We start by assessing $\alpha w_1 = 0.9^*3 = 2.7 > \underline{Q^p}$ for initial wind power generation. Following Eq. (25.2), we determine the following optimal reference points:

$$\begin{cases} E_2^{(1)^*} = argmax_{\underline{E} \le E_2 \le \overline{E}} \{ -P_1((E_2/\eta - E_1)/\alpha - w_1)/\rho - c(E_2/\eta - E_1)/\alpha\rho + V_2^* \} = 3 \\ E_2^{(2)^*} = argmax_{\underline{E} \le E_2 \le \overline{E}} \{ P_1(w_1 - (E_2/\eta - E_1)/\alpha)\rho - c(E_2/\eta - E_1)/\alpha\rho + V_2^* \} = 3 \\ E_2^{(3)^*} = argmax_{\underline{E} \le E_2 \le \overline{E}} \{ P_1(w_1 - (E_2/\eta - E_1)\beta)\rho + c(E_2/\eta - E_1)\beta\rho + V_2^* \} = 3 \end{cases}$$

Leveraging Proposition 4, we determine the optimal strategy at stage 1.

$$\alpha_1^* = \begin{cases} min\{3 - E_1, 7\} & E_1 \in [0,0.3) \\ min\{3 - E_1, 2.7, 1\} & E_1 \in [0.3,2.53) \\ 0 & E_2 \in [2.53,4.49] \\ max\{3 - E_1, -2\} & E_1 \in (4.49,5] \\ max\{3 - E_1, -12\} & E_1 \in (5,10] \end{cases}$$

$$= \begin{cases} 3 - E_1 & E_1 \in [0,2) \\ 1 & E_1 \in [2,2.53) \\ 0 & E_1 \in [2.53,4.49] \\ -2 & E_1 \in (4.49,5] \\ 3 - E_1 & E_1 \in (5,10] \end{cases}$$

The scenarios unfold as follows, given the initial SOC in storage: $E_1 = 1$.

Stage 1: If $E_1 = 1$, (Action 1: buy and pump, $\alpha_1^* = 2$), then $E_2 = 3$;

Stage 2: If $E_2 = 3$, (Action 2: buy and pump, $\alpha_2^* = 7$), then $E_3 = 10$;

Stage 3: If $E_3 = 10$, (Action 3: generate and sell, $\alpha_3^* = 10$), then $E_4 = 0$.

The optimal value at Stage 1 is shown as: $V_1^* = 5.12E_1 + 71.33 = 76.45$.

Based on the established analytical solutions and these two cases, we find that profit-maximizing electricity merchants utilizing a linear reward function typically choose between two options when charging: the upper limit of the maximum charging capacity or the available space in the storage for full charging. Similarly, for discharging, they select between the maximum discharging capacity per unit period and the available energy in the storage for full discharging. This indicates that they generally operate at maximum capacity when purchasing electricity to charge the storage or discharging stored electricity for sale. However, the significance of non-zero lower bound constraints become evident only when the available storage capacity or stored energy falls below these minimum thresholds. In such cases, these constraints directly impact the optimal dispatch strategy, necessitating adjustments to accommodate reduced operational flexibility.

This finding reflects a fundamental principle in energy storage operations: when resources are abundant (in this case, available storage capacity), merchants can operate at maximum efficiency without constraints. However, as these resources approach their limits, strategic adjustments become crucial to maintain operational efficiency. The introduction of non-zero lower bound constraints require the merchant to consider minimum operational levels, which alters their ability to act freely in the market. Thus, while under optimal conditions they can charge or discharge fully, these constraints necessitate more complex strategies when capacity is limited, influencing operational behavior and market strategies.

Moreover, traditional studies that assume zero lower bounds for charging and discharging capacities are a special case of our model. When the generating and charging lower bounds are set to zero, our results align with these traditional models, demonstrating the generality and scalability of our approach.

6. Real data case simulation

This section extends our synthetic scenario analysis by applying the methodology to real data concerning electricity prices and wind power generation from the MISO.

6.1. PSH storage only

In this section, we sourced electricity price data from MISO (price data available at https://www.misoenergy.org/) for a two-week period (10/01/2023–10/14/2023). Using hourly intervals as optimization periods, the electricity price series $P = \{P_1, P_2, \ldots, P_T\}$($/M.W.) consists of 336 stages (i.e., $T = 336$). The maximum and minimum SOC ($\overline{E}$ and $\underline{E}$) for the PSH are set at 22000 MWh and 2000 MWh, respectively. This configuration ensures that the storage cannot be fully depleted, reflecting typical operational constraints of PSH systems. The charging and discharging rates are capped at $\overline{Q^p} = \overline{Q^g} = 2100$(MW), and it takes approximately $\left(\overline{E} - \underline{E}\right)/\overline{Q^g} = 9.52$ hours to fully recharge the upper reservoir (i.e., $\left(\overline{E} - \underline{E}\right)/\overline{Q^p} = 9.52$), showcasing the system's robust

Table 3
Profit for PSH with Wind ($).

	0	30 %Q_max	50 %Q_max	80 %Q_max	Q_max
10/01–14/2023	15,341,000	15,341,000	15,341,000	15,339,000	15,331,000
11/06–19/2023	11,361,000	11,361,000	11,360,000	11,359,000	11,356,000
12/04–17/2023	10,562,000	10,562,000	10,562,000	10,561,000	10,557,000

operational capabilities. These data correspond approximately to the Ludington PSH facility in Michigan (Details available at https://www.consumersenergy.com). The operating cost for charging/pumping and discharging/generating activities are $c^p = c^g = 1(\$/\text{MWh})$. The self-discharge of energy storage is not considered, which is $\eta = 1$. We also assume that the transmission efficiency is $\rho = 1$, and the pumping and generating efficiencies are $\alpha = \beta = 0.8$. Notably, we do not consider the residual value of energy storage at the end of the period (i.e., $VOT_{T+1} = 0$). According to Propositions 1 and 2, the levels of minimum dispatching values $\underline{Q^p}$ and $\underline{Q^g}$ play an important role in determining the optimal strategy of the merchant.

Figs. 2 and 3 illustrate the optimal dispatch strategy and the SOC trajectory under an initial SOC of 2000 MWh. Considering the given optimization periods (336), the primary figures are densely plotted. For enhanced clarity, subfigures Fig. 2 (a) and Fig. 2 (b) selectively zoom into specific continuous segments of the operational timeline. These visual aids underscore the substantial impact of non-zero minimum generating and pumping capacities on the dispatch strategy. By showcasing these effects, the figures vividly demonstrate how operational constraints influence storage economic dispatch strategies in a real-world setting, thus validating the practical significance and applicability of our proposed optimization framework under real-world market conditions.

In Figs. 2 and 3, we analyze five different levels of minimum generating and pumping capacities. All other parameters across the four curves remain constant. The minimum capacities analyzed are 0 % (i.e., 0 MW), 30 % (i.e., 630 MW), 50 % (i.e., 1050 MW), 80 % (i.e., 1680 MW), and 100 % (i.e., 2100 MW) of the maximum capacity (i.e., 2100 MW), respectively. Given that the upper limits for generating and pumping capacities are the same in this study, for simplicity, we use '*Q_max*' to represent the upper boundary values for both PSH generating and pumping capacities.

In practice, the characteristics of a PSH system—including the reservoir size, terrain conditions, and the types of turbines and pumps—directly impact the minimum operational capacities for both pumping and generation. As a result, different lower-bound constraints reflect different PSH design specifications and operational limitations [67]. To capture this diversity, we consider five different levels of minimum generation and pumping capacities in our model, ranging from a lower limit of zero (representing a storage behavior similar to batteries with high flexibility) to 100 % of the maximum power capacity (i.e., where the system can only operate at full power when charging or discharging). In our numerical simulations, we fixed the upper boundaries and varied the lower boundaries to assess the scalability of our model and to highlight how it adapts under different operational constraints. This approach offers valuable insights into the adaptability of energy storage systems in different operational settings. It also demonstrates how our model provides flexibility that accommodates different types of PSH plants, which are often constrained by their physical and technical characteristics. Traditional studies often assume a lower limit of zero to generate analytical solutions [26–30], which ensures model continuity but does not fully reflect the operational complexities of PSH systems. Therefore, traditional research represents a special case within the broader spectrum analyzed in our study.

To ensure the robustness of our results, we selected three sets of electricity price data for numerical simulations, each covering a full two-week period: 10/01–14/2023, 11/06–19/2023, and 12/04–17/2023. These data sets allow us to test the effectiveness of our model under different market conditions. Due to space constraints, Figs. 2 and 3 present the energy storage economic dispatch optimal results (optimal actions) derived from the 10/01–14/2023 data set. The corresponding results for the other two data sets are provided in Appendix C. The results from the other two data sets are consistent and reinforce our findings, and their key outcomes are summarized in Tables 1 and 2. Tables 1 and 2 illustrate the profit and the number of full charges/discharges in different cases, respectively.

In Tables 1 and 2, we can see that when these minimum capacities are low, their influence on the optimal strategy is minimal, resulting in a dispatch approach that closely resembles scenarios where minimum dispatch values are assumed to be zero. However, as the level of minimum generating and pumping capacities increases, the optimal strategy changes significantly, often leading to reduced arbitrage opportunities and, hence, lower merchant profits. Moreover, the frequency of merchants choosing to "full charge" or "full discharge" shows a decreasing trend. It should be noted that when Q_min = 100 %Q_max, it means that the merchant can only make a full charge or full discharge decision, resulting in an increase in frequency. It should be noted that when Q_min = 100 %Q_max, the charging and discharging options are only at one identical value. Therefore, in all optimization periods, the merchant's charging and discharging actions can only be "full charge" and "full discharge." As a result, compared to other scenarios, the number of full charges and discharges increases, which is why in Table 2 we see an increase in frequency in this special case. However, as observed in Table 1, this increase in charging and discharging frequency does not lead to higher profits for the merchants.

These constraints are particularly influential under conditions of limited available storage capacity. With ample storage capacity, the constraints are less impactful because the merchant has sufficient flexibility to either charge or discharge power. Conversely, when storage capacity is nearing its lower limit, the constraints become crucial in decision-making. For instance, if the storage is nearly full, the merchant might be unable to discharge sufficient power without breaching the minimum capacity threshold, thus directly impacting the feasible dispatch actions.

Table 4
Total full charge and discharge times for PSH with wind.

		0	30 %Q_max	50 %Q_max	80 %Q_max	Q_max
10/01–14/2023	Full Charge Times	65	65	64	58	67
	Full Discharge Times	66	66	64	63	67
11/06–19/2023	Full Charge Times	50	50	50	48	50
	Full Discharge Times	49	49	48	43	50
12/04–17/2023	Full Charge Times	40	40	40	38	40
	Full Discharge Times	40	40	40	38	40

6.2. PSH storage with wind generation plant

In this section, we focus on the joint optimization of a PSH system integrated with a wind power plant over a two-week period from 10/01/2023 to 10/14/2023. This analysis utilizes hourly data for both electricity prices and wind power generation (accessible at https://www.misoenergy.org/), covering 336 stages. For this scenario, operational costs for charging and discharging are $c^p = c^g = 1(\$/\text{MWh})$ considered alongside the cost of wind generation $c_w = 1(\$/\text{MWh})$, contributing to the dynamic interplay of energy resources. Key parameters such as maximum and minimum SOC levels, rate constraints, and efficiency metrics remain consistent with those detailed in Section 6.1, emphasizing the similarities in basic operational characteristics despite the integration of wind energy.

Figs. 4 and 5 illustrate the outcomes of this joint optimization, showing how the integration of wind power influences the operational strategies of the PSH system, especially when starting with an initial SOC of 2000 MWh. Subfigures included in these visuals clarify how non-zero minimum generating and pumping capacity constraints impact the merchant's strategic decisions. This detailed visualization highlights the strategic depth and operational intricacies necessary for effectively managing such integrated energy systems.

Figs. 4 and 5 display five different levels of minimum generating and pumping capacities. As in the previous figures, all other parameters across these curves are identical. The minimum capacities analyzed are 0 % (i.e., 0 MW), 30 % (i.e., 630 MW), 50 % (i.e., 1050 MW), 80 % (i.e., 1680 MW), and 100 % (i.e., 2100 MW) of the maximum capacity (i.e., 2100 MW), respectively. That the joint optimization of PSH with wind power generation distinctly modifies the value function due to non-zero minimum generating and pumping capacities constraints. In scenarios with limited available capacity, these constraints significantly influence the merchant's decisions, impacting the overall dispatch strategy for the energy storage system.

Using the same three sets of electricity price data as in Section 6.1, we performed numerical simulations and summarized the results in Tables 3 and 4, which indicate that both profit and the frequency of full charge and discharge decrease as the minimum capacities increase.

Similarly to the PSH-only case, when $Q\text{min} = Q\text{max}$, the merchant's charging and discharging actions are limited to "full charge" and "full discharge" at identical power levels. This restriction results in an increase in the number of full charges and discharges compared to other scenarios, as shown in Table 4. However, as observed in Table 3, this increased frequency does not translate into higher profits for the merchants.

When the minimum generating and pumping capacities are low, their impact on the optimal strategy is minimal, allowing dispatch strategies to resemble traditional approaches where minimum dispatch values are assumed to be zero. However, as these minimum capacity levels increase, they necessitate adjustments in the strategy. This demonstrates that integrating wind power with PSH storage adds a layer of complexity to energy management, requiring sophisticated strategies to optimize operations.

In conclusion, by fixing the upper boundary and varying the lower boundaries, we have demonstrated the scalability of our research conclusions in both the PSH-only and the PSH with wind farm scenarios. Our model encompasses traditional studies as a special case when the lower bounds are zero, highlighting the general applicability of our approach. These simulations reinforce the practical value of our analytical solutions, providing actionable insights for decision-makers in the energy sector.

7. Conclusion

In this study, we developed an analytical framework to optimize the economic dispatch of PSH systems, addressing the challenges posed by non-zero minimum generating and pumping capacities—constraints often overlooked in traditional analytical models. By employing dynamic programming and introducing auxiliary decision variables, we transformed the discontinuous multi-period optimization problem into a continuous one, enabling us to derive analytically optimal dispatch strategies for both standalone PSH systems and PSH systems co-optimized with wind power plants.

Our findings reveal that the optimal dispatch strategy depends critically on State of Charge (SOC) reference points, which are determined by the current SOC level, forecasted electricity prices, wind generation (in the co-optimization scenario), and the minimum pumping and generating capacity limits. In the PSH-only scenario, four SOC reference points divide the SOC range into five sub-ranges, each corresponding to a specific dispatch decision. In the co-optimization scenario, up to five SOC reference points divide the SOC range into six sub-ranges. By comparing the current SOC with these reference points, merchants can make unique, analytically optimal decisions in each period.

We discovered that while the minimum pumping and generating capacity constraints alter the reward function, they do not fundamentally change the relationships between SOC reference points compared to existing research. However, as these minimum capacities increase, they significantly impact the optimal strategy, often leading to reduced arbitrage profits due to the increased likelihood of remaining idle. These constraints are especially influential when the available storage capacity is insufficient, necessitating strategic adjustments to ensure operational flexibility and profitability. These findings provide additional managerial insights for managing the economic dispatch of energy storage systems.

A case study with three periods of synthesized data are used to showcase the solution steps comprehensively. This detailed illustration confirms the effectiveness of our methodology and provides clear guidance on implementing the optimal dispatch strategies derived from our analytical solutions. By validating our proposed methodologies using real-world data from the MISO, we demonstrated the practical applicability and effectiveness of our analytical solutions. Our approach offers substantial advantages over traditional numerical simulations by enabling rapid adjustments in response to market fluctuations, reducing computational demands, and enhancing the scalability and efficiency of decision-making tools essential for managing the dynamics of energy markets.

Our study offers several significant strengths and contributions. By providing analytically optimal solutions for multi-period economic dispatch problems with discontinuous reward functions, we make a substantial advancement over traditional methods that rely on numerical simulations. The practical applicability of our approach is demonstrated through validation with both synthetic and real-world data, confirming its effectiveness in actual market conditions. Moreover, our methodology enhances scalability and efficiency by allowing rapid adjustments to changes in initial parameters, thereby improving decision-making tools in energy storage management. By addressing both standalone PSH systems and those co-optimized with wind farms, we offer a comprehensive framework adaptable to various energy storage scenarios.

To reduce complexity and create a reasonable and tractable framework, we have made simplifying assumptions. In future research, we can relax these assumptions or add more constraints to obtain a model that is closer to the real-world conditions, thereby expanding our understanding of the problem. The energy storage system studied in our research utilizes PSH as the energy storage facility; in fact, most existing PSH systems are large-scale energy storage with market impact. When PSH owners engage in market arbitrage, their dispatch decisions will affect electricity prices. One direction for future work is to consider the market impact of energy storage in modeling and determine how it changes the optimal strategy. Another direction for future work is to consider the co-optimization of PSH with wind power plants under government subsidy policies for wind power generation. Additionally, exploring the role of capacity markets and capacity payments and their impact on energy

storage operations—particularly in providing capacity reserves and ensuring system reliability—is another important direction for future research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Appendix

A.1. Proof of proposition 1

The current payoff rewards are defined as follows:

$$R(y_t,P_t)=\begin{cases} -\dfrac{(P_t+c)\left(y_t+\underline{Q^p}\right)}{\alpha\rho} & \left(\overline{Q^p}-\underline{Q^p}\geq y_t\geq 0\right)\ \text{(buying-and-pumping)}\\ -(P_t-c)\left(y_t-\underline{Q^g}\right)\beta\rho & \left(-\overline{Q^g}+\underline{Q^g}\leq y_t\leq 0\right)\ \text{(generating-and-selling)}\end{cases}$$

Where, y_t is an auxiliary function defined as:

$$y_t=\begin{cases}\alpha_t-\underline{Q^p} & \left(\overline{Q^p}\geq\alpha_t\geq\underline{Q^p}\right)\\ \alpha_t+\underline{Q^g} & \left(-\overline{Q^g}\leq\alpha_t\leq-\underline{Q^g}\right)\end{cases}$$

Here, α_t represents the energy change from period t to period $t+1$ before accounting for energy loss. By using $y_t=\begin{cases}\dfrac{E_{t+1}}{\eta_t}-E_t-\underline{Q^p} & \left(\overline{Q^p}-\underline{Q^p}\geq y_t\geq 0\right)\\ \dfrac{E_{t+1}}{\eta_t}-E_t+\underline{Q^g} & \left(-\overline{Q^g}+\underline{Q^g}\leq y_t\leq 0\right)\end{cases}$, we derive the following rewards function at time t.

$$R(y_t,P_t)=\begin{cases} -\dfrac{(P_t+c)\left(\dfrac{E_{t+1}}{\eta_t}-E_t\right)}{\alpha\rho} & \left(\overline{Q^p}-\underline{Q^p}\geq y_t\geq 0\right)\\ -(P_t-c)\left(\dfrac{E_{t+1}}{\eta_t}-E_t\right)\beta\rho_t & \left(-\overline{Q^g}+\underline{Q^g}\leq y_t\leq 0\right)\end{cases}\tag{A1}$$

We also define the following value function:

$$V_t(E_tP_t)=E[R(y_tP_t)+V_{t+1}(S(t+1)\,|S(t))]=\begin{cases} -\dfrac{(P_t+c)}{\alpha\rho}\left(\dfrac{E_{t+1}}{\eta_t}-E_t\right)+E[V_{t+1}(S(t+1)\,|S(t))] & \left(\overline{Q^p}-\underline{Q^p}\geq y_t\geq 0\right)\\ (c-P_t)\beta\rho\left(\dfrac{E_{t+1}}{\eta_t}-E_t\right)+E[V_{t+1}(S(t+1)\,|S(t))] & \left(-\overline{Q^g}+\underline{Q^g}\leq y_t\leq 0\right)\end{cases}$$

By optimizing the value function $V_t(E_t,P_t)$, subject to $max\left\{-\overline{Q^g}+\underline{Q^g},\underline{E}-E_t+\underline{Q^g}\right\}\leq y_t\leq min\left\{\overline{Q^p}-\underline{Q^p},\overline{E}-E_t-\underline{Q^p}\right\}$, we obtain the following result based on the Bellman Equation:

$$\begin{cases} V_t^{p^*}(S(t))=max_{\underline{E}\leq E_{t+1}\leq\overline{E}}\left\{-\dfrac{(P_t+c)}{\alpha\rho}\left(\dfrac{E_{t+1}}{\eta_t}-E_t\right)+E\left[V_{t+1}^*\left(S(t+1)\,|S(t))\right]\right\}\\ or\ V_t^{p^*}(S(t))=max_{\underline{E}\leq E_{t+1}\leq\overline{E}}\left\{-\dfrac{(P_t+c)}{\alpha\rho}\dfrac{E_{t+1}}{\eta_t}+E\left[V_{t+1}^*\left(S(t+1)\,|S(t))\right]\right\}\\ V_t^{g^*}(S(t))=max_{\underline{E}\leq E_{t+1}\leq\overline{E}}\left\{(c-P_t)\beta\rho\left(\dfrac{E_{t+1}}{\eta_t}-E_t\right)+E\left[V_{t+1}^*\left(S(t+1)\,|S(t))\right]\right\}\\ or\ V_t^{g^*}(S(t))=max_{\underline{E}\leq E_{t+1}\leq\overline{E}}\left\{(c-P_t)\beta\rho\dfrac{E_{t+1}}{\eta_t}+E\left[V_{t+1}^*\left(S(t+1)\,|S(t))\right]\right\}\end{cases}\tag{A2}$$

It is known that for any $t \in \{1, 2, \ldots, T\}$, in every stage t, the value functions $V_t(S(t))$ and $E[V_{t+1}(S(t+1)|S(t))]$ are concave in $E_t \in \left[\underline{E}, \overline{E}\right]$ for each given stage $S(t) = S_t(E_t, P_t)$ and $P_t < \infty$ holds. Consequently, the following equivalence relations hold:

$$\begin{cases} V_t^{p''}(S(t)) = \dfrac{\partial^2 E\left[V_{t+1}^*(S(t+1)|S(t))\right]}{\partial {E_{t+1}}^2} \le 0 \\ V_t^{g''}(S(t)) = \dfrac{\partial^2 E\left[V_{t+1}^*(S(t+1)|S(t))\right]}{\partial {E_{t+1}}^2} \le 0 \end{cases} \tag{A3}$$

Since the second-order derivative is negative, we can find the unique optimal solutions through the first-order condition. This yields the following results:

$$\begin{cases} E_{t+1}^{p^*} = argmax_{\underline{E} \le E_{t+1} \le \overline{E}} \left\{ -\dfrac{(P_t + c)}{\alpha\rho} \dfrac{E_{t+1}}{\eta_t} + E\left[V_{t+1}^*(S(t+1)|S(t))\right] \right\} \\ E_{t+1}^{g^*} = argmax_{\underline{E} \le E_{t+1} \le \overline{E}} \left\{ (c - P_t)\beta\rho \dfrac{E_{t+1}}{\eta_t} + E\left[V_{t+1}^*(S(t+1)|S(t))\right] \right\} \end{cases} \tag{A4}$$

Obviously, the corresponding first-order functions are shown as follows:

$$\begin{cases} \dfrac{\partial E\left[V_{t+1}^*(S(t+1)|S(t))\right]}{\partial E_{t+1}}\Big|_{E_{t+1}=E_{t+1}^{p^*}} = \dfrac{(P_t + c)}{\alpha\rho} \dfrac{1}{\eta_t} \\ \dfrac{\partial E\left[V_{t+1}^*(S(t+1)|S(t))\right]}{\partial E_{t+1}}\Big|_{E_{t+1}=E_{t+1}^{g^*}} = (P_t - c)\beta\rho \dfrac{1}{\eta_t} \end{cases} \tag{A5}$$

Based on the given parameters, we can derive $\frac{(P_t+c)}{\alpha\rho} \frac{1}{\eta_t} \ge (P_t - c)\beta\rho \frac{1}{\eta_t}$.

The following conclusion is reached:

$$\frac{\partial E\left[V_{t+1}^*(S(t+1)|S(t))\right]}{\partial E_{t+1}}\Big|_{E_{t+1}=E_{t+1}^{p^*}} \ge \frac{\partial E\left[V_{t+1}^*(S(t+1)|S(t))\right]}{\partial E_{t+1}}\Big|_{E_{t+1}=E_{t+1}^{g^*}} \tag{A6}$$

According to the Eq. (A3), $\frac{\partial E\left[V_{t+1}^*(S(t+1)|S(t))\right]}{\partial E_{t+1}}$ is monotonically decreasing on $E_{t+1} \in \left[\underline{E}, \overline{E}\right]$. Combining with Eq. (A6), we obtain:

$$\underline{E} \le E_{t+1}^{p^*} \le E_{t+1}^{g^*} \le \overline{E} \tag{A7}$$

In summary, we get the following results:

$$y_t = \begin{cases} min\left\{\overline{Q^p} - \underline{Q^p}, E_{t+1}^{p^*} - E_t - \underline{Q^p}\right\} & E_t \in \left[\underline{E}, E_{t+1}^{p^*} - \underline{Q^p}\right) \\ 0 & E_t \in \left[E_{t+1}^{p^*} - \underline{Q^p}, E_{t+1}^{g^*} + \underline{Q^g}\right] \\ max\left\{-\overline{Q^g} + \underline{Q^g}, E_{t+1}^{g^*} - E_t + \underline{Q^g}\right\} & E_t \in \left(E_{t+1}^{g^*} + \underline{Q^g}, \overline{E}\right] \end{cases} \tag{A8}$$

A.2. Proof of proposition 2

Before discussing the optimal dispatch decision α_t^*, we make an assumption to ensure that the dispatch proceeds smoothly. We assume that $E_{t+1}^{g^*} \ge \underline{E} + \underline{Q^g}$ and $E_{t+1}^{p^*} \le \overline{E} - \underline{Q^p}$, and we will later discuss cases where the assumption does not hold.

Due to the presence of a non-zero lower limit constraint on dispatch, when $E_t \in \left[E_{t+1}^{p^*} - \underline{Q^p}, E_{t+1}^{g^*} + \underline{Q^g}\right]$, there are three choices: $\alpha_t = \underline{Q^p}$, $\alpha_t = -\underline{Q^g}$, and $\alpha_t = 0$. We need to select the optimal one among these three decisions.

(1) When $E_t \in \left[E_{t+1}^{p^*} - \underline{Q^p}, E_{t+1}^{p^*}\right)$, there is $y_t = 0$, corresponds to two decisions:

① If $\alpha_t = \underline{Q^p}$, the corresponding value equation is:

$$V_t^p(S(t)) = -\frac{(P_t + c)\underline{Q^p}}{\alpha\rho} + E\left[V_{t+1}\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right)|S(t)\right)\right]$$

② If $\alpha_t = 0$, the corresponding value equation is:

$$V_t^i(S(t)) = E[V_{t+1}(S_{t+1}(E_t, P_{t+1})|S(t))]$$

We can determine the reference point $E_{t+1}^{p^{**}}$ by $V_t^p(S(t)) = V_t^i(S(t))$:

$$\frac{(P_t+c)\underline{Q^p}}{\alpha\rho}+E[V_{t+1}(S_{t+1}(E_t,P_{t+1})\,|S(t)\,)]\Big|_{E_{t+1}=E_{t+1}^{p^{**}}}-E\left[V_{t+1}\left(S_{t+1}\left(E_t+\underline{Q^p},P_{t+1}\right)|S(t)\right)\right]|_{E_{t+1}=E_{t+1}^{p^{**}}}=0$$

When $E_t\in\left[E_{t+1}^{p^*}-\underline{Q^p},E_{t+1}^{p^{**}}\right)$, the merchant should choose $\alpha_t=\underline{Q^p}$.

When $E_t\in\left[E_{t+1}^{p^{**}},E_{t+1}^{p^*}\right)$, the merchant should choose $\alpha_t=0$.

(2) When $E_t\in\left(E_{t+1}^{g^*},E_{t+1}^{g^*}+\underline{Q^g}\right]$, there is $y_t=0$, corresponds to two decisions:

① If $\alpha_t=-\underline{Q^g}$, the corresponding value equation is:

$$V_t^g(S(t)\,)=(P_t-c)\underline{Q^g}\beta\rho+E\left[V_{t+1}\left(S_{t+1}\left(E_t-\underline{Q^g},P_{t+1}\right)|S(t)\right)\right]$$

② If $\alpha_t=0$, the corresponding value equation is:

$$V_t^i(S(t)\,)=E[V_{t+1}(S_{t+1}(E_t,P_{t+1})\,|S(t)\,)]$$

We can determine the reference point $E_{t+1}^{g^{**}}$ by $V_t^g(S(t)\,)=V_t^i(S(t)\,)$:

$$(P_t-c)\underline{Q^g}\beta\rho+E\left[V_{t+1}\left(S_{t+1}\left(E_t-\underline{Q^g},P_{t+1}\right)|S(t)\right)\right]\Big|_{E_{t+1}=E_{t+1}^{g^{**}}}-E[V_{t+1}(S_{t+1}(E_t,P_{t+1})\,|S(t)\,)]\,\Big|_{E_{t+1}=E_{t+1}^{g^{**}}}=0$$

When $E_t\in(E_{t+1}^{g^*},E_{t+1}^{g^{**}})$, the merchant should choose $\alpha_t=0$.

When $E_t\in\left[E_{t+1}^{g^{**}},E_{t+1}^{g^*}+\underline{Q^g}\right)$, the merchant should choose $\alpha_t=-\underline{Q^g}$.

(3) When $E_t\in\left[E_{t+1}^{p^*},E_{t+1}^{g^*}\right]$, the optimal decision is $\alpha_t=0$.

To summarize, the following results are obtained:

$$\alpha_t^*=\begin{cases}min\left\{max\left\{E_{t+1}^{p^*}-E_t,\underline{Q^p}\right\},\overline{Q^p}\right\} & E_t\in\left[\underline{E},E_{t+1}^{p^{**}}\right)\\ 0 & E_t\in[E_{t+1}^{p^{**}},E_{t+1}^{g^{**}}]\\ max\left\{min\left\{E_{t+1}^{g^*}-E_t,-\underline{Q^g}\right\},-\overline{Q^g}\right\} & E_t\in(E_{t+1}^{g^{**}},\overline{E}]\end{cases}\tag{A9}$$

In the above discussion, we assumed that $E_{t+1}^{g^*}\geq\underline{E}+\underline{Q^g}$ and $E_{t+1}^{p^*}\leq\overline{E}-\underline{Q^p}$, which ensures that the energy storage capacity constraints are not violated.

Therefore, due to the existence of non-zero lower limit constraints in power scheduling, there are three additional special case considered in this paper:

Case 1: The reference points $E_{t+1}^{p^*}$ and $E_{t+1}^{g^*}$ satisfy the following relationship:

$$\underline{E}\leq E_{t+1}^{p^*}\leq E_{t+1}^{g^*}\leq\underline{E}+\underline{Q^g}$$

When $E_t\in\left[E_{t+1}^{g^*},\underline{E}+\underline{Q^g}\right]$, we can only choose $\alpha_t=0$, because choosing to discharge would result in $E_{t+1}<\underline{E}$, thereby violating the lower storage capacity constraint.

We will find the reference point $E_{t+1}^{g^{**}}$ within the range $\left[\underline{E}+\underline{Q^g},E_{t+1}^{g^*}+\underline{Q^g}\right]$. Nevertheless, we can still obtain the optimal policy for α_t^* as follows:

$$\alpha_t^*=\begin{cases}min\left\{max\left\{E_{t+1}^{p^*}-E_t,\underline{Q^p}\right\},\overline{Q^p}\right\} & E_t\in\left[\underline{E},E_{t+1}^{p^{**}}\right)\\ 0 & E_t\in[E_{t+1}^{p^{**}},E_{t+1}^{g^{**}}]\\ max\left\{min\left\{E_{t+1}^{g^*}-E_t,-\underline{Q^g}\right\},-\overline{Q^g}\right\} & E_t\in(E_{t+1}^{g^{**}},\overline{E}]\end{cases}$$

Case 2: The reference points $E_{t+1}^{p^*}$ and $E_{t+1}^{g^*}$ satisfy the following relationship:

$$\overline{E}-\underline{Q^p}\leq E_{t+1}^{p^*}\leq E_{t+1}^{g^*}\leq\overline{E}$$

When $E_t\in\left[\overline{E}-\underline{Q^p},E_{t+1}^{p^*}\right]$, we can only choose $\alpha_t=0$, because choosing to charge would results in $E_{t+1}>\overline{E}$, there by violating the upper storage

capacity constraint.

We will find the reference point $E_{t+1}^{p^{**}}$ within the range $\left[E_{t+1}^{p^{*}}-\underline{Q^p},\overline{E}-\underline{Q^p}\right]$. Nevertheless, we can still obtain the optimal policy for α_t^* as follows:

$$\alpha_t^*=\begin{cases} min\left\{max\left\{E_{t+1}^{p^{*}}-E_t,\underline{Q^p}\right\},\overline{Q^p}\right\} & E_t\in\left[\underline{E},E_{t+1}^{p^{**}}\right) \\ 0 & E_t\in[E_{t+1}^{p^{**}},E_{t+1}^{g^{**}}] \\ max\left\{min\left\{E_{t+1}^{g^{*}}-E_t,-\underline{Q^g}\right\},-\overline{Q^g}\right\} & E_t\in(E_{t+1}^{g^{**}},\overline{E}] \end{cases}$$

Case 3: The reference points $E_{t+1}^{p^*}$ and $E_{t+1}^{g^*}$ satisfy the following relationship:

$E_{t+1}^{g^{*}}\le\underline{E}+\underline{Q^g}$ and $E_{t+1}^{p^{*}}\ge\overline{E}-\underline{Q^p}$.

At this point, the merchant can only choose $\alpha_t=0$ on $E_t=\left[\underline{E},\overline{E}\right]$, since either charging or discharging will violate the storage capacity constraints.

The optimal policy for α_t^* is shown below:

$$\alpha_t^*=0;\quad E_t=\left[\underline{E},\overline{E}\right]$$

Through our discussion, we find that the form of the optimal policy for energy storage dispatch highly consistent even in these special cases. Thus, we can obtain a generalized optimal solution to the problem:

$$\alpha_t^*=\begin{cases} min\left\{max\left\{E_{t+1}^{p^{*}}-E_t,\underline{Q^p}\right\},\overline{Q^p}\right\} & E_t\in\left[\underline{E},E_{t+1}^{p^{**}}\right) \\ 0 & E_t\in[E_{t+1}^{p^{**}},E_{t+1}^{g^{**}}] \\ max\left\{min\left\{E_{t+1}^{g^{*}}-E_t,-\underline{Q^g}\right\},-\overline{Q^g}\right\} & E_t\in(E_{t+1}^{g^{**}},\overline{E}] \end{cases} \tag{A10}$$

A.3. Proof of proposition 3

The current payoff rewards are defined as follows:

Case 1 for $\alpha w_t<\underline{Q^p}$

$$R(y_t,P_t)=\begin{cases} -P_t\left(\dfrac{\left(y_t+\underline{Q^p}\right)}{\alpha}-w_t\right)\Big/\rho-c\dfrac{\left(y_t+\underline{Q^p}\right)}{\alpha\rho}-c_w w_t & \left(\overline{Q^p}-\underline{Q^p}\ge y_t\ge 0\right) \\ P_t\left(w_t-\left(y_t-\underline{Q^g}\right)\beta\right)\rho+c\left(y_t-\underline{Q^g}\right)\beta\rho-c_w w_t & \left(-\overline{Q^g}+\underline{Q^g}\le y_t\le 0\right) \end{cases} \tag{A11.1}$$

Case 2 for $\alpha w_t\ge\underline{Q^p}$

$$R(y_t,P_t)=\begin{cases} -P_t\left(\dfrac{\left(y_t+\underline{Q^p}\right)}{\alpha}-w_t\right)\Big/\rho-c\dfrac{\left(y_t+\underline{Q^p}\right)}{\alpha\rho}-c_w w_t & \left(\overline{Q^p}-\underline{Q^p}\ge y_t\ge \alpha w_t-\underline{Q^p}\right) \\ P_t\left(w_t-\dfrac{\left(y_t+\underline{Q^p}\right)}{\alpha}\right)\rho-c\dfrac{\left(y_t+\underline{Q^p}\right)}{\alpha\rho}-c_w w_t & \left(\alpha w_t-\underline{Q^p}>y_t\ge 0\right) \\ P_t\left(w_t-\left(y_t-\underline{Q^g}\right)\beta\right)\rho+c\left(y_t-\underline{Q^g}\right)\beta\rho-c_w w_t & \left(-\overline{Q^g}+\underline{Q^g}\le y_t\le 0\right) \end{cases} \tag{A11.2}$$

Where, y_t is an auxiliary function defined as:

$$y_t=\begin{cases} \alpha_t-\underline{Q^p} & \left(\overline{Q^p}\ge\alpha_t\ge\underline{Q^p}\right) \\ \alpha_t+\underline{Q^g} & \left(-\overline{Q^g}\le\alpha_t\le-\underline{Q^g}\right) \end{cases}$$

Here, α_t represents the energy change from period t to period $t+1$ before accounting for energy loss.

By using $y_t = \begin{cases} \frac{E_{t+1}}{\eta_t} - E_t - \underline{Q^p} & \left(\overline{Q^p} - \underline{Q^p} \geq y_t \geq 0\right) \\ \frac{E_{t+1}}{\eta_t} - E_t + \underline{Q^g} & \left(-\overline{Q^g} + \underline{Q^g} \leq y_t \leq 0\right) \end{cases}$, we derive the following rewards function at time t.

Case 1 for $\alpha w_t < \underline{Q^p}$

$$R(y_t, P_t) = \begin{cases} -P_t\left(\left(\frac{E_{t+1}}{\eta_t} - E_t\right) \Big/ \alpha - w_t\right) \Big/ \rho - \frac{c}{\alpha\rho}\left(\frac{E_{t+1}}{\eta_t} - E_t\right) - c_w w_t & \left(\overline{Q^p} - \underline{Q^p} \geq y_t \geq 0\right) \\ P_t\left(w_t - \left(\frac{E_{t+1}}{\eta_t} - E_t\right)\beta\right)\rho + c\left(\frac{E_{t+1}}{\eta_t} - E_t\right)\beta\rho - c_w w_t & \left(-\overline{Q^g} + \underline{Q^g} \leq y_t \leq 0\right) \end{cases} \tag{A.12.1}$$

Case 2 for $\alpha w_t \geq \underline{Q^p}$

$$R(y_t, P_t) = \begin{cases} -P_t\left(\left(\frac{E_{t+1}}{\eta_t} - E_t\right) \Big/ \alpha - w_t\right) \Big/ \rho - \frac{c}{\alpha\rho}\left(\frac{E_{t+1}}{\eta_t} - E_t\right) - c_w w_t & \left(\overline{Q^p} - \underline{Q^p} \geq y_t \geq \alpha w_t - \underline{Q^p}\right) \\ P_t\left(w_t - \left(\frac{E_{t+1}}{\eta_t} - E_t\right) \Big/ \alpha\right)\rho - \frac{c}{\alpha\rho}\left(\frac{E_{t+1}}{\eta_t} - E_t\right) - c_w w_t & \left(\alpha w_t - \underline{Q^p} > y_t \geq 0\right) \\ P_t\left(w_t - \left(\frac{E_{t+1}}{\eta_t} - E_t\right)\beta\right)\rho + c\left(\frac{E_{t+1}}{\eta_t} - E_t\right)\beta\rho - c_w w_t & \left(-\overline{Q^g} + \underline{Q^g} \leq y_t \leq 0\right) \end{cases} \tag{A.12.2}$$

We also have the following value function:

Case 1 for $\alpha w_t < \underline{Q^p}$

$$\begin{cases} V_t^{(1)}(S(t)) = -P_t\left(\left(\frac{E_{t+1}}{\eta_t} - E_t\right) \Big/ \alpha - w_t\right) \Big/ \rho - \frac{c}{\alpha\rho}\left(\frac{E_{t+1}}{\eta_t} - E_t\right) - c_w w_t + E[V_{t+1}(S(t+1)\,|S(t))] & \left(\overline{Q^p} - \underline{Q^p} \geq y_t \geq 0\right) \\ V_t^{(3)}(S(t)) = P_t\left(w_t - \left(\frac{E_{t+1}}{\eta_t} - E_t\right)\beta\right)\rho + c\left(\frac{E_{t+1}}{\eta_t} - E_t\right)\beta\rho - c_w w_t + E[V_{t+1}(S(t+1)\,|S(t))] & \left(-\overline{Q^g} + \underline{Q^g} \leq y_t \leq 0\right) \end{cases} \tag{A.13.1}$$

Case 2 for $\alpha w_t \geq \underline{Q^p}$

$$\begin{cases} V_t^{(1)}(S(t)) = -P_t\left(\left(\frac{E_{t+1}}{\eta_t} - E_t\right) \Big/ \alpha - w_t\right) \Big/ \rho - \frac{c}{\alpha\rho}\left(\frac{E_{t+1}}{\eta_t} - E_t\right) - c_w w_t + E[V_{t+1}(S(t+1)\,|S(t))] & \left(\overline{Q^p} - \underline{Q^p} \geq y_t \geq \alpha w_t - \underline{Q^p}\right) \\ V_t^{(2)}(S(t)) = P_t\left(w_t - \left(\frac{E_{t+1}}{\eta_t} - E_t\right) \Big/ \alpha\right)\rho - \frac{c}{\alpha\rho}\left(\frac{E_{t+1}}{\eta_t} - E_t\right) - c_w w_t + E[V_{t+1}(S(t+1)\,|S(t))] & \left(\alpha w_t - \underline{Q^p} > y_t \geq 0\right) \\ V_t^{(3)}(S(t)) = P_t\left(w_t - \left(\frac{E_{t+1}}{\eta_t} - E_t\right)\beta\right)\rho + c\left(\frac{E_{t+1}}{\eta_t} - E_t\right)\beta\rho - c_w w_t + E[V_{t+1}(S(t+1)\,|S(t))] & \left(0 > y_t \geq -\overline{Q^g} + \underline{Q^g}\right) \end{cases} \tag{A.13.2}$$

By optimizing the value function $V_t(E_t, P_t, w_t)$, subject to $max\left\{-\overline{Q^g} + \underline{Q^g}, \underline{E} - E_t + \underline{Q^g}\right\} \leq y_t \leq min\left\{\overline{Q^p} - \underline{Q^p}, \overline{E} - E_t - \underline{Q^p}\right\}$, we obtain the following result based on the Bellman Equation:

Case 1 for $\alpha w_t < \underline{Q^p}$

$$\begin{cases} V_t^{(1)^*}(S(t)) = max_{\underline{E} \leq E_{t+1} \leq \overline{E}}\left\{-P_t\left(\left(\frac{E_{t+1}}{\eta_t} - E_t\right) \Big/ \alpha - w_t\right) \Big/ \rho - \frac{c}{\alpha\rho}\left(\frac{E_{t+1}}{\eta_t} - E_t\right) - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ \text{or } V_t^{(1)^*}(S(t)) = max_{\underline{E} \leq E_{t+1} \leq \overline{E}}\left\{-P_t\left(\frac{E_{t+1}}{\alpha\eta_t} - w_t\right) \Big/ \rho - \frac{c}{\alpha\rho}\frac{E_{t+1}}{\eta_t} - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ V_t^{(3)^*}(S(t)) = max_{\underline{E} \leq E_{t+1} \leq \overline{E}}\left\{P_t\left(w_t - \left(\frac{E_{t+1}}{\eta_t} - E_t\right)\beta\right)\rho + c\left(\frac{E_{t+1}}{\eta_t} - E_t\right)\beta\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ \text{or } V_t^{(3)^*}(S(t)) = max_{\underline{E} \leq E_{t+1} \leq \overline{E}}\left\{P_t\left(w_t - \frac{E_{t+1}}{\eta_t}\beta\right)\rho + c\frac{E_{t+1}}{\eta_t}\beta\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \end{cases} \tag{A.14.1}$$

Case 2 for $\alpha w_t \geq \underline{Q^p}$

$$\begin{cases} V_t^{(1)^*}(S(t)) = max_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ -P_t\left(\left(\frac{E_{t+1}}{\eta_t}-E_t\right)\Big/\alpha - w_t\right)\Big/\rho - \frac{c}{\alpha\rho}\left(\frac{E_{t+1}}{\eta_t}-E_t\right) - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ \text{or } V_t^{(1)^*}(S(t)) = max_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ -P_t\left(\frac{E_{t+1}}{\alpha\eta_t}-w_t\right)\Big/\rho - \frac{c}{\alpha\rho}\frac{E_{t+1}}{\eta_t} - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ V_t^{(2)^*}(S(t)) = max_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ P_t\left(w_t - \left(\frac{E_{t+1}}{\eta_t}-E_t\right)\Big/\alpha\right)\rho - \frac{c}{\alpha\rho}\left(\frac{E_{t+1}}{\eta_t}-E_t\right) - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ \text{or } V_t^{(2)^*}(S(t)) = max_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ P_t\left(w_t - \frac{E_{t+1}}{\alpha\eta_t}\right)\rho - \frac{c}{\alpha\rho}\frac{E_{t+1}}{\eta_t} - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ V_t^{(3)^*}(S(t)) = max_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ P_t\left(w_t - \left(\frac{E_{t+1}}{\eta_t}-E_t\right)\beta\right)\rho + c\left(\frac{E_{t+1}}{\eta_t}-E_t\right)\beta\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ \text{or } V_t^{(3)^*}(S(t)) = max_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ P_t\left(w_t - \frac{E_{t+1}}{\eta_t}\beta\right)\rho + c\frac{E_{t+1}}{\eta_t}\beta\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \end{cases} \tag{A.14.2}$$

It is known that for any $t \in \{1, 2, \ldots, T\}$, in each stage t, the value functions $V_t(S(t))$ and $E[V_{t+1}(S(t+1)\,|S(t))]$ are concave in $E_t \in \left[\underline{E}, \overline{E}\right]$ for each given stage $S(t) = S_t(E_t, P_t, w_t)$ and $P_t < \infty$ holds. Consequently, the following equivalence relations hold:

Case 1 for $\alpha w_t < \underline{Q^p}$

$$\begin{cases} V_t^{(1)^{\cdot\cdot}}(S(t)) = \dfrac{\partial^2 E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]}{\partial {E_{t+1}}^2} \leq 0 \\ V_t^{(3)^{\cdot\cdot}}(S(t)) = \dfrac{\partial^2 E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]}{\partial {E_{t+1}}^2} \leq 0 \end{cases} \tag{A.15.1}$$

Case 2 for $\alpha w_t \geq \underline{Q^p}$

$$\begin{cases} V_t^{(1)^{\cdot\cdot}}(S(t)) = \dfrac{\partial^2 E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]}{\partial {E_{t+1}}^2} \leq 0 \\ V_t^{(2)^{\cdot\cdot}}(S(t)) = \dfrac{\partial^2 E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]}{\partial {E_{t+1}}^2} \leq 0 \\ V_t^{(3)^{\cdot\cdot}}(S(t)) = \dfrac{\partial^2 E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]}{\partial {E_{t+1}}^2} \leq 0 \end{cases} \tag{A.15.2}$$

Since the second-order derivative is negative, we can find the unique optimal solutions by applying the first-order condition. We can get the following results:

Case 1 for $\alpha w_t < \underline{Q^p}$

$$\begin{cases} E_{t+1}^{(1)^*} = argmax_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ -P_t\left(\frac{E_{t+1}}{\alpha\eta_t}-w_t\right)\Big/\rho - \frac{c}{\alpha\rho}\frac{E_{t+1}}{\eta_t} - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ E_{t+1}^{(3)^*} = argmax_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ P_t\left(w_t - \frac{E_{t+1}}{\eta_t}\beta\right)\rho + c\frac{E_{t+1}}{\eta_t}\beta\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \end{cases} \tag{A.16.1}$$

Case 2 for $\alpha w_t \geq \underline{Q^p}$

$$\begin{cases} E_{t+1}^{(1)^*} = argmax_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ -P_t\left(\frac{E_{t+1}}{\alpha\eta_t}-w_t\right)\Big/\rho - \frac{c}{\alpha\rho}\frac{E_{t+1}}{\eta_t} - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ E_{t+1}^{(2)^*} = argmax_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ P_t\left(w_t - \frac{E_{t+1}}{\alpha\eta_t}\right)\rho - \frac{c}{\alpha\rho}\frac{E_{t+1}}{\eta_t} - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \\ E_{t+1}^{(3)^*} = argmax_{\underline{E}\leq E_{t+1}\leq \overline{E}}\left\{ P_t\left(w_t - \frac{E_{t+1}}{\eta_t}\beta\right)\rho + c\frac{E_{t+1}}{\eta_t}\beta\rho - c_w w_t + E\left[V_{t+1}^*(S(t+1)\,|S(t))\right]\right\} \end{cases} \tag{A.16.2}$$

Clearly, the corresponding first-order functions are shown as follows:

Case 1 For $\alpha w_t < \underline{Q^p}$

$$\begin{cases} \left.\dfrac{\partial E\left[V_{t+1}^{*}(S(t+1)\,|S(t))\right]}{\partial E_{t+1}}\right|_{E_{t+1}=E_{t+1}^{(1)^{*}}} = \dfrac{(P_t+c)}{\alpha\rho}\dfrac{1}{\eta_t} \\ \left.\dfrac{\partial E\left[V_{t+1}^{*}(S(t+1)\,|S(t))\right]}{\partial E_{t+1}}\right|_{E_{t+1}=E_{t+1}^{(3)^{*}}} = (P_t-c)\beta\rho\dfrac{1}{\eta_t} \end{cases} \tag{A.17.1}$$

Case 2 For $\alpha w_t \geq \underline{Q^p}$

$$\begin{cases} \left.\dfrac{\partial E\left[V_{t+1}^{*}(S(t+1)\,|S(t))\right]}{\partial E_{t+1}}\right|_{E_{t+1}=E_{t+1}^{(1)^{*}}} = \dfrac{(P_t+c)}{\alpha\rho}\dfrac{1}{\eta_t} \\ \left.\dfrac{\partial E\left[V_{t+1}^{*}(S(t+1)\,|S(t))\right]}{\partial E_{t+1}}\right|_{E_{t+1}=E_{t+1}^{(2)^{*}}} = \dfrac{(c+P_t\rho^2)}{\alpha\rho}\dfrac{1}{\eta_t} \\ \left.\dfrac{\partial E\left[V_{t+1}^{*}(S(t+1)\,|S(t))\right]}{\partial E_{t+1}}\right|_{E_{t+1}=E_{t+1}^{(3)^{*}}} = (P_t-c)\beta\rho\dfrac{1}{\eta_t} \end{cases} \tag{A.17.2}$$

By observation, we can see that essentially Case 1 is a special form of Case2, so we will only discuss Case 2 here. Based on the given parameters, we can get $\frac{(P_t+c)}{\alpha\rho}\frac{1}{\eta_t} \geq \frac{(c+P_t\rho^2)}{\alpha\rho}\frac{1}{\eta_t} \geq (P_t-c)\beta\rho\frac{1}{\eta_t}$.

The following conclusion is reached:

$$\left.\frac{\partial E\left[V_{t+1}^{*}(S(t+1)\,|S(t))\right]}{\partial E_{t+1}}\right|_{E_{t+1}=E_{t+1}^{(1)^{*}}} \geq \left.\frac{\partial E\left[V_{t+1}^{*}(S(t+1)\,|S(t))\right]}{\partial E_{t+1}}\right|_{E_{t+1}=E_{t+1}^{(2)^{*}}} \geq \left.\frac{\partial E\left[V_{t+1}^{*}(S(t+1)\,|S(t))\right]}{\partial E_{t+1}}\right|_{E_{t+1}=E_{t+1}^{(3)^{*}}} \tag{A18}$$

According to Eq. (A15), $\frac{\partial E\left[V_{t+1}^{*}(S(t+1)\,|S(t))\right]}{\partial E_{t+1}}$ is monotonically decreasing on $E_{t+1} \in \left[\underline{E}, \overline{E}\right]$. Combining this with Eq. (A18), we obtain:

$$\underline{E} \leq E_{t+1}^{(1)^{*}} \leq E_{t+1}^{(2)^{*}} \leq E_{t+1}^{(3)^{*}} \leq \overline{E} \tag{A19}$$

In summary, we get the following results:

Case 1: For $\alpha w_t < \underline{Q^p}$.

Case 1.1: If $\alpha w_t < min\left\{E_{t+1}^{(1)^{*}}, \overline{Q^p} - \underline{Q^p}\right\}$, which means there is less wind power generation. At this point, even if all the wind power is stored, the SOC cannot reach $E_{t+1}^{(1)^{*}}$. Therefore, there is the possibility of deciding to store all the wind power and purchase electricity from the market.

Case 1.1.1: If $E_{t+1}^{(1)^{*}} < \underline{Q^p} \Rightarrow \alpha w_t < E_{t+1}^{(1)^{*}} < \underline{Q^p}$, which means the minimum dispatch value is large, and whenever a charging operation is performed, it will definitely lead to the SOC exceeding $E_{t+1}^{(1)^{*}}$. Thus, the strategy of storing all wind power and buying power from the market is not valid. The optimal policy for y_t in this case can be expressed as:

$$y_t^{*} = \begin{cases} 0 & E_t \in \left[\underline{E}, E_{t+1}^{(3)^{*}} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^{*}} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^{*}} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 1.1.2: If $E_{t+1}^{p^{*}} \geq \underline{Q^p} \Rightarrow E_{t+1}^{p^{*}} \geq \underline{Q^p} > \alpha w_t$, which means the minimum dispatch value is small, the strategy of storing all wind power and buying power from the market is valid. The optimal policy for y_t in this case can be expressed as:

$$y_t^{*} = \begin{cases} min\left\{E_{t+1}^{(1)^{*}} - E_t - \underline{Q^p}, \overline{Q^p} - \underline{Q^p}\right\} & E_t \in \left[\underline{E}, E_{t+1}^{(1)^{*}} - \underline{Q^p}\right) \\ 0 & E_t \in \left[E_{t+1}^{(1)^{*}} - \underline{Q^p}, E_{t+1}^{(3)^{*}} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^{*}} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^{*}} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 1.2: If $\alpha w_t \geq min\left\{E_{t+1}^{p^{*}}, \overline{Q^p} - \underline{Q^p}\right\}$, which means the wind power generation is so large that storing all of it would cause the SOC to exceed $E_{t+1}^{(1)^{*}}$. Thus, the optimal policy for y_t can be expressed as:

$$y_t^{*} = \begin{cases} 0 & E_t \in \left[\underline{E}, E_{t+1}^{(3)^{*}} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^{*}} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^{*}} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 2: For $\alpha w_t \geq \underline{Q^p}$.

Case 2.1: If $\alpha w_t < min\left\{E_{t+1}^{(1)^{*}}, \overline{Q^p} - \underline{Q^p}\right\}$, which means there is less wind power generation. At this point, even if all the wind power is stored, the SOC cannot reach $E_{t+1}^{(1)^{*}}$. We get this following relationship:

$\underline{Q^p} \le \alpha w_t < E_{t+1}^{(1)^*} \Rightarrow \underline{Q^p} < E_{t+1}^{(1)^*}$.

This implies that the minimum dispatch value is small, and the strategy of storing all wind power and buying power from the market is valid. The optimal policy for y_t in this case can be expressed as:

$$y_t^* = \begin{cases} min\left\{E_{t+1}^{(1)^*} - E_t - \underline{Q^p}, \overline{Q^p} - \underline{Q^p}\right\} & E_t \in \left[\underline{E}, E_{t+1}^{(1)^*} - \underline{Q^p} - \alpha w_t\right) \\ min\left\{E_{t+1}^{(2)^*} - E_t - \underline{Q^p}, \alpha w_t\right\} & E_t \in \left[E_{t+1}^{(1)^*} - \underline{Q^p} - \alpha w_t, E_{t+1}^{(2)^*} - \underline{Q^p}\right) \\ 0 & E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 2.2: If $\alpha w_t \ge min\left\{E_{t+1}^{(1)^*}, \overline{Q^p} - \underline{Q^p}\right\}$, which means the wind power generation is so large that storing all of it would cause the SOC to exceed $E_{t+1}^{(1)^*}$. Thus, the optimal policy for y_t in this case can be expressed as:

$$y_t^* = \begin{cases} min\left\{E_{t+1}^{(2)^*} - E_t - \underline{Q^p}, \alpha w_t\right\} & E_t \in \left[0, E_{t+1}^{(2)^*} - \underline{Q^p}\right) \\ 0 & E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

A.4. Proof of proposition 4

Due to the existence of the non-zero lower limit constrain of dispatch, when $y_t^* = 0$, we have three choices: $\alpha_t = \underline{Q^p}$, $\alpha_t = -\underline{Q^g}$, and $\alpha_t = 0$. We need to choose the optimal one from these three decisions.

Case 1

(1) When $E_t \in \left[E_{t+1}^{(1)^*} - \underline{Q^p}, E_{t+1}^{(1)^*}\right)$, there is $y_t^* = 0$, corresponds to two decisions:

① Decision $\alpha_t = \underline{Q^p}$: The corresponding value equation is:

$$V_t^p(S(t)) = -P_t\left(\frac{\alpha_t}{\alpha} - w_t\right)\Big/\rho - c\frac{\alpha_t}{\alpha\rho} - c_w w_t + E\left[V_{t+1}\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right)|S(t)\right)\right]$$

② Decision $\alpha_t = 0$: The corresponding value equation is:

$$V_t^i(S(t)) = P_t w_t \rho - c_w w_t + E[V_{t+1}(S_{t+1}(E_t, P_{t+1})|S(t))]$$

We can get the reference point $E_{t+1}^{(1)^{**}}$ by $V_t^p(S(t)) = V_t^i(S(t))$:

$$\left\{\begin{array}{l} -P_t\left(\frac{\alpha_t}{\alpha} - w_t\right)\Big/\rho - c\frac{\alpha_t}{\alpha\rho} - P_t w_t \rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right)|S(t)\right)\right] \\ -E[V_{t+1}(S_{t+1}(E_t, P_{t+1})|S(t))] \end{array}\right\}\Bigg|_{E_{t+1}^* = E_{t+1}^{(1)^{**}}} = 0$$

When $E_t \in \left[E_{t+1}^{(1)^*} - \underline{Q^p}, E_{t+1}^{(1)^{**}}\right)$, the merchant should choose $\alpha_t = \underline{Q^p}$.

When $E_t \in \left[E_{t+1}^{(1)^{**}}, E_{t+1}^{(1)^*}\right)$, the merchant should choose $\alpha_t = 0$.

(2) When $E_t \in \left(E_{t+1}^{(3)^*}, E_{t+1}^{(3)^*} + \underline{Q^g}\right]$, there is $y_t^* = 0$, corresponds to two decisions:

① Decision $\alpha_t = -\underline{Q^g}$: The corresponding value equation is:

$$V_t^g(S(t)) = P_t(w_t - \alpha_t\beta)\rho + c\alpha_t\beta\rho - c_w w_t + E\left[V_{t+1}\left(S_{t+1}\left(E_t - \underline{Q^g}, P_{t+1}\right)|S(t)\right)\right]$$

② Decision $\alpha_t = 0$: The corresponding value equation is:

$$V_t^i(S(t)) = P_t w_t \rho - c_w w_t + E[V_{t+1}(S_{t+1}(E_t, P_{t+1})|S(t))]$$

We can get the reference point $E_{t+1}^{(3)^{**}}$ by $V_t^g(S(t)) = V_t^i(S(t))$:

$$\left\{ \begin{array}{l} P_t(w_t-\alpha_t\beta)\rho + c\alpha_t\beta\rho - P_t w_t\rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t-\underline{Q^g},P_{t+1}\right)|S(t)\right)\right] \\ -E[V_{t+1}(S_{t+1}(E_t,P_{t+1})\,|S(t)\,)] \end{array} \right\}\Bigg|_{E^*_{t+1}=E^{(3)^{**}}_{t+1}} = 0.$$

When $E_t \in \left(E^{(3)^*}_{t+1}, E^{(3)^{**}}_{t+1}\right)$, the merchant should choose $\alpha_t = 0$.

When $E_t \in \left[E^{(3)^{**}}_{t+1}, E^{(3)^*}_{t+1} + \underline{Q^g}\right)$, the merchant should choose $\alpha_t = -\underline{Q^g}$.

(3) When $E_t \in \left[E^{(1)^*}_{t+1}, E^{(3)^*}_{t+1}\right]$, the optimal decision is $\alpha_t = 0$.

Case 2

(1) When $E_t \in \left[E^{(2)^*}_{t+1} - \underline{Q^p}, E^{(2)^*}_{t+1}\right)$, there is $y^*_t = 0$, corresponds to two decisions:

① Decision $\alpha_t = \underline{Q^p}$: The corresponding value equation is:

$$V^p_t(S(t)\,) = P_t\left(w_t - \frac{\alpha_t}{\alpha}\right)\rho - c\frac{\alpha_t}{\alpha\rho} - c_w w_t + E\left[V_{t+1}\left(S_{t+1}\left(E_t+\underline{Q^p},P_{t+1}\right)|S(t)\right]\right.$$

② Decision $\alpha_t = 0$: The corresponding value equation is:

$$V^i_t(S(t)\,) = P_t w_t\rho - c_w w_t + E[V_{t+1}(S_{t+1}(E_t,P_{t+1})\,|S(t)\,)]$$

We can get the reference point $E^{(2)^{**}}_{t+1}$ by $V^p_t(S(t)\,) = V^i_t(S(t)\,)$:

$$\left\{ \begin{array}{l} P_t\left(w_t - \frac{\alpha_t}{\alpha}\right)\rho - c\frac{\alpha_t}{\alpha\rho} - P_t w_t\rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t+\underline{Q^p},P_{t+1}\right)|S(t)\right]\right. \\ -E[V_{t+1}(S_{t+1}(E_t,P_{t+1})\,|S(t)\,)] \end{array} \right\}\Bigg|_{E^*_{t+1}=E^{(2)^{**}}_{t+1}} = 0$$

When $E_t \in \left[E^{(2)^*}_{t+1} - \underline{Q^p}, E^{(2)^{**}}_{t+1}\right)$, the merchant should choose $\alpha_t = \underline{Q^p}$.

When $E_t \in \left[E^{(2)^{**}}_{t+1}, E^{(2)^*}_{t+1}\right)$, the merchant should choose $\alpha_t = 0$.

(2) When $E_t \in \left(E^{(3)^*}_{t+1}, E^{(3)^*}_{t+1} + \underline{Q^g}\right]$, there is $y^*_t = 0$, corresponds to two decisions:

① If $\alpha_t = -\underline{Q^g}$, the corresponding value equation is:

$$V^g_t(S(t)\,) = P_t(w_t-\alpha_t\beta)\rho + c\alpha_t\beta\rho - c_w w_t + E\left[V_{t+1}\left(S_{t+1}\left(E_t-\underline{Q^g},P_{t+1}\right)|S(t)\right)\right]$$

② If $\alpha_t = 0$, the corresponding value equation is:

$$V^i_t(S(t)\,) = P_t w_t\rho - c_w w_t + E[V_{t+1}(S_{t+1}(E_t,P_{t+1})\,|S(t)\,)]$$

We can get the reference point $E^{(3)^{**}}_{t+1}$ by $V^g_t(S(t)\,) = V^i_t(S(t)\,)$:

$$\left\{ \begin{array}{l} P_t(w_t-\alpha_t\beta)\rho + c\alpha_t\beta\rho - P_t w_t\rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t-\underline{Q^g},P_{t+1}\right)|S(t)\right)\right] \\ -E[V_{t+1}(S_{t+1}(E_t,P_{t+1})\,|S(t)\,)] \end{array} \right\}\Bigg|_{E^*_{t+1}=E^{(3)^{**}}_{t+1}} = 0.$$

When $E_t \in \left(E^{(3)^*}_{t+1}, E^{(3)^{**}}_{t+1}\right)$, the merchant should choose $\alpha_t = 0$.

When $E_t \in \left[E^{(3)^{**}}_{t+1}, E^{(3)^*}_{t+1} + \underline{Q^g}\right)$, the merchant should choose $\alpha_t = -\underline{Q^g}$.

(3) When $E_t \in \left[E^{(2)^*}_{t+1}, E^{(3)^*}_{t+1}\right]$, the optimal decision is $\alpha_t = 0$.

To summarize, we get the following results:

Case 1: For $\alpha w_t < \underline{Q^p}$.

Case 1.1: If $\alpha w_t < min\left\{E^{(1)^*}_{t+1}, \overline{Q^p}\right\}$, the optimal policy needs to be further discussed.

Case 1.1.1: If $E^{(1)^*}_{t+1} \le \underline{Q^p} \Rightarrow \alpha w_t < E^{(1)^*}_{t+1} \le \underline{Q^p}$, the optimal policy for α^*_t can be expressed as:

$$\alpha_t^* = \begin{cases} 0 & E_t \in \left[\underline{E}, E_{t+1}^{(1)^{**}}\right) \\ -\underline{Q^g} & E_t \in \left[E_{t+1}^{(1)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t, -\overline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 1.1.2: If $E_{t+1}^{p^*} \geq \underline{Q^p} \Rightarrow E_{t+1}^{p^*} \geq \underline{Q^p} > \alpha w_t$, the optimal policy for α_t^* can be expressed as:

$$\alpha_t^* = \begin{cases} min\left\{E_{t+1}^{(1)^*} - E_t, \overline{Q^p}\right\} & E_t \in \left[\underline{E}, E_{t+1}^{(1)^*} - \underline{Q^p}\right) \\ \underline{Q^p} & E_t \in \left[E_{t+1}^{(1)^*} - \underline{Q^p}, E_{t+1}^{(1)^{**}}\right) \\ 0 & E_t \in \left[E_{t+1}^{(1)^{**}}, E_{t+1}^{(3)^{**}}\right) \\ -\underline{Q^g} & E_t \in \left[E_{t+1}^{(3)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 1.2: If $\alpha w_t \geq min\{E_{t+1}^{p^*}, \overline{Q^p}\}$, the optimal policy for α_t^* in this case can be expressed as:

$$\alpha_t^* = \begin{cases} 0 & E_t \in \left[\underline{E}, E_{t+1}^{(1)^{**}}\right) \\ -\underline{Q^g} & E_t \in \left[E_{t+1}^{(1)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t, -\overline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 2: For $\alpha w_t \geq \underline{Q^p}$.

Case 2.1: If $\alpha w_t < min\left\{E_{t+1}^{(1)^*}, \overline{Q^p}\right\}$, the optimal policy for α_t^* in this case can be expressed as:

$$\alpha_t^* = \begin{cases} min\left\{E_{t+1}^{(1)^*} - E_t, \overline{Q^p}\right\} & E_t \in \left[\underline{E}, E_{t+1}^{(1)^*} - \alpha w_t\right) \\ min\left\{E_{t+1}^{(2)^*} - E_t, \alpha w_t\right\} & E_t \in \left[E_{t+1}^{(1)^*} - \alpha w_t, E_{t+1}^{(2)^*} - \underline{Q^p}\right) \\ \underline{Q^p} & E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(2)^{**}}\right) \\ 0 & E_t \in \left[E_{t+1}^{(2)^{**}}, E_{t+1}^{(3)^{**}}\right) \\ -\underline{Q^g} & E_t \in \left[E_{t+1}^{(3)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t, -\overline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Case 2.2: If $\alpha w_t \geq min\left\{E_{t+1}^{(1)^*}, \overline{Q^p}\right\}$, the optimal policy for α_t^* in this case can be expressed as:

$$\alpha_t^* = \begin{cases} min\left\{E_{t+1}^{(2)^*} - E_t, \alpha w_t\right\} & E_t \in \left[0, E_{t+1}^{(2)^*} - \underline{Q^p}\right) \\ \underline{Q^p} & E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(2)^{**}}\right) \\ 0 & E_t \in \left[E_{t+1}^{(2)^{**}}, E_{t+1}^{(3)^{**}}\right) \\ -\underline{Q^g} & E_t \in \left[E_{t+1}^{(3)^{**}}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t, -\overline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases}$$

Appendix B. Appendix

B.1. Proof of Case 1 in Section 6

In Stage 3:

Action 3: Sell power to market and make the storage level down to $\underline{E} = 0$.

$$V_3 = -(P_3 - c)\left(\underline{E} - E_3\right)\beta\rho + V_4 = 8.1E_3 \quad E_3 \in [0, 10]$$

In Stage 2:

$$\begin{cases} E_3^{g^*} = \underset{\underline{E} \le E_3 \le \overline{E}}{argmax}\left\{-(P_2 - c)E_3\beta\rho + V_3^*\right\} = \underset{E_3 \in [010]}{argmax}\left\{-(2-1)0.9E_3 + 8.1E_3\right\} = \underset{E_3 \in [010]}{argmax}\{7.2E_3\} = \overline{E} = 10 \\ E_3^{p^*} = \underset{\underline{E} \le E_3 \le \overline{E}}{argmax}\left\{-\frac{(P_2 + c)E_3}{\alpha\rho} + V_3^*\right\} = \underset{E_3 \in [010]}{argmax}\left\{-\frac{3}{0.9}E_3 + 8.1E_3\right\} = \underset{E_3 \in [010]}{argmax}\left\{\frac{143}{30}E_3\right\} = \overline{E} = 10 \end{cases}$$

Thus, we get $E_3^{p^*} = E_3^{g^*} = 10$, from which we deduce y_2^*:

$$y_2^* = \begin{cases} min\left[\overline{Q^p} - \underline{Q^p}\overline{E} - E_2 - \underline{Q^p}\right] & E_2 \in \left[\underline{E}E_3^{p^*} - \underline{Q^p}\right] \\ 0 & E_2 \in \left(E_3^{p^*} - \underline{Q^p}\overline{E}\right] \end{cases} = \begin{cases} \overline{Q^p} - \underline{Q^p} & E_2 \in [03] \\ \overline{E} - E_2 - \underline{Q^p} & E_2 \in (39] \\ 0 & E_2 \in (910] \end{cases} = \begin{cases} 6 & E_2 \in [03] \\ 9 - E_2 & E_2 \in (39] \\ 0 & E_2 \in (910] \end{cases}$$

From Proposition 2, we can determine the following optimal actions at Stage 2:

$$\alpha_2^* = \begin{cases} 7 & E_2 \in [0, 3] \\ 10 - E_2 & E_2 \in (3, 9] \\ 0 & E_2 \in (9, 10] \end{cases}$$

The reward payoff functions at Stage 2 are shown as follows:

$$R_2 = \begin{cases} -\frac{(2+1)7}{0.9} & E_2 \in [0, 3] \\ -\frac{(2+1)(10 - E_2)}{0.9} & E_2 \in (3, 9] \\ 0 & E_2 \in (9, 10] \end{cases} = \begin{cases} -\frac{70}{3} & E_2 \in [0, 3] \\ -\frac{10(10 - E_2)}{3} & E_2 \in (3, 9] \\ 0 & E_2 \in (9, 10] \end{cases}$$

The optimal value functions at Stage 3 can be rewritten as:

$$V_3^* = \begin{cases} 8.1(E_2 + 7) & E_2 \in [0, 3) \\ 81 & E_2 \in [3, 9) \\ 8.1E_2 & E_2 \in [9, 10] \end{cases}$$

Thus, the optimal value functions at Stage 2 are shown as follows:

$$V_2^* = max(R_2 + V_3^*) = \begin{cases} 8.1E_2 + 33.37 & E_2 \in [0, 3) \\ \frac{10}{3}E_2 + \frac{143}{3} & E_2 \in [3, 9) \\ 8.1E_2 & E_2 \in [9, 10] \end{cases}$$

In Stage 1:

$$E_2^{g^*} = \underset{\underline{E} \le E_2 \le \overline{E}}{argmax}\left\{-(P_1 - c)E_2\beta\rho + V_2^*\right\} = \begin{cases} 4.5E_2 + 33.37 & E_2 \in [0, 3) \\ -\frac{4}{15}E_2 + \frac{143}{3} & E_2 \in [3, 9) \\ 4.5E_2 & E_2 \in [9, 10] \end{cases}$$

(1) If $E_2 \in [0, 3]$, in the first equation, we get $E_2^{g^*} = 3$.
(2) If $E_2 \in (3, 9)$, in the second equation, we have $E_2^{g^*} = 3$.
(3) If $E_2 \in [9, 10]$, in the third equation, we obtain $E_2^{g^*} = 10$.

We also find for any $E_1 \in [0, 10]$, there are:

$$\left[4.5E_2+33.37|_{E_2=3}\right]=\left[-\frac{4}{15}E_2+\frac{143}{3}|_{E_2=3}]>[4.5E_2|_{E_2=10}\right]$$

Therefore, we derive $E_2^{g^*}=3$.

$$E_2^{p^*}=\underset{\underline{E}\le E_2\le\overline{E}}{argmax}\left\{-\frac{(P_1+c)E_2}{\alpha\rho}+V_2^*\right\}=\begin{cases}\frac{43}{30}E_2+33.37 & E_2\in[0,3)\\ -\frac{10}{3}E_2+\frac{143}{3} & E_2\in[3,9)\\ \frac{43}{30}E_2 & E_2\in[9,10)\end{cases}$$

1) If $E_2\in[0,3]$, in the first Eq., we get $E_2^{p^*}=3$.
2) If $E_2\in(3,9)$, in the second Eq., we get $E_2^{p^*}=3$.
3) If $E_2\in[9,10]$, in the third Eq., we get $E_2^{p^*}=10$.

We also find for any $E_1\in[0,10]$, there are:

$$\left[\frac{43}{30}E_2+33.37\Big|_{E_2=3}\right]=\left[-\frac{10}{3}E_2+\frac{143}{3}\Big|_{E_2=3}\right]>\left[\frac{43}{30}E_2\Big|_{E_2=10}\right]$$

Therefore, we will get $E_2^{p^*}=3$.

From the above analysis, we can obtain $E_2^{g^*}=E_2^{p^*}=3=\overline{E}-\overline{Q^p}$, from which we deduce y_1^*:

$$y_1^*=\begin{cases}min\left[E_2^{p^*}-E_1-\underline{Q^p}\overline{Q^p}-\underline{Q^p}\right] & E_1\in\left[\underline{E}E_2^{p^*}-\underline{Q^p}\right)\\ 0 & E_1\in\left[E_2^{p^*}-\underline{Q^p}E_2^{g^*}+\underline{Q^g}\right)\\ max\left[E_2^{g^*}-E_1+\underline{Q^g}-\overline{Q^g}+\underline{Q^p}\right] & E_1\in\left[E_2^{g^*}+\underline{Q^g}\overline{E}\right]\end{cases}=\begin{cases}3-E_1-\underline{Q^p} & E_1\in[02)\\ 0 & E_1\in[\overline{25}]\\ 3-E_1+\underline{Q^g} & E_1\in(510]\end{cases}=\begin{cases}2-E_1 & E_1\in[02)\\ 0 & E_1\in[25]\\ 1-E_1 & E_1\in(510]\end{cases}$$

When $E_1\in\left[E_2^{p^*}-\underline{Q^p},E_2^{p^*}\right]$, we need to compare the two decisions $\alpha_1=0$ and $\alpha_1=\underline{Q^p}$.

When $V_1^p(S(1))=V_1^i(S(1))$, we can get $E_2^{p^{**}}$:

$$V_1^p(S(1))=V_1^i(S(1))$$
$$\Rightarrow-\frac{(P_1+c)\underline{Q^p}}{\alpha\rho}+V_2^*\left(S_2\left(E_2^{p^{**}}+\underline{Q^p},P_2\right)\right)=V_2^*(S_2(E_2^{p^{**}},P_2))$$
$$\Rightarrow-\frac{(P_1+c)\underline{Q^p}}{\alpha\rho}+V_2(S_2(E_2^{p^{**}}+1,2))-V_2(S_2(E_2^{p^{**}},2))=0$$
$$\Rightarrow-\frac{(5+1)^*1}{0.9}+\left[\frac{10}{3}(E_2^{p^{**}}+1)+\frac{143}{3}\right]+8.1E_2^{p^{**}}+33.37=0$$
$$\Rightarrow E_2^{p^{**}}=2.3$$

Similarly, we need to get $E_2^{g^{**}}$ at $E_t\in\left(E_{t+1}^{g^*},E_{t+1}^{g^*}+\underline{Q^g}\right]$ by comparing $\alpha_1=0$ with $\alpha_1=-\underline{Q^g}$.

$$V_1^g(S(1))=V_1^i(S(1))$$
$$\Rightarrow(P_1-c)\underline{Q^g}\beta\rho+V_2^*\left(S_2\left(E_2^{p^{**}}-\underline{Q^g},P_2\right)\right)=V_2^*(S_2(E_2^{g^{**}},P_2))$$
$$\Rightarrow(P_1-c)\underline{Q^g}\beta\rho+V_2^*(S_2(E_2^{p^{**}}-2,2))-V_2^*(S_2(E_2^{g^{**}},2))=0$$
$$\Rightarrow(5-1)^*2^*0.9+[8.1(E_2^{g^{**}}-2)+33.37]-\left[\frac{10}{3}E_2^{g^{**}}+\frac{143}{3}\right]=0$$
$$\Rightarrow E_2^{g^{**}}=4.88$$

In summary, we can get the following optimal actions at Stage 1.

$$\alpha_1^* = \begin{cases} min\left\{max\left\{E_2^{p^*} - E_1, \underline{Q^p}\right\}, \overline{Q^p}\right\} & E_1 \in \left[\underline{E}, E_2^{p^{**}}\right) \\ 0 \quad E_t \in [E_2^{p^{**}}, E_2^{g^{**}}] \\ max\left\{min\left\{E_2^{g^*} - E_1, -\underline{Q^g}\right\}, -\overline{Q^g}\right\} & E_1 \in (E_2^{g^{**}}, \overline{E}] \end{cases}$$

$$= \begin{cases} min\{max\{3 - E_1, 1\}, 12\} & E_1 \in [0, 2.3) \\ 0 \quad E_t \in [2.3, 4.88] \\ max\{min\{3 - E_1, -2\}, -7\} & E_1 \in (4.88, 10] \end{cases} = \begin{cases} 3 - E_1 & \mathrm{E}_1 \in [0, 2) \\ 1 & \mathrm{E}_1 \in [2, 2.3) \\ 0 & \mathrm{E}_1 \in [2.3, 4.88) \\ -2 & \mathrm{E}_1 \in [4.88, 5) \\ 3 - E_1 & \mathrm{E}_1 \in [5, 10] \end{cases}$$

The reward payoff functions at Stage 1 are shown as follows:

$$R_1 = \begin{cases} -\dfrac{(P_1 + c)(3 - E_1)}{\alpha\rho} & E_1 \in [0, 2) \\ -\dfrac{(P_1 + c)^*1}{\alpha\rho} & E_1 \in [2, 2.3) \\ 0 \quad E_1 \in [2.3, 4.88) \\ (P_1 - c)^*2^*\beta\rho & E_1 \in [4.88, 5) \\ (P_1 - c)(E_1 - 3)\beta\rho & E_1 \in [5, 10] \end{cases} = \begin{cases} -\dfrac{20(3 - E_1)}{3} & E_1 \in [0, 2) \\ -\dfrac{20}{3} & E_1 \in [2, 2.3) \\ 0 \quad E_1 \in [2.3, 4.88) \\ 7.2 \quad E_1 \in [4.88, 5) \\ 3.6^*(E_1 - 3) & E_1 \in [5, 10] \end{cases}$$

The optimal value functions at Stage 2 can be rewritten as:

$$V_2^* = \begin{cases} 8.1E_2 + 33.37 & E_2 \in [0, 3) \\ \dfrac{10}{3}E_2 + \dfrac{143}{3} & E_2 \in [3, 9) \\ 8.1E_2 & E_2 \in [9, 10] \end{cases} = \begin{cases} 57.67 & \mathrm{E}_1 \in [0, 2) \\ \dfrac{10}{3}E_1 + 51 & \mathrm{E}_1 \in [2, 2.3) \\ 8.1E_1 + 33.37 & \mathrm{E}_1 \in [2.3, 3) \\ \dfrac{10}{3}E_1 + \dfrac{143}{3} & \mathrm{E}_1 \in [3, 4.88) \\ 8.1E_1 + 17.17 & \mathrm{E}_1 \in [4.88, 5) \\ \dfrac{173}{3} & \mathrm{E}_1 \in [5, 10] \end{cases}$$

Thus, the optimal value functions at Stage 1 are expressed as:

$$V_1^* = max(R_1 + V_2^*) = \begin{cases} \dfrac{20}{3}E_1 + 37.67 & \mathrm{E}_1 \in [0, 2) \\ \dfrac{10}{3}E_1 + \dfrac{133}{3} & \mathrm{E}_1 \in [2, 2.3) \\ 8.1E_1 + 33.37 & \mathrm{E}_1 \in [2.3, 3) \\ \dfrac{10}{3}E_1 + \dfrac{143}{3} & \mathrm{E}_1 \in [3, 4.88) \\ 8.1E_1 + 24.37 & \mathrm{E}_1 \in [4.88, 5) \\ 3.6E_1 + 46.87 & \mathrm{E}_1 \in [5, 10] \end{cases}$$

B.2. Proof of Case 2 in Section 6

B.2.1. In Stage 3:

Action 3: Sell power to market and reduce the storage level to $\underline{E} = 0$. We get the following:

$$V_3 = P_3\left(w_3 - \left(\frac{\underline{E}}{\eta} - E_3\right)\beta\right)\rho + c\left(\frac{\underline{E}}{\eta} - E_3\right)\beta\rho + V_4 = 8.02E_3$$

In Stage 2:

First, we need to compare αw_2 and $\underline{Q^p}$:

$$\alpha w_2 = 0.9^*5 = 4.5 > \underline{Q^p}$$

Therefore, the problem at Stage 2 is consistent with Case 2.

$$\begin{cases} E_3^{(1)^*} = \underset{\underline{E} \le E_3 \le \overline{E}}{argmax} \left\{ -P_2\left(\left(\frac{E_3}{\eta} - E_2\right)\Big/\alpha - w_2\right)\Big/\rho - \frac{c}{\alpha\rho}\left(\frac{E_3}{\eta} - E_2\right) + V_3^* \right\} = \underset{E_3 \in [010]}{argmax} \{2.59E_2 + 5.43E_3 + 11.1\} = 10 = \overline{E} \\ E_3^{(2)^*} = \underset{\underline{E} \le E_3 \le \overline{E}}{argmax} \left\{ P_2\left(w_2 - \left(\frac{E_3}{\eta} - E_2\right)\Big/\alpha\right)\rho - \frac{c}{\alpha\rho}\left(\frac{E_3}{\eta} - E_2\right) + V_3^* \right\} = \underset{E_3 \in [010]}{argmax} \{2.12E_2 + 5.9E_3 + 9\} = 10 = \overline{E} \\ E_3^{(3)^*} = \underset{\underline{E} \le E_3 \le \overline{E}}{argmax} \left\{ P_2\left(w_2 - \left(\frac{E_3}{\eta_3} - E_2\right)\beta\right)\rho + c\left(\frac{E_3}{\eta_3} - E_2\right)\beta\rho + V_3^* \right\} = \underset{E_3 \in [010]}{argmax} \{1.24E_2 + 6.48E_3 + 9\} = 10 = \overline{E} \end{cases}$$

Thus, we get $E_3^{(1)^*} = E_3^{(2)^*} = E_3^{(3)^*} = 10$.

At the same time, we can find the following relationship:

$$\alpha w_2 < min\left\{E_3^{(1)^*}, \overline{Q^p} - \underline{Q^p}\right\}$$

Therefore, we can get the optimal strategy for y_2^*:

$$y_2^* = \begin{cases} min\left\{E_{t+1}^{(1)^*} - E_t - \underline{Q^p}, \overline{Q^p} - \underline{Q^p}\right\} & E_t \in \left[\underline{E}, E_{t+1}^{(1)^*} - \underline{Q^p} - \alpha w_t\right) \\ min\left\{E_{t+1}^{(2)^*} - E_t - \underline{Q^p}, \alpha w_t\right\} & E_t \in \left[E_{t+1}^{(1)^*} - \underline{Q^p} - \alpha w_t, E_{t+1}^{(2)^*} - \underline{Q^p}\right) \\ 0 & E_t \in \left[E_{t+1}^{(2)^*} - \underline{Q^p}, E_{t+1}^{(3)^*} + \underline{Q^g}\right) \\ max\left\{E_{t+1}^{(3)^*} - E_t + \underline{Q^g}, -\overline{Q^g} + \underline{Q^g}\right\} & E_t \in \left[E_{t+1}^{(3)^*} + \underline{Q^g}, \overline{E}\right] \end{cases} = \begin{cases} min\{9 - E_2, 6\} & E_2 \in [0, 4.5) \\ min\{9 - E_2, 4.5\} & E_2 \in [4.5, 9) \\ 0 & E_2 \in [9, 10] \end{cases}$$

From Proposition 4, we can determine the following optimal actions at Stage 2:

$$\alpha_2^* = \begin{cases} min\{10 - E_2, 7\} & E_2 \in [0, 5.5) \\ min\{10 - E_2, 4.5\} & E_2 \in [5.5, 9) \\ 0 & E_2 \in [9, 10] \end{cases} = \begin{cases} 7 & E_2 \in [0, 3) \\ 10 - E_2 & E_2 \in [3, 9) \\ 0 & E_2 \in [9, 10] \end{cases}$$

The reward payoff functions at Stage 2 are defined as follows:

$$R_2 = \begin{cases} -P_t\left(\frac{q_t^p}{\alpha} - w_t\right)\Big/\rho - c\frac{q_t^p}{\alpha\rho} & E_2 \in [0, 3) \\ -P_t\left(\frac{q_t^p}{\alpha} - w_t\right)\Big/\rho - c\frac{q_t^p}{\alpha\rho} & E_2 \in [3, 5.5) \\ P_t\left(w_t - \frac{q_t^p}{\alpha}\right)\rho - c\frac{q_t^p}{\alpha\rho} & E_2 \in [5.5, 9) \\ P_t w_t \rho & E_2 \in [9, 10] \end{cases} = \begin{cases} -7.014 & E_2 \in [0, 3) \\ 2.59E_2\text{-}14.8 & E_2 \in [3, 5.5) \\ 2.12E_2\text{-}12.2 & E_2 \in [5.5, 9) \\ 9 & E_2 \in [9, 10] \end{cases}$$

Thus, the optimal value functions at Stage 3 can be rewritten as:

$$V_3^* = \begin{cases} 8.02(E_2 + 7) = 8.02E_2 + 56.14 & E_2 \in [0, 3] \\ 8.02^*10 = 80.2 & E_2 \in (3, 9] \\ 8.02E_2 & E_2 \in (9, 10] \end{cases}$$

Thus, the optimal value functions at Stage 2 are expressed as:

$$V_2^* = max(R_2 + V_3^*) = \begin{cases} 8.02E_2 + 56.14 - 7.014 & E_2 \in [0, 3) \\ 2.59E_2\text{-}14.8 + 80.2 & E_2 \in [3, 5.5) \\ 2.12E_2\text{-}12.2 + 80.2 & E_2 \in [5.5, 9) \\ 9 + 8.02E_2 & E_2 \in [9, 10] \end{cases} = \begin{cases} 8.02E_2 + 49.13 & E_2 \in [0, 3) \\ 2.59E_2 + 65.4 & E_2 \in [3, 5.5) \\ 2.12E_2 + 68 & E_2 \in [5.5, 9) \\ 8.02E_2 + 9 & E_2 \in [9, 10] \end{cases}$$

B.2.2. In stage 1:

First, we need to compare αw_1 and $\underline{Q^p}$:

$$\alpha w_2 = 0.9^*3 = 2.7 > \underline{Q^p}$$

Therefore, the problem at Stage 1 is consistent with Case 2.

$$E_2^{(1)^*} = \underset{\underline{E} \le E_2 \le \overline{E}}{argmax} \left\{ -P_1 \left(\left(\frac{E_2}{\eta} - E_1 \right) \Big/ \alpha - w_1 \right) \Big/ \rho - \frac{c}{\alpha\rho} \left(\frac{E_2}{\eta} - E_1 \right) + V_2^* \right\}$$

$$= \begin{cases} 6.29E_1 + 1.73E_2 + 66.8 & E_2 \in [0,3) \\ 6.29E_1 - 3.7E_2 + 82.07 & E_2 \in [3,5.5) \\ 6.29E_1 - 4.17E_2 + 84.67 & E_2 \in [5.5,9) \\ 6.29E_1 + 1.73E_2 + 25.67 & E_2 \in [9,10] \end{cases}$$

1) If $E_2 \in [0,3)$, in the first equation, we obtain $E_2^{(1)^*} = 3$.
2) If $E_2 \in (3,5.5]$, in the secondequation, we get $E_2^{(1)^*} = 3$.
3) If $E_2 \in [5.5,9)$, in the third equation, we have $E_2^{(1)^*} = 5.5$.
4) If $E_2 \in [9,10]$, in the fourth equation, there is $E_2^{(1)^*} = 10$.

We also find that for any $E_1 \in [0,10]$, there are:

$$[6.29E_1 + 1.73E_2 + 66.8_{E_2=3}] = [6.29E_1 - 3.7E_2 + 82.07\,|_{E_2=3}] > \left[6.29E_1 - 4.17E_2 + 84.67|_{E_2=5.5}\right] > \left[6.29E_1 + 1.73E_2 + 25.67|_{E_2=10}\right]$$

Therefore, we derive $E_2^{(1)^*} = 3$.

$$E_2^{(2)^*} = \underset{\underline{E} \le E_2 \le \overline{E}}{argmax} \left\{ P_1 \left(w_1 - \left(\frac{E_2}{\eta} - E_1 \right) \Big/ \alpha \right) \rho - \frac{c}{\alpha\rho} \left(\frac{E_2}{\eta} - E_1 \right) + V_2^* \right\}$$

$$= \begin{cases} 5.12E_1 + 2.9E_2 + 62.63 & E_2 \in [0,3) \\ 5.12E_1 - 2.53E_2 + 78.9 & E_2 \in [3,5.5) \\ 5.12E_1 - 3E_2 + 81.5 & E_2 \in [5.5,9) \\ 5.12E_1 + 2.9E_2 + 22.5 & E_2 \in [9,10] \end{cases}$$

1) If $E_2 \in [0,3)$, in the first equation, we get $E_2^{(2)^*} = 3$.
2) If $E_2 \in (3,5.5]$, in the second equation, we get $E_2^{(2)^*} = 3$.
3) If $E_2 \in [5.5,9)$, in the third equation, we get $E_2^{(2)^*} = 5.5$.
4) If $E_2 \in [9,10]$, in the fourth equation, we get $E_2^{(2)^*} = 10$.

We also find that for any $E_1 \in [0,10]$, there are:

$$[5.12E_1 + 2.9E_2 + 62.63_{E_2=3}] = [5.12E_1 - 2.53E_2 + 78.9\,|_{E_2=3}] > \left[5.12E_1 - 3E_2 + 81.5|_{E_2=5.5}\right] > \left[5.12E_1 + 2.9E_2 + 22.5|_{E_2=10}\right]$$

Therefore, we derive $E_2^{(2)^*} = 3$.

$$E_3^{(3)^*} = \underset{\underline{E} \le E_2 \le \overline{E}}{argmax} \left\{ P_1 \left(w_1 - \left(\frac{E_2}{\eta} - E_1 \right) \beta \right) \rho + c \left(\frac{E_2}{\eta} - E_1 \right) \beta\rho + V_2^* \right\}$$

$$= \begin{cases} 3.97E_1 + 4.05E_2 + 62.63 & E_2 \in [0,3) \\ 3.97E_1 - 1.38E_2 + 78.9 & E_2 \in [3,5.5) \\ 3.97E_1 - 1.85E_2 + 81.5 & E_2 \in [5.5,9) \\ 3.97E_1 + 4.05E_2 + 22.5 & E_2 \in [9,10] \end{cases} = 3$$

1) If $E_2 \in [0,3)$, in the first equation, we get $E_2^{(3)^*} = 3$.
2) If $E_2 \in (3,5.5]$, in the second equation, we get $E_2^{(3)^*} = 3$.
3) If $E_2 \in [5.5,9)$, in the third equation, we get $E_2^{(3)^*} = 5.5$.
4) If $E_2 \in [9,10]$, in the fourth equation, we get $E_2^{(3)^*} = 10$.

We also find that for any $E_1 \in [0,10]$, there are:

$$[3.97E_1 + 4.05E_2 + 62.63_{E_2=3}] = [3.97E_1 - 1.38E_2 + 78.9\,|_{E_2=3}] > \left[3.97E_1 - 1.85E_2 + 81.5|_{E_2=5.5}\right] > \left[3.97E_1 + 4.05E_2 + 22.5|_{E_2=10}\right]$$

Therefore, we have $E_2^{(3)^*} = 3$.

From the above analysis, we can obtain $E_2^{g^*} = E_2^{p^*} = 3 = \overline{E} - \overline{Q^p}$.

At the same time, we establish the following relationship:

$$\alpha w_1 < min\left\{E_2^{(1)^*}, \overline{Q^p} - \underline{Q^p}\right\}$$

All in all, we can get the optimal strategy for y_1^*:

$$y_1^* = \begin{cases} min\{2 - E_1, 6\} & E_1 \in [0, 0.3) \\ min\{2 - E_1, 2.7\} & E_1 \in [0.3, 2) \\ 0 & E_1 \in [2, 5] \\ max\{5 - E_1, -10\} & E_1 \in (5, 12] \end{cases}$$

When $E_1 \in \left[E_2^{(2)^*} - \underline{Q^p}, E_2^{(2)^*}\right]$, we need to compare the two decisions $\alpha_1 = 0$ and $\alpha_1 = \underline{Q^p}$.

When $V_1^p(S(1)) = V_1^i(S(1))$, we can get $E_2^{(2)^{**}}$:

$$V_1^p(S(1)) = V_1^i(S(1))$$

$$\Rightarrow P_1\left(w_1 - \frac{\alpha_1}{\alpha}\right)\rho - c\frac{\alpha_1}{\alpha\rho} + E\left[V_2\left(S_{t+1}\left(E_t + \underline{Q^p}, P_{t+1}\right)|S(t)\right)\right] = P_t w_t \rho + E[V_{t+1}(S_{t+1}(E_t, P_{t+1})\,|S(t))]$$

$$\Rightarrow [76.37 + 2.59E_1 - 8.02E_1 - 62.63]\Big|_{E_1 = E_2^{(2)^{**}}} = 0$$

$$\Rightarrow E_2^{(2)^{**}} = 2.53$$

Similarly, we need to get $E_2^{(3)^{**}}$ at $E_1 \in \left(E_2^{(3)^*}, E_2^{(3)^*} + \underline{Q^g}\right]$ by comparing $\alpha_1 = 0$ with $\alpha_1 = -\underline{Q^g}$.

$$V_1^g(S(1)) = V_1^i(S(1))$$

$$\Rightarrow P_t(w_t - \alpha_t\beta)\rho + c\alpha_t\beta\rho + E\left[V_{t+1}\left(S_{t+1}\left(E_t - \underline{Q^g}, P_{t+1}\right)|S(t)\right)\right] = P_t w_t \rho + E[V_{t+1}(S_{t+1}(E_t, P_{t+1})\,|S(t))]$$

$$\Rightarrow [54.53 + 8.02E_1 - 2.59E_1\text{-}78.9]\Big|_{E_1 = E_2^{(3)^{**}}} = 0$$

$$\Rightarrow E_2^{(3)^{**}} = 4.49$$

Summary of Optimal Actions at Stage 1.

We can get the following optimal actions at Stage 1.

$$\alpha_1^* = \begin{cases} min\{3 - E_1, 7\} & E_1 \in [0, 0.3) \\ min\{3 - E_1, 2.7, 1\} & E_1 \in [0.3, 2.53) \\ 0 & E_2 \in [2.53, 4.49] \\ max\{3 - E_1, -2\} & E_1 \in (4.49, 5] \\ max\{3 - E_1, -12\} & E_1 \in (5, 10] \end{cases} = \begin{cases} 3 - E_1 & E_1 \in [0, 2) \\ 1 & E_1 \in [2, 2.53) \\ 0 & E_1 \in [2.53, 4.49] \\ \text{-2} & E_1 \in (4.49, 5] \\ 3 - E_1 & E_1 \in (5, 10] \end{cases}$$

The reward payoff functions at Stage 1 are shown as follows:

$$R_1 = \begin{cases} -P_1\left(\frac{q_1^p}{\alpha} - w_t\right)\Big/\rho - c\frac{q_1^p}{\alpha\rho} & E_1 \in [0, 0.3) \\ P_1\left(w_1 - \frac{q_1^p}{\alpha}\right)\rho - c\frac{q_1^p}{\alpha\rho} & E_1 \in [0.3, 2) \\ P_1\left(w_1 - \frac{Q^p}{\alpha}\right)\rho - c\frac{Q^p}{\alpha\rho} & E_1 \in [2, 2.53) \\ P_1 w_1 \rho & E_1 \in [2.53, 4.49] \\ P_1\left(w_1 - \underline{Q^g}\beta\right)\rho + c\underline{Q^g}\beta\rho & E_1 \in (4.49, 5] \\ P_1(w_1 - q_1^g\beta)\rho + cq_1^g\beta\rho & E_1 \in (5, 10] \end{cases} = \begin{cases} 6.29E_1 - 2.2 & E_1 \in [0, 0.3) \\ 5.12E_1\text{-}1.86 & E_1 \in [0.3, 2) \\ 8.38 & E_1 \in [2, 2.53) \\ 13.5 & E_1 \in [2.53, 4.49] \\ 21.44 & E_1 \in (4.49, 5] \\ 3.97E_1 + 1.59 & E_2 \in (5, 10] \end{cases}$$

The optimal value functions at Stage 2 can be rewritten as:

$$V_2^* = \begin{cases} 73.19 & E_1 \in [0,2) \\ 2.59(E_1+1)+65.4 & E_1 \in [2,2.53) \\ 8.02E_1+49.13 & E_1 \in [2.53,3) \\ 2.59E_1+65.4 & E_1 \in [3,4.49) \\ 8.02(E_1-2)+49.13 & E_1 \in (4.49,5] \\ 73.19 & E_1 \in (5,10] \end{cases}$$

Thus, the optimal value functions at Stage 1 are shown as follows:

$$V_1^* = max(R_1 + V_2^*) = \begin{cases} 6.29E_1+70.99 & E_1 \in [0,0.3) \\ 5.12E_1+71.33 & E_1 \in [0.3,2) \\ 2.59E_1+76.37 & E_1 \in [2,2.53) \\ 8.02E_1+62.63 & E_1 \in [2.53,3) \\ 2.59E_1+78.9 & E_1 \in [3,4.49) \\ 8.02E_1+54.53 & E_1 \in (4.49,5] \\ 3.97E_1+74.78 & E_1 \in (5,10] \end{cases}$$

Appendix C. Appendix

C.1. To ensure the robustness of our proposed method, we selected two additional real-world data sets for numerical simulation

- **Data Set 1:** November 6–19, 2023
- **Data Set 2:** December 4–17, 2023

These data sets, like the one used in Section 6, are sourced from the MISO and include hourly electricity prices and wind power generation over two-week periods. By applying our optimization framework to these additional data sets, we aim to validate the consistency and reliability of our results under different market conditions.

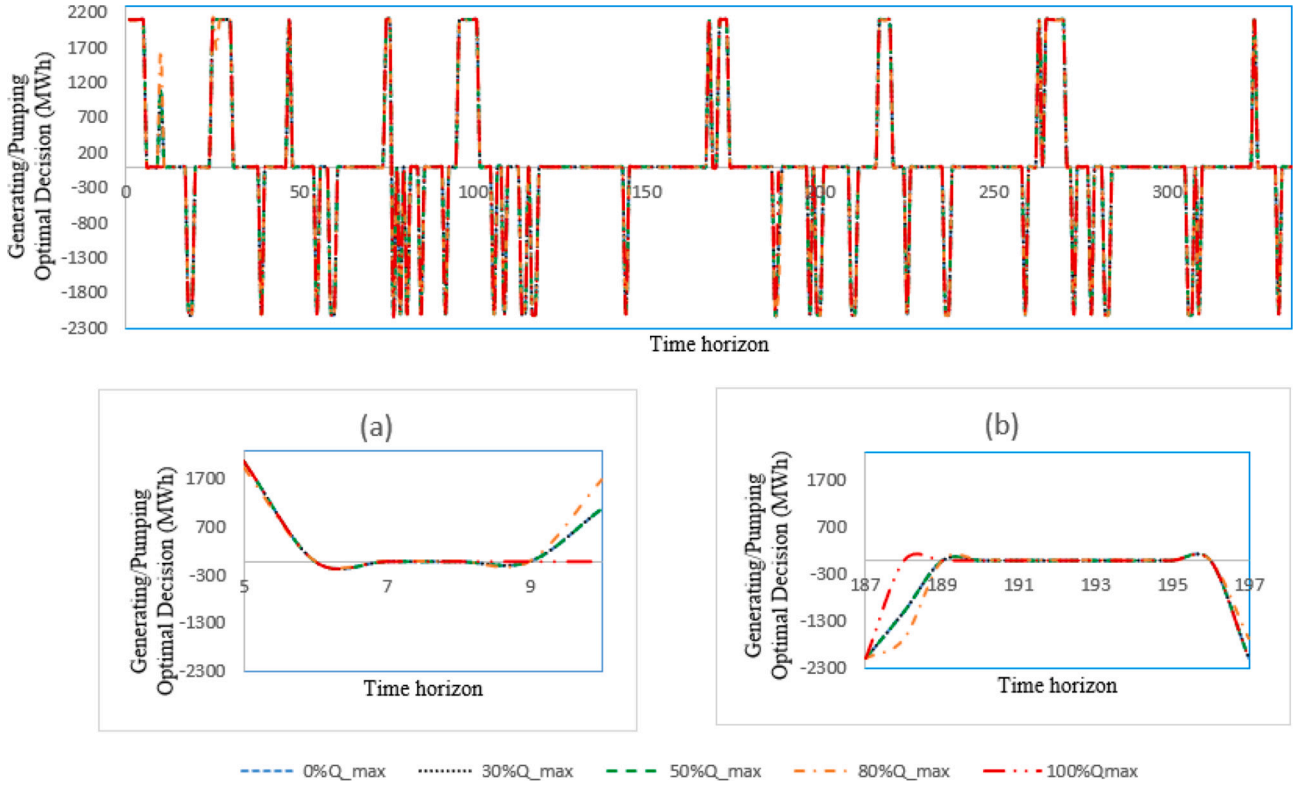


Fig. 1C. The optimal action with PSH only (11/06–19/2023).

Fig. 1C illustrates the optimal dispatch actions for a PSH system without wind integration over the specified period. The results demonstrate consistent operational patterns with those observed in Section 6.1, confirming that the optimal strategy adapts effectively to varying electricity prices while considering non-zero minimum generating and pumping capacities.

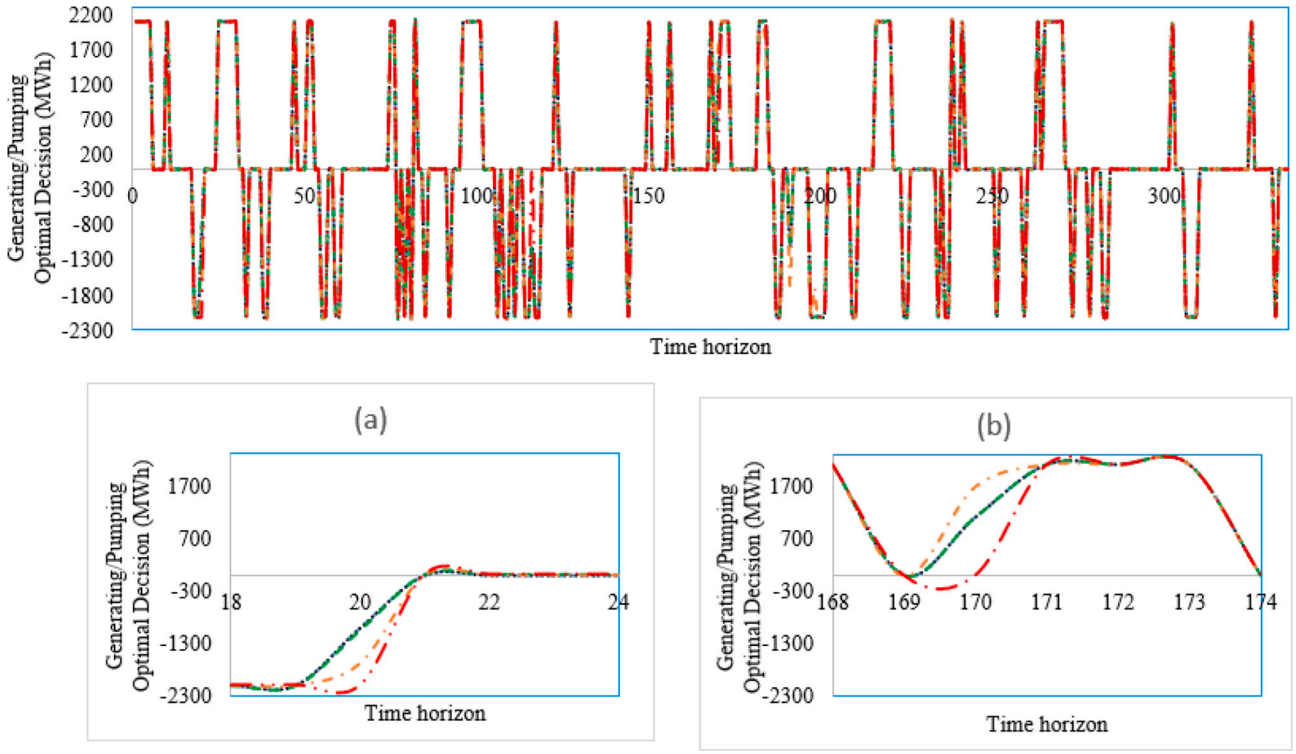


Fig. 2C. The optimal action with PSH and wind power plants (11/06–19/2023).

Fig. 2C shows the optimal dispatch actions when co-optimizing PSH with wind power generation. The integration of wind power introduces additional variability, but the optimization framework successfully accommodates these changes, maintaining profitability and operational efficiency.

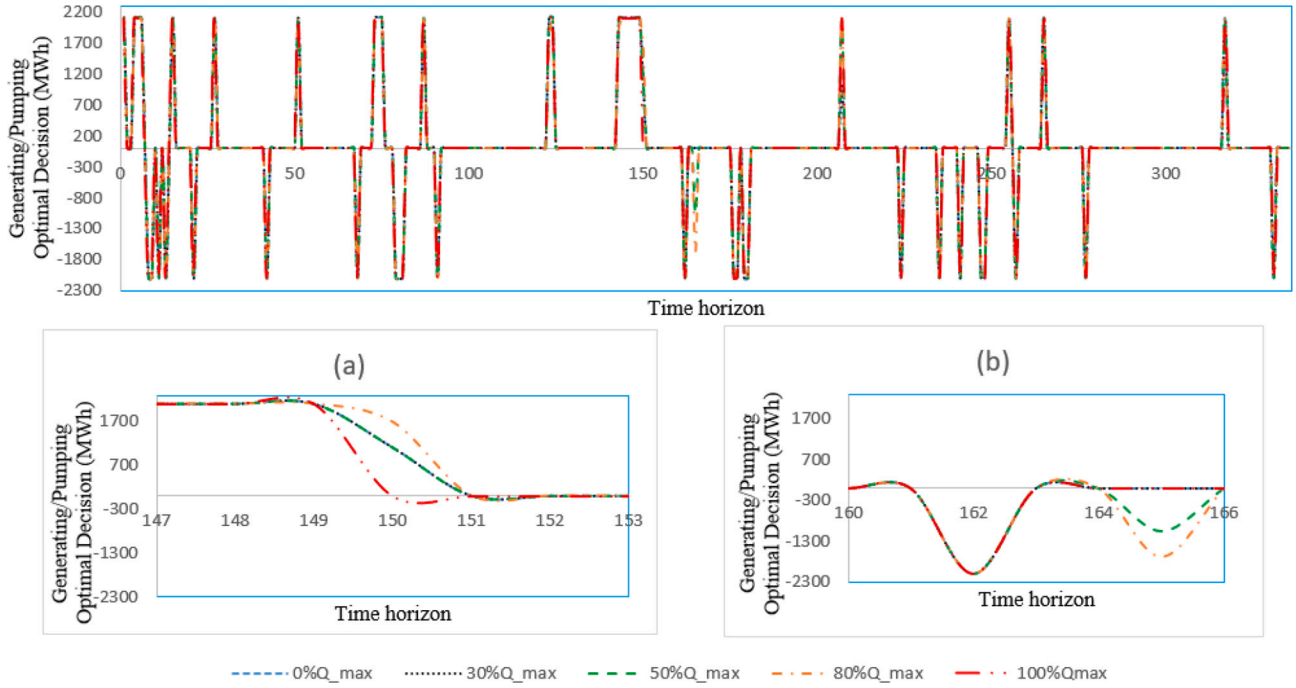


Fig. 3C. The optimal action with PSH only (12/04–17/2023).

Like Fig. 1C, Fig. 3C presents the optimal actions for a PSH-only system during a different two-week period. The consistent application of the optimal strategy across different time frames reinforces the robustness of our method.

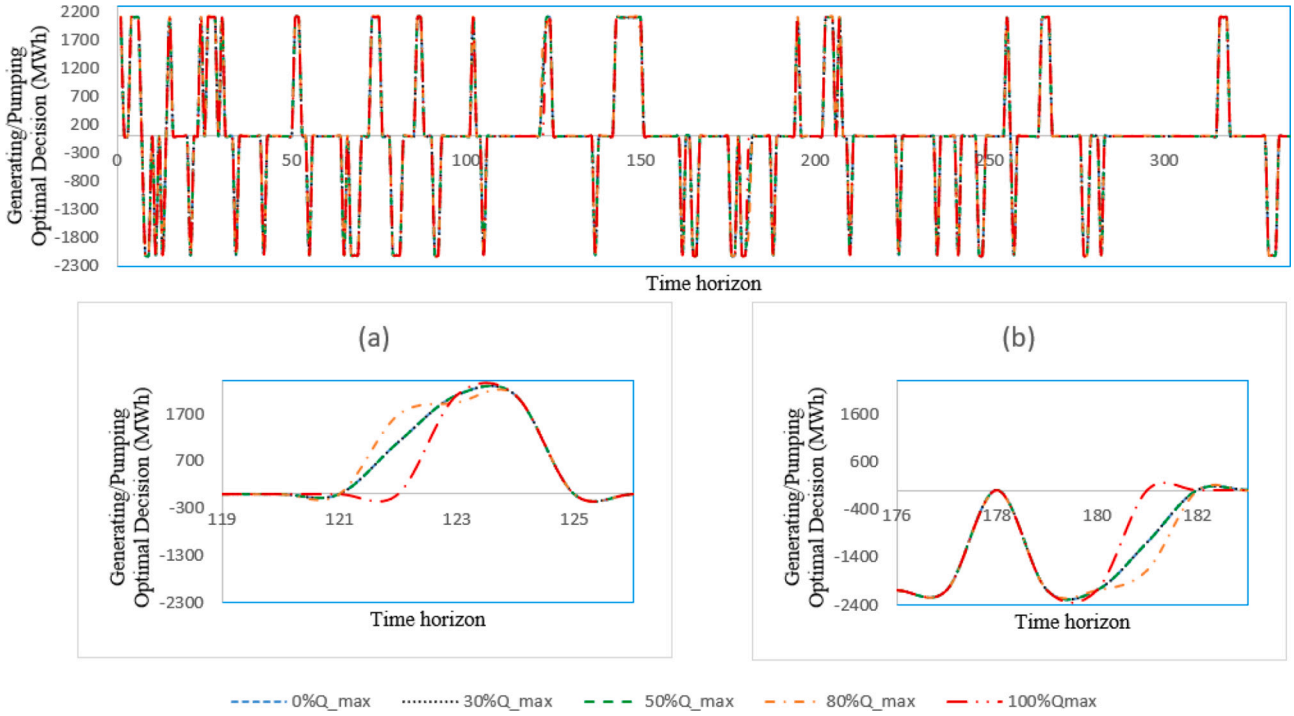


Fig. 4C. The optimal action with PSH and wind power plants (12/04–17/2023).

The results in Fig. 4C further validate the adaptability of our optimization approach to different market conditions and wind generation profiles.

The simulations using these additional data sets confirm that our proposed optimization framework consistently delivers effective dispatch strategies for both PSH-only systems and those integrated with wind power plants. The results across all three data sets (including the one in Section 6) show variations in optimal economic dispatch decisions, even when subjected to different electricity price fluctuations and wind generation levels.

References

[1] Guttikunda SK, Jawahar P. Atmospheric emissions and pollution from the coal-fired thermal power plants in India. Atmos Environ 2014;92:449–60.

[2] Wang YJ, Wang R, Tanaka K, Ciais P, Penuelas J, Balkanski Y, et al. Accelerating the energy transition towards photovoltaic and wind in China. Nature 2023;619:761–7.

[3] Martinot E. Grid integration of renewable energy: flexibility, innovation, and experience. In: Gadgil A, Gadgil TP, editors. Annual Review of Environment and Resources. Vol 41; 2016. p. 223–51.

[4] Williams JC, Wright BD. Storage and commodity markets. Cambridge University Press; 1991.

[5] Jarvinen J, Goldsworthy M, White S, Pudney P, Belusko M, Bruno F. Evaluating the utility of passive thermal storage as an energy storage system on the Australian energy market. Renew Sustain Energy Rev 2021;137:110615.

[6] Benato A, Stoppato A. Pumped thermal electricity storage: a technology overview. Therm Sci Eng Progress 2018;6:301–15.

[7] Gilmore N, Britz T, Maartensson E, Orbegoso-Jordan C, Schroder S, Malerba M. Continental-scale assessment of micro-pumped hydro energy storage using agricultural reservoirs. Appl Energy 2023;349:121715.

[8] Huang B, Chen YH, Baldick R. A configuration based pumped storage hydro model in the MISO day-ahead market. IEEE Trans Power Syst 2022;37:132–41.

[9] Lu J, Dong Y, Wang YR, Wang WL, Ding J. High efficient thermochemical energy storage of methane reforming with carbon dioxide in cavity reactor with novel catalyst bed under concentrated sun simulator. Renew Energy 2022;188:361–71.

[10] Saghafifar M, Schnellmann MA, Scott SA. Chemical looping electricity storage. Appl Energy 2020;279:115553.

[11] Bashi MH, De Tommasi L, Lyons P. Electricity market integration of utility-scale battery energy storage units in Ireland, the status and future regulatory frameworks. J Energy Storage 2022;55:105442.

[12] Nguyen MY, Yoon YT. Battery modeling for economic operation of energy storage systems in electricity markets. Int Rev Modell Simul (IREMOS) 2013;8:1000–7.

[13] Water Power Technologies Office 2022–2023 Accomplishments Report. Water Power Technologies Office; 2024.

[14] IHA. World Hydropower Outlook 2024. https://www.hydropower.org/publications/2024-world-hydropower-outlook; 2024.

[15] Nasir J, Javed A, Ali M, Ullah K, Kazmi SAA. Capacity optimization of pumped storage hydropower and its impact on an integrated conventional hydropower plant operation. Appl Energy 2022;323:119561.

[16] Nascimento J, Powell WB. An optimal approximate dynamic programming algorithm for concave, scalar storage problems with vector-valued controls. IEEE Trans Autom Control 2013;58:2995–3010.

[17] Liu J, Bo R, Wang SY, Chen HT. Optimal scheduling for profit maximization of energy storage merchants considering market impact based on dynamic programming. Comput Ind Eng 2021;155:107212.

[18] Zolfaghari M, Ghaffarzadeh N, Ardakani AJ. Optimal sizing of battery energy storage systems in off-grid micro grids using convex optimization. J Energy Storage 2019;23:44–56.

[19] Nagarajan A, Ayyanar R. Design and strategy for the deployment of Energy storage Systems in a Distribution Feeder with Penetration of renewable resources. IEEE Trans Sustain Energy 2015;6:1085–92.

[20] Diao R, Hu Z, Song Y. Subgradient of cycle-based aging cost function and its application in optimal operation of battery Energy storage system with multiple subsystems. IEEE Trans Energy Convers 2024;39:625–43.

[21] Vögelin P, Koch B, Georges G, Boulouchos K. Heuristic approach for the economic optimisation of combined heat and power (CHP) plants: operating strategy, heat storage and power. Energy 2017;121:66–77.

[22] Kim H, Sioshansi R, Lannoye E, Ela E. A stochastic-dynamic-optimization approach to estimating the capacity value of Energy storage. IEEE Trans Power Syst 2022;37:1809–19.

[23] Ntube N, Li HY. Stochastic multi-objective optimal sizing of battery energy storage system for a residential home. J Energy Storage 2023;59:106403.

[24] Xu G, Zhang Y, Sun J, Chen G, Wu C. Enable microgrid Energy management: a graph based aggregative game approach. IEEE Trans Netw Sci Eng 2024;11:5099–113.

[25] Paudel A, Chaudhari K, Long C, Gooi HB. Peer-to-peer Energy trading in a prosumer-based community microgrid: a game-theoretic model. IEEE Trans Ind Electron 2019;66:6087–97.

[26] Liu J, Ou M, Sun XY, Chen J, Mi CM, Bo R. Implication of production tax credit on economic dispatch for electricity merchants with storage and wind farms. Appl Energy 2022;308:118318.

[27] Liu J, Dai ZN, Bo R, Meng FL, Ou M. Optimal economic dispatch policy for prosumer with energy storage considering self-consumption demand. Comput Ind Eng 2023;176:108853.

[28] Zhou Y, Scheller-Wolf A, Secomandi N, Smith S. Managing wind-based electricity generation in the presence of storage and transmission capacity. Prod Oper Manag 2019;28:970–89.

[29] Zhou YF, Scheller-Wolf A, Secomandi N, Smith S. Electricity trading and negative prices: storage vs. Disposal. Manag Sci 2016;62:880–98.

[30] Liu J, Sun XY, Bo R, Wang SY, Ou M. Economic dispatch for electricity merchant with energy storage and wind plant: state of charge based decision making considering market impact and uncertainties. J Energy Storage 2022;53:104816.
[31] Felix B, Woll O, Weber C. Gas storage valuation under limited market liquidity: an application in Germany. Eur J Financ 2013;19:715–33.
[32] Wei L, Guan YP. Optimal control of plug-in hybrid electric vehicles with market impact and risk attitude. Transp Sci 2014;48:467–82.
[33] Gangwar T, Padhy NP, Jena P. Energy management approach to battery Energy storage in unbalanced distribution networks. IEEE Trans Ind Appl 2024;60:1345–56.
[34] Garcia-Gonzalez J, de la Muela RMR, Santos LM, Gonzalez AM. Stochastic joint optimization of wind generation and pumped-storage units in an electricity market. IEEE Trans Power Syst 2008;23:460–8.
[35] Braff WA, Mueller JM, Trancik JE. Value of storage technologies for wind and solar energy. Nat Clim Chang 2016;6:964–9.
[36] Tucson Electric Power. TEP's Largest Solar Plus Storage System Now in Service. https://www.tdworld.com/distributed-energy-resources/article/21163964/teps-largest-solar-plus-storage-system-now-in-service; 2024.
[37] Consumers Energy. Lake winds energy park fast facts. https://ourworldindata.org/renewable-energy; 2024.
[38] Cerejo A, Mariano S, Carvalho PMS, Calado MRA. Hydro-wind optimal operation for joint bidding in day-ahead market: storage efficiency and impact of wind forecasting uncertainty. J Mod Power Syst Clean Energy 2020;8:142–9.
[39] Zhang C, Xia P, Zhang XX. Multi-attribute decision-making method of pumped storage capacity planning considering wind power uncertainty. J Clean Prod 2024;449:141655.
[40] Karhinen S, Huuki H. Private and social benefits of a pumped hydro energy storage with increasing amount of wind power. Energy Econ 2019;81:942–59.
[41] Shi LJ, Duanmu C, Wu F, He SM, Lee KY. Optimal allocation of energy storage capacity for hydro-wind-solar multi-energy renewable energy system with nested multiple time scales. J Clean Prod 2024;446:141357.
[42] Kim JH, Powell WB. Optimal Energy commitments with storage and intermittent supply. Oper Res 2011;59:1347–60.
[43] Secomandi N. Optimal commodity trading with a capacitated storage asset. Manag Sci 2010;56:449–67.
[44] Cao D, Hu WH, Xu X, Dragicevic T, Huang Q, Liu Z, et al. Bidding strategy for trading wind energy and purchasing reserve of wind power producer - a DRL based approach. Int J Electr Power Energy Syst 2020;117:105648.
[45] Toubeau JF, Bottieau J, De Grève Z, Vallée F, Bruninx K. Data-driven scheduling of energy storage in day-ahead energy and reserve markets with probabilistic guarantees on real-time delivery. IEEE Trans Power Syst 2021;36:2815–28.
[46] Martinez-Rico J, Zulueta E, de Argandoña IR, Fernandez-Gamiz U, Armendia M. Multi-objective optimization of production scheduling using particle swarm optimization algorithm for hybrid renewable power plants with battery Energy storage system. J Mod Power Syst Clean Energy 2021;9:285–94.
[47] Favaro P, Gobert M, Toubeau J-F. Multi-fidelity optimization for the day-ahead scheduling of pumped hydro Energy storage. J Energy Storage 2024;103:114096.
[48] Ochoa T, Gil E, Angulo A, Valle C. Multi-agent deep reinforcement learning for efficient multi-timescale bidding of a hybrid power plant in day-ahead and real-time markets. Appl Energy 2022;317:119067.
[49] Li X, Yang W, Liao Y, Zhang S, Zheng Y, Zhao Z, et al. Short-term risk-management for hydro-wind-solar hybrid energy system considering hydropower part-load operating characteristics. Appl Energy 2024;360:122818.
[50] Tian MW, Yan SR, Tian XX, Nojavan S, Jermsittiparsert K. Risk and profit-based bidding and offering strategies for pumped hydro storage in the energy market. J Clean Prod 2020;256:120715.
[51] Topalović Z, Haas R, Sayer M. Economic benefits of PHS and Li-ion storage. Study cases: Austria and Bosnia and Herzegovina. Appl Energy 2024;362:122988.
[52] Wang H, Liao S, Cheng C, Liu B, Fang Z, Wu H. Short-term scheduling strategies for hydro-wind-solar-storage considering variable-speed unit of pumped storage. Appl Energy 2025;377:124336.
[53] Li ZF, Zhang Q, Guo Q, Nojavan S. Pumped hydro energy storage arbitrage in the day-ahead market in smart grid using stochastic p-robust optimization method. Sustain Cities Soc 2021;75:103274.
[54] Zhang S, Xiang Y, Liu JY, Liu JC, Yang JX, Zhao X, et al. A regulating capacity determination method for pumped storage hydropower to restrain PV generation fluctuations. Csee J Power Energy Syst 2022;8:304–16.
[55] Gong Y, Liu T, Liu P, Tong X. Joint unit commitment model for hydro and hydrogen power to adapt to large-scale photovoltaic power. Energy Convers Manag 2024;317:118794.
[56] Agarwal A, Pileggi L. Large scale multi-period optimal power flow with Energy storage systems using differential dynamic programming. IEEE Trans Power Syst 2022;37:1750–9.
[57] Khaloie H, Vallée F. Day-ahead dispatch of liquid air Energy storage coupled with LNG regasification in electricity and LNG markets. IEEE Trans Power Syst 2024;39:5177–90.
[58] Liang WK, Lin SJ, Liu MB, Wang Q, Xie YQ, Sheng X. Stochastic economic dispatch of regional integrated energy system considering the pipeline dynamics using improved approximate dynamic programming. Int J Electr Power Energy Syst 2022;141:108190.
[59] Lauinger D, Vuille F, Kuhn D. Frequency regulation with storage: on losses and profits. Eur J Oper Res 2024;319:442–55.
[60] Alvarez M, Rönnberg SK, Bermúdez J, Zhong J, Bollen MHJ. A generic storage model based on a future cost piecewise-linear approximation. IEEE Transactions on Smart Grid 2019;10:878–88.
[61] Toufani P, Nadar E, Kocaman AS. Operational benefit of transforming cascade hydropower stations into pumped hydro energy storage systems. J Energy Storage 2022;51:104444.
[62] Pan Y, Lin SJ, Liang WK, Feng XY, Sheng X, Liu MB. Stochastic optimal dispatch of offshore-onshore regional integrated energy system based on improved state-space approximate dynamic programming. Int J Electr Power Energy Syst 2024;155:109661.
[63] Zheng NK, Jaworski JJ, Xu BL. Arbitraging variable efficiency energy storage using analytical stochastic dynamic programming. IEEE Trans Power Syst 2022;37:4785–95.
[64] Baldick R, Chen YH, Huang B. Optimization formulations for storage devices with disjoint operating modes. Oper Res 2023;71:1978–96.
[65] Porteus EL. Foundations of stochastic inventory theory. Stanford University Press; 2002.
[66] Puterman ML. Markov decision processes: Discrete stochastic dynamic programming. John Wiley & Sons; 2014.
[67] Vladimir K, Quentin P, Jonghwan K, Patrick B. A review of technology innovations for pumped storage hydropower. United States: N. p.; 2022. https://doi.org/10.2172/1867238. Web.